Reading the Mind of the Machine

The Desk Volume

A Course in Mechanistic Interpretability — the season's science in full paperwork: definitions, derivations, protocols, exhibits, and problems

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Edition v2.3.2

How to Use This Volume

Each chapter of this volume is the desk companion to one episode of the season — and each is written to stand on its own, to be *reread* rather than merely consulted. A chapter walks its episode's ground in the same order, one narrated pass under the episode's own part titles — the same story, told for the page, with less chat and more proof — and the formal matter (the definitions and propositions, the derivations the narration can only gesture at, the protocol boxes and the trace exhibits) is set *inline*, at the exact story beat that wants it. The episodes refer to this desk; the desk rarely returns the call — what the season says, the page states outright. The season's spoken furniture — the shared notebook and its clerks, the fold, the microscope, the keyhole, the leash — is standing vocabulary here, introduced beside its formal twins where each first appears and worn thereafter without ceremony. After the narrated body comes the Bench — the chapter's laboratory briefs, for the reader who wishes to make his own machine produce the figures printed here; every exhibit of real model internals in this volume was produced by a named Bench lab, and its caption says so. Then the problems, in three tiers: Tier I is calisthenics; Tier II is the real thing, including the problems that finish arguments the chapter sketches; Tier III is starred and optional. Hints are segregated and spoiler-safe; solutions are complete for Tier I, complete for the Tier II problems that finish such arguments, and withheld (hints only) for Tier III. Each chapter closes with The Reading: the primary sources, with instructions for reading them against the desk's own objects. Nothing here appears unmotivated by the audio, and nothing in the audio pretends to a rigour that is not here. The volume closes with its back matter: a Debt Register, an Exhibit Index, and a Protocol-Box Index. The companion treatise on evaluation, which Chapter 14 converses with, is planned to join a later edition as a chapter of its own; until it does, Chapter 14 stands alone and reconstructs its method where needed.

Contents

Chapter 1

Reading the Mind of the Machine

Companion to Episode One. The panorama, with the paperwork done: the grown object and the confession of its opacity; the residual stream as a shared notebook, written as an equation, and the lenses that read it mid-thought; features as directions; the board caught in the machine, and the probe that reads it; superposition, its packing argument a proposition with a proof, and the instrument that unfolds it; circuits from the curve detectors to the induction pair; the honest borders of the science; and the bench never far away.

1.1 The Grown, Not Built, Object

The confession that opens the course is an engineering fact with an exact statement. We possess the complete construction of the model — architecture, objective, optimiser, data — and we do not possess the program the construction produced, because the program is the settled values of the weights, and those were *found* by optimisation in a space no mind can survey. The distinction between a written artifact and a grown one is the distinction between an object whose explanation is its source text and an object whose explanation must be *recovered by experiment.* Everything in this volume is downstream of that one asymmetry: the chapters that follow are, without exception, descriptions of instruments for the recovery, and of the discipline that keeps their readings honest.

1.2 Part One — The Residual Stream as a Shared Notebook

One image above all others: not a bucket brigade but a shared notebook. Here it is as arithmetic.

Definition 1.1 (The residual stream)**.** Fix a transformer with L layers and model width d_{model}. At each token position, the *residual stream* is the sequence of vectors $x_0, x_1, \dots, x_L \in \mathbb{R}^{d_{\text{model}}}$, where x_0 is the token's embedding (plus any positional term the architecture adds there), and each layer contributes additively:

$$x_{\ell+1} \;=\; x_\ell \;+\; a_\ell(x_\ell) \;+\; m_\ell\big(x_\ell + a_\ell(x_\ell)\big),$$

with a_ℓ the layer's attention sublayer and m_ℓ its MLP sublayer (normalisation absorbed into the sublayers for this statement). Unrolling,

$$x_L \;=\; x_0 \;+\; \sum_\ell a_\ell \;+\; \sum_\ell m_\ell,$$

the final state is the *sum of every note ever added.*

The unrolled form is the notebook, and the sublayers a_ℓ and m_ℓ are its clerks: no term is ever erased, every later reader receives every earlier writer's contribution intact, and any question of the form "where did this behaviour come from?" is, in principle, a question about which terms of the sum carry it. That the two arrangements — relay and notebook — have entirely different failure modes is now visible in the formula: nothing in the sum *obliges* an intermediate layer to touch a contribution written below it.

The season's first instrument reads the notebook mid-thought; its tuned companion allows for the handwriting that drifts — the layerwise change of representational basis.

Definition 1.2 (Logit lens; tuned lens)**.** For an intermediate state x_ℓ, the *logit lens* is the readout $W_U\,\mathrm{LN}(x_\ell)$ — the summit's own conversion of stream to next-token scores, applied early. The *tuned lens* replaces this with $W_U\,\mathrm{LN}(A_\ell x_\ell + b_\ell)$, where the affine translator (A_ℓ, b_ℓ) is trained, per layer, to minimise the divergence between the early readout and the model's final distribution.

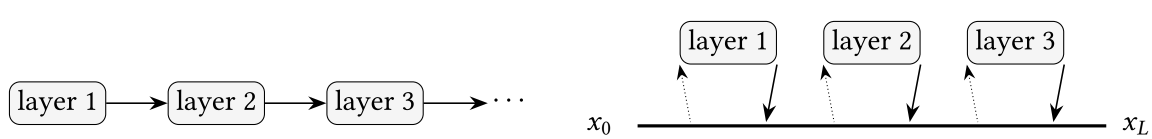
Remark 1.3. The tuned lens exists because the naive lens fails on many model families, and the *reason* it fails is the chapter's first moral in miniature: the notebook's notational conventions drift as it ascends, and the summit's reading apparatus is fitted to the summit's hand. Every reading is a joint product of the specimen and the lens — an instrument-relative fact, not a fact about the stream alone. The season meets this moral again at every price point, most expensively in Chapter 5.

Protocol — Running a logit lens, then a tuned lens (checked against the tuned-lens paper's setup)

1: Load a small model in TransformerLens (GPT-2 small suffices; it is the fruit fly of this volume).
2: Run a prompt; cache the residual stream x_ℓ at every layer at the final position.
3: For each ℓ, apply the model's own final LayerNorm and unembedding to x_ℓ; record the top token and the probability assigned to the eventually-chosen token.
4: Plot rank and probability against ℓ: easy tokens settle early; hard tokens churn late.
5: Repeat on a documented failure family — GPT-Neo or BLOOM, the tuned-lens paper's own dramatic specimens — and observe the naive lens produce fluent nonsense (on both, its top prediction is often the *input* token, layer after layer).
6: Install the *tuned-lens* package, load its published translators for the same model, and repeat. The legible picture returns.

This protocol is Lab 1.1's skeleton; the lab adds the measurement harness and the figures printed as Exhibit 1.2.

The two exhibits below fix the chapter's two pictures: the wrong one and the right one, then the lens's view of a thought converging.



the bucket brigade (wrong) *the shared notebook (right)*

Exhibit 1.1 (Relay against notebook). The wrong picture beside its replacement. In the brigade, each layer's output is the next layer's whole input; in the notebook, layers *read from* (dotted) and *add* *to* (arrowed) one persistent stream, and nothing obliges a layer to disturb what an earlier layer wrote. Schematic.

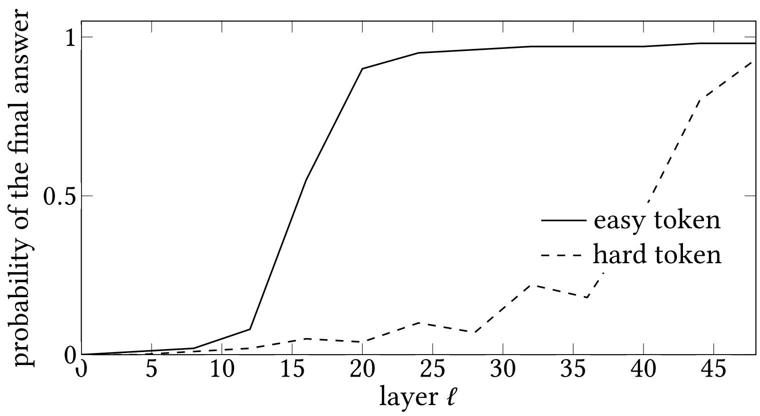


Exhibit 1.2 (A thought converging, through the lens). The logit lens's characteristic signature: an easy, predictable token is settled less than halfway up the stack and the remaining layers scarcely touch it; a genuinely hard token churns — lurching between candidates — into the final layers. *Schematic. Lab 1.1 produces the real figure from GPT-2 measurements; a future edition prints it in this frame, naming the prompt.*

1.3 Part Two — Features and Directions

What is written in the notebook? Directions that mean something — so runs the field's central working assumption, and it deserves a statement precise enough to be doubted.

Definition 1.4 (Feature; the linear representation hypothesis)**.** A *feature* is a property of the input together with a direction $v \in \mathbb{R}^{d_{\text{model}}}$ (conventionally unit-norm) such that the model represents the property's presence, with strength s, by adding $s\,v$ into the residual stream, and downstream components read it by the inner product $\langle x, v\rangle$. The *linear representation hypothesis* is the working assumption that human-interpretable concepts are represented as features in this sense, and that co-occurring concepts compose by *addition*: $x \approx \sum_i s_i\,v_i$ over the concepts active at that position.

Remark 1.5 (Scope of validity). Definition 1.4 is an empirical hypothesis, not an axiom, and this volume treats it the way the season does: extraordinarily productive, well-evidenced in many places, and *known to be incomplete* — Chapter 4 exhibits concepts that live on curved, multi-dimensional structures for which a single direction is the wrong container, and Chapter 4's privileged-basis exception locates the one corner of the model where individual coordinates carry meaning after all. Instruments built atop the hypothesis (probes, autoencoders, steering vectors) inherit its scope; when a later chapter's instrument misbehaves, the first suspect is always this definition's fine print.

The hypothesis's ancestry is the word-embedding arithmetic — *king* − *man* + *woman* ≈ *queen* — and the desk owes the folklore its footnote-sized audit: in the classic evaluations the input words were excluded from the candidate answers, without which the nearest neighbour of the composed vector is very often an input word itself. The arithmetic is real but weaker than the party trick suggests; what survives scrutiny is exactly what the hypothesis needs — early, unsupervised models encoding semantic relations as *consistent directions* — and Problem 1.5 has you measure both the trick and its inflation on your own machine.

The counting argument, stated once and bluntly: a frontier model demonstrably deploys concepts numbering in the millions; its stream has some tens of thousands of dimensions at most; therefore one-concept-one-coordinate is not parsimony but arithmetic impossibility; the fold — superposition, more features than dimensions — is forced; and *if* concepts are directions, most pairs of those directions cannot be orthogonal. How many merely-distinguishable directions a space affords is Part Four's proposition, and the answer is the reason the whole subject exists.

1.4 Part Three — The Board in the Machine

The evidence before the hypothesis is a laboratory case: a world small enough to check. The instrument introduced along the way is used in every chapter that follows.

Definition 1.6 (Linear probe)**.** Given internal states $x \in \mathbb{R}^d$ and a labelled property y, a *linear probe* is a classifier of the form $\hat{y} = \sigma(\langle x, w\rangle + b)$ (or its multiclass analogue) trained by us — not the model — to predict y from x. High held-out probe accuracy certifies the information is *linearly present* in the representation; it does not, by itself, certify the model *uses* it. Use is established by intervention: edit x along w and observe whether downstream behaviour follows the edit.

The Othello result, in this vocabulary: a small transformer trained only on legal-move sequences was probed for the board state. Nonlinear probes read the board; linear probes read it poorly — apparent trouble for Definition 1.4 — until the labels were re-expressed from *black/white* to *mine/theirs* relative to the player to move, whereupon plain linear probes read the board cleanly, and activation-level edits along the probe directions changed the model's play in accordance with the edited board, not the actual history. Three findings ride on that story: world models condense out of next-token prediction; the linear hypothesis survived its sternest available test; and the probe's initial failure was a failure of *our* vocabulary, not the model's representation.

Remark 1.7 (The wrong-vocabulary lesson, generalised). An empty-handed probe admits two readings — nothing is there, or the question was asked in the wrong parameterisation — and nothing in the probe's output distinguishes them. The mine/theirs resolution is this volume's standing counterexample to "we probed for it and found nothing." Every negative probing claim in later chapters carries an implicit *modulo vocabulary* qualifier, and the honest report of a failed probe names the vocabularies tried.

Protocol — Training and validating a linear probe (checked against the Othello-GPT line of work)

1: Choose the property y and the site: layer, position, and which stream (residual, attention out, MLP out).
2: Collect (x, y) pairs from a corpus the model processes naturally; hold out a test split *by game/document*, not by token, so near-duplicates cannot leak.
3: Fit the probe; report held-out accuracy against the trivial baselines (majority class; probing the *embedding* layer, which knows the input but no computation).
4: Run the negative control: probe an untrained model of the same architecture. Structure found there is structure the probe brought with it.
5: If accuracy is poor, re-parameterise the labels before concluding absence (Remark 1.7) — the model's ontology, not yours.
6: To claim *use*, intervene: move x along the probe direction to encode a counterfactual value

of y; verify downstream behaviour follows the counterfactual, not the history.

Lab 1.2 executes this protocol end to end on Othello-GPT, including the re-parameterisation and the intervention.

1.5 Part Four — Superposition: the Packing Proposition

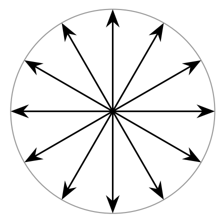
An intuition first, payment after: high-dimensional space is indecently roomy for directions merely distinguishable, not perpendicular — pairwise $|\cos\theta| \le \epsilon$, near-orthogonality. The payment:

Proposition 1.8 (Exponentially many near-orthogonal directions)**.** *Fix $\epsilon \in (0,1)$. In $\mathbb{R}^d$ there exist $N = \lfloor e^{d\epsilon^2/4} \rfloor$ unit vectors that are pairwise ϵ-near-orthogonal: $|\langle v_i, v_j \rangle| \le \epsilon$ for all $i \ne j$.*

Proof. Draw N vectors independently and uniformly from the unit sphere. For a fixed pair, the inner product concentrates: by the standard spherical measure bound, $\Pr\big[|\langle v_i, v_j\rangle| > \epsilon\big] \le 2e^{-d\epsilon^2/2}$. There are $N(N-1)/2$ pairs, so the probability that *any* pair violates the bound is at most $N(N-1)\, e^{-d\epsilon^2/2}$, which is strictly below $N^2 e^{-d\epsilon^2/2} \le e^{d\epsilon^2/2}\, e^{-d\epsilon^2/2} = 1$ for our $N \le e^{d\epsilon^2/4}$ — so a configuration with every pair compliant exists. (Random directions *are* the construction; no cleverness is required, which is itself the lesson; and Problem 1.4 redoes the accounting with every constant kept.) □

Remark 1.9. For d_{model} in the thousands and tolerances a model can live with, $e^{d\epsilon^2/4}$ dwarfs any concept inventory a model could want — the librarian shelves the million volumes after all. The cost is interference: by the calculation of Problem 1.2, a feature active at strength s contaminates an ϵ-overlapping neighbour's readout by exactly $s\epsilon$. Superposition is therefore a wager on *sparsity* — that few features are active at once, so the contaminations rarely compound — and Chapter 4 prices the wager precisely: when it is taken, what geometry it produces, and what happens when the sparsity assumption thins. This proposition sits in the neighbourhood of the Johnson–Lindenstrauss lemma; "in the neighbourhood of" is a phrase chosen with care, and the distinction is Problem 1.7's business.

Remark 1.10 (Distinguishable is not near-orthogonal). A caution about the chapter's own favourite picture. The clock face of Exhibit 1.3 packs directions by *minimum angle* — thirty degrees of separation — which is a different criterion from near-orthogonality's $|\cos\theta| \le \epsilon$: adjacent hands overlap at $|\cos\theta| \approx 0.87$, and opposite hands at exactly 1 — maximal interference, not near-orthogonality at all. In two dimensions no more than two directions can be near-orthogonal at any small tolerance; the exponential generosity of Proposition 1.8 is strictly a high-dimensional purchase, and the clock illustrates the *relaxation*, not the payoff. The antipodal pairs are not thereby useless: a direction and its negative can house two features that rarely co-occur — Chapter 4's toy models fold features into exactly such *antipodal pairs* as their first arrangement — but that is a different economy (sharing under sparsity) from the one this proposition prices (many almost-private directions).



twelve distinguishable directions in $d = 2$

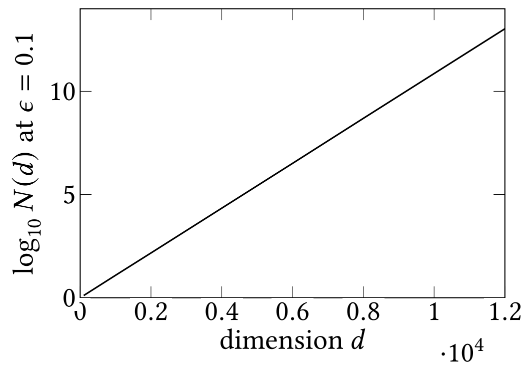


Exhibit 1.3 (The clock face, and the exponential). Left: in two dimensions, relaxing perpendicular to distinguishable buys a clock face — a dozen directions instead of two, though under the weaker minimum-angle criterion of Remark 1.10, not near-orthogonality. Right: the relaxation's true payoff in high dimension, via Proposition 1.8's bound ($\log_{10}$ of the guaranteed packing count at tolerance $\epsilon = 0.1$): the guarantee alone reaches trillions of directions well before frontier widths. Mathematical schematic; nothing here was measured on a model.

1.6 Part Five — The Sparse Autoencoder, in Outline

The panorama gives the unfolding instrument one part; Chapter 5 gives it an entire chapter, and the desk discipline is to state here only what the panorama actually uses: the shape of the objective, so that its two constraints can be seen to mirror the two properties superposition forces on the model's true vocabulary.

$$f = \mathrm{ReLU}(W_{\mathrm{enc}}\, x + b_{\mathrm{enc}}), \qquad \hat{x} = W_{\mathrm{dec}}\, f + b_{\mathrm{dec}}, \qquad \mathcal{L} = \|x - \hat{x}\|_2^2 + \lambda\, \|f\|_1,$$

with f living in a space far *wider* than d_{model} (the million-slot middle) and the ℓ_1 term holding it *sparse*. Wide and sparse are precisely the two properties the counting argument and the sparsity wager assign to the model's own feature vocabulary; the autoencoder is a machine whose only route to a good score is to rediscover that vocabulary. The Golden Gate demonstration — locate the one slot, clamp it, watch the model declare itself a bridge — matters to this chapter for one reason: it showed a recovered feature to be a *causal lever*, not a correlate. Everything else about the instrument, including the reasons to distrust it and the reckoning that repriced it, is Chapter 5's business — the season's most consequential.

1.7 Part Six — Circuits: the Grammar

Definition 1.11 (Circuit)**.** A *circuit* is a subgraph of the model's computation — specific heads, MLP units or features, and the weights connecting them — together with a claimed algorithm, such that the subgraph provably implements the algorithm and the algorithm accounts for a measured behaviour. The proof standard is interventional: components are individually perturbed and the behaviour follows the account (Chapter 6 makes the standard precise; Chapter 9 mechanises the search).

The vision anatomy supplies the evidence standard the definition presupposes, and it is worth recording as a checklist, because it recurs in every dissection the season performs: converging independent evidence (synthetic stimuli that maximally excite the unit; natural stimuli sorted by response; smooth parametric variation of the stimulus handing the response from unit to unit), then the flourish —

build the part by hand from the recovered blueprint and watch the hand-made organ behave like the grown one. And the finding that gives the programme its hope, *universality*: the same features and some of the same circuits — the same entries of the dictionary, the same turns of the grammar — recur across networks of different architectures and upbringings, so anatomy learned on one specimen transfers — tentatively, but transferably — to the next.

The induction pair — the season's first solved language circuit — is deliberately *not* formalised here: this chapter carries only the sketch (a previous-token head that annotates; an induction head that hunts the annotation), and the full trace, with the two scores that certify membership and the path expansion that isolates the mechanism, is Chapter 3's centrepiece, where it can be done once and properly. What belongs to this chapter is the moral the sketch already licenses: two individually trivial components, composing through the shared notebook of Definition 1.1, implement a genuinely general algorithm nobody wrote — and that composition through the stream, not any single part's cleverness, is where the machine's sophistication lives.

The Engineer's Dividend

The panorama's cash value, in three registers. *Vocabulary:* when a fine-tune regresses or a quantised model degrades in one narrow respect, "which feature directions were damaged?" and "which circuit's components were disturbed?" are askable, testable questions where "the weights changed" is not — and the outlier-dimension asterisk of Chapter 2 is the quantisation engineer's first worked example. *Instruments:* the logit lens (Definition 1.2) and the linear probe (Definition 1.6) are an afternoon's code apiece and both run on the models you already serve; the protocol boxes above are written to be executed, not admired. *Posture:* the difference between a mechanistic hypothesis and a vague one is the difference between engineering and superstition — a mechanistic hypothesis names the component, predicts the intervention's result, and can be wrong in public, which is precisely what makes it useful.

The Honest Limits

The four borders of the science, each drawn with its precise scope. *Scale:* the fully-traced circuits live in small models; a frontier model's circuit count is unknown and large; the validated methodology has been applied exhaustively to a sliver of any real model — a beautiful map of a few streets. *The instrument:* the autoencoder's dictionary is lossy and non-unique (Chapter 5 measures both defects), so feature inventories are readings through a microscope known to be slightly out of focus. *The hypothesis:* Definition 1.4 is not universal — Chapter 4's curved features bound it from below — and instruments built on it mischaracterise what falls outside it, silently. *The whole:* a complete parts-list is not an understanding; whether a full mechanistic account of a frontier model would be humanly comprehensible — or merely a better-organised bewilderment — is open, and it is the deepest open question the field owns.

1.8 The Bench

Chapter 1's laboratory briefs; full write-ups in `lab/`*. Statuses per the Bench Manual.*

1.1 *The logit lens, then the tuned lens* (T1, an evening; authored). GPT-2 small per the protocol box, then a documented failure family (GPT-Neo or BLOOM) naive against tuned. Produces Exhibit 1.2's real data. Expected: early settling for easy tokens; naive-lens gibberish on the failure family repaired by the translators.

1.2 *Othello-GPT: probe, re-probe, intervene* (T2, a weekend; authored). The probe protocol end to end on the published Othello-GPT checkpoint: black/white probes fail where mine/theirs probes succeed; a probe-direction edit flips a disc in the model's mind and the continuation obeys the edited board. Every claim of Part Three becomes a cell you ran.

1.3 *Superposition in a histogram* (T1, an hour; authored). Sample $k \gg d$ random unit vectors for $d \in \{10, 100, 10{,}000\}$; histogram pairwise cosines; watch concentration produce Proposition 1.8 empirically.

1.4 *Feature safari with a clamp* (T0–T2, an evening; authored). Neuronpedia on Gemma-2-2B: find a crisp feature, clamp it during generation via the steering interface, reproduce the obsession and find the dose at which it becomes incoherence — previewing Chapter 5's honesty about the instrument.

1.5 *Residual arithmetic, audited* (T1, an evening; authored). Reproduce *king* − *man* + *woman* on word2vec with and without excluding input words from the candidates; then attempt the analogous arithmetic in GPT-2's embedding space. Produces the numbers for Problem 1.5.

1.9 Problems

Tier I — Calisthenics

Problem 1.1 (The notebook, unrolled)**.** Starting from the layerwise update in Definition 1.1, derive the unrolled form, and state exactly what had to be assumed about the normalisation layers for the derivation to go through unchanged. *(Full solution.)*

Problem 1.2 (Reading a direction)**.** A feature is claimed to live along the unit direction v. Write the readout a later layer would use to measure it; then show that if a second feature lives along w with $\langle v, w\rangle = \epsilon$, an activation of strength s on the second feature contaminates the first's reading by exactly $s\epsilon$. *(Full solution.)*

Problem 1.3 (The clock face counted)**.** On the plane, how many unit directions can be placed so that every pair is separated by at least thirty degrees (the minimum-angle criterion of Remark 1.10)? Verify your count against the clock face of Exhibit 1.3, and explain in one sentence why the analogous packing question in $d_{\text{model}} = 12{,}288$ dimensions cannot be answered by any mental picture. *(Full solution.)*

Tier II — The Real Thing

Problem 1.4 (The packing constants, earned)**.** Prove the spherical concentration bound used in Proposition 1.8: for u, v independent uniform unit vectors in $\mathbb{R}^d$, $\Pr[|\langle u, v\rangle| > \epsilon] \le 2e^{-d\epsilon^2/2}$. (Fix v; bound the spherical cap's measure.) Then redo the union bound keeping constants explicit, and compute the guaranteed packing count for $d = 12{,}288$, $\epsilon = 0.05$. Compare it to any plausible concept inventory. *(Full solution — this problem completes the chapter's sketch.)*

Problem 1.5 (Word2vec, deflated and vindicated)**.** Using Lab 1.5's setup: measure the top-1 analogy accuracy of the classic arithmetic on a standard analogy set, first excluding input words from the candidate set (the classic protocol), then including them. Report both numbers; explain what each protocol actually measures; and state precisely which claim of Definition 1.4 survives the weaker number intact. *(Hints only.)*

Problem 1.6 (The vocabulary trap, constructed)**.** Build the smallest synthetic example you can of the mine/theirs phenomenon: a data-generating process and representation in which a natural labelling is not linearly decodable but an equivalent re-parameterised labelling is. (Two features and an alternating "player to move" bit suffice.) Prove both halves. *(Full solution — this problem is Remark 1.7 made portable.)*

Tier III — For the Ambitious (starred)

Problem 1.7 (Johnson–Lindenstrauss, at arm's length)**.** ⋆ State the Johnson–Lindenstrauss lemma precisely, and explain — in prose an engineer would sign — why Proposition 1.8 is its neighbour and not its corollary: what JL preserves that packing does not require, and what packing guarantees that JL does not mention. Then show that a JL projection of a sparse feature family into $\mathbb{R}^d$ yields a near-orthogonal packing, recovering a weaker form of the proposition. *(Hints only.)*

Problem 1.8 (Decoration or working state)**.** ⋆ Design, in full experimental detail for a model and property of your choosing, an intervention protocol that would distinguish a representation the model *consults* from one it merely *carries*. State the confound your design defeats that a plain probing study does not, and the confound it still cannot defeat. *(Hints only.)*

1.10 Hints

Hint (Problem 1.1)**.** The update as stated has already absorbed normalisation into the sublayers; ask what property of the *absorbed* maps the unrolling uses — and notice it uses none.

Hint (Problem 1.4)**.** The cap measure: condition on $v = e_1$; then $\langle u, v\rangle$ is the first coordinate of a uniform point on the sphere, and its tail is bounded by the Gaussian-like estimate $\Pr[u_1 > \epsilon] \le e^{-d\epsilon^2/2}$ (prove it via the Beta distribution of u_1^2, or geometrically by comparing cap area to hemisphere area).

Hint (Problem 1.5)**.** The candidate-exclusion switch is one argument in every standard analogy evaluator; the interesting quantity is the *gap* between the two accuracies, and the interesting question is which of the two protocols a believer in Definition 1.4 is entitled to cite.

Hint (Problem 1.6)**.** Let the representation store $m = c \oplus p$ (colour XOR player) in one coordinate and p in another; ask which of c and m a linear readout can recover, and from what.

Hint (Problem 1.7)**.** JL is a statement about *preserving* the geometry of a given point set under projection; packing is a statement about the *existence* of a geometry. One is a map, the other a census.

Hint (Problem 1.8)**.** The Othello editing experiment is the template; the confound a probe cannot defeat is decoration, and the confound your edit cannot defeat is off-distribution damage — Chapter 6's ablation section names it and prices it.

1.11 Solutions

Policy: full for Tier I and for Problems 1.4 and 1.6 (they complete arguments the chapter sketches); hints only for Problems 1.5, 1.7, and 1.8.

Solution (Problem 1.1)**.** Substitute repeatedly: each application replaces $x_{\ell+1}$ with x_ℓ plus the two contribution terms; after L substitutions the sum contains x_0 and every contribution once. Nothing about the contributions was used — linearity is *not* required, only that each layer's output is *added* to its input rather than replacing it. The normalisation caveat: the statement absorbed LayerNorm into a_ℓ and m_ℓ; the unrolling is exact under that absorption, and what is *lost* is only the ability to read a single fixed basis across layers without accounting for the per-layer normalisation — the tuned lens of Definition 1.2 exists precisely to pay that account.

Solution (Problem 1.2)**.** The readout is $\langle x, v\rangle$. With $x = s\,w$ (the second feature active at strength s, the first silent), the first feature's reading is $\langle s\,w, v\rangle = s\,\langle w, v\rangle = s\epsilon$: interference is bilinear in activation strength and overlap. This two-line calculation is the seed of the entire superposition economy of Chapter 4 — the cost side of its ledger.

Solution (Problem 1.3)**.** Twelve: the plane affords 360/30 directions at exactly thirty-degree spacing, and no thirteenth fits. The mental picture fails in high dimension because the count grows as $e^{c\epsilon^2 d}$ (Proposition 1.8 — under the stricter near-orthogonality criterion, at that; Remark 1.10) while every picture we can hold is two- or three-dimensional, where the count is linear in the inverse angle; the imagination underestimates the sphere by a factor that is itself exponential.

Solution (Problem 1.4)**.** *Cap bound.* Condition on v; by symmetry take $v = e_1$, so $\langle u, v\rangle = u_1$. For a uniform point on the sphere, $u_1^2 \sim \mathrm{Beta}(\frac{1}{2}, \frac{d-1}{2})$, whence $\Pr[u_1 > \epsilon] = \frac{1}{2} I_{1-\epsilon^2}(\frac{d-1}{2}, \frac{1}{2}) \le e^{-d\epsilon^2/2}$ for $d \ge 3$ (bound the incomplete Beta by its integrand's maximum; or geometrically, the cap of angular radius $\arccos \epsilon$ has area at most that of a hemisphere scaled by $(1-\epsilon^2)^{(d-1)/2} \le e^{-(d-1)\epsilon^2/2}$). Doubling for both signs gives the stated two-sided bound. *Union bound with constants.* With N random directions, $\Pr[\text{some pair violates}] \le \binom{N}{2} \cdot 2e^{-d\epsilon^2/2} < N^2 e^{-d\epsilon^2/2}$; this is below one whenever $N \le e^{d\epsilon^2/4}$, at which point a compliant configuration exists (and in fact random draws succeed with probability approaching one for N half that size). *The number.* For $d = 12{,}288$ and $\epsilon = 0.05$: $d\epsilon^2/4 = 12{,}288 \times 0.0025/4 = 7.68$ — in natural-log units, $e^{7.68} \approx 2{,}170$ directions *guaranteed by this crude bound alone*; at $\epsilon = 0.1$ the same arithmetic gives $e^{30.7} \approx 2\times10^{13}$. The bound is loose — sharper constants and non-random constructions do far better — but even the loose guarantee at a five-percent tolerance already exceeds a dictionary of two thousand concepts per *completely non-interfering* family, and at ten percent it exceeds any concept inventory ever proposed by orders of magnitude. The librarian's bookcase was never the constraint.

Solution (Problem 1.6)**.** Generating process: a two-square board; each turn, square colours $c_1, c_2 \in \{+1, -1\}$ and player-to-move $p \in \{+1, -1\}$ alternating. Representation: $x = (c_1 p,\ c_2 p,\ p)$ — the model stores each square as *mine/theirs* ($m_i = c_i p$) plus whose turn it is. *Colour is not linearly decodable from the first two coordinates alone*: $c_i = m_i p$ is a product of two stored quantities, and over the uniform distribution $\mathbb{E}[c_i \cdot (\text{any linear function of } m_1, m_2)] = 0$, so every linear readout of the mine/theirs coordinates has chance accuracy on colour — while m_i itself is read off coordinate i exactly. (With the p coordinate included, colour becomes a *bilinear* function, $c_i = m_i p$ — still outside a linear probe's class.) Re-parameterised labels succeed where natural labels fail, though the representation is two honest linear features plus a turn bit: the probe measured our ontology, not the model's. This is the Othello resolution reduced to three coordinates and one XOR.

1.12 The Reading

Citations verified against the campaign's register and hyperlinked; reading instructions stable.

nostalgebraist, "interpreting GPT: the logit lens," 2020. The instrument of Definition 1.2 in its original informal dress. Read it after running the protocol box, not before; the surprise survives better that way. LessWrong.

Belrose et al., "Eliciting Latent Predictions from Transformers with the Tuned Lens," 2023. The repair and its training objective; §3 against Definition 1.2, and the model-family failure cases against the protocol's step five. arXiv:2303.08112.

Li et al., "Emergent World Representations" (Othello-GPT), 2023; and Nanda, "Actually, Othello-GPT Has A Linear Emergent World Representation," 2023. The board, the probes, the interventions; then the re-parameterisation. Read the pair in order, against Part Three and Remark 1.7, and notice that the second piece's correction is a *strengthening* of the first's claim. arXiv:2210.13382; neelnanda.io.

Mikolov et al., word2vec, 2013; with any later analogy- evaluation audit. The ancestry, and the deflation Problem 1.5 measures. The desk's rule: cite the trick only with its asterisk. arXiv:1301.3781.

Elhage et al., "A Mathematical Framework for Transformer Circuits," 2021. The residual-stream-as-communication-channel picture beneath Definition 1.1; deferred in earnest to Chapters 2–3, but its §1 reads naturally against Part One. transformer-circuits.pub.

Olah et al., the Distill circuits thread ("Zoom In," curve detectors, curve circuits), 2020. The evidence standard of Part Six in its original habitat, hand-written weights and all. Zoom In; Curve Detectors; Curve Circuits.

Elhage et al., "Toy Models of Superposition," 2022. Deferred to Chapter 4; its §2 intuition sections pair with Proposition 1.8 tonight if impatience demands. transformer-circuits.pub.

Chapter 2

The Attention Mechanism, Read Closely

Companion to Episode Two. Chapter 1's query/key loan, paid in full: the head as four matrices that are really two; the positional hole patched by the rotary proposition, with its two-line proof; the symmetry argument with its honest asterisk; and the master discovery — the seam — spent three ways, on the census of head species, on the economics of the key-value archive, and on the ladder of theorems that proves depth is a necessity rather than a luxury.

2.1 The Clerk We Have Not Yet Met

Two species of clerk share the table: the feed-forward specialist, sealed in his room with a single notebook entry (and two-thirds of the payroll — his turn is the supplement's), and the attention clerk, the only component in the architecture through which information at one position can influence computation at another. One fact governs everything below: the head that presents itself as one operation factors into two independent machines — *where to look* and *what to write* — and the factorisation is the master key to reading attention circuits at all.

2.2 Parts One and Two — The Mechanism, in Full Notation

The head arrives as four matrices, and the course's names for them do honest work: the *three lenses* W_Q, W_K, W_V, which turn a position's state into its *question* q_i, its *advertisement* k_j, and its *goods* v_j; and the *output formatting* W_O, which dresses whatever the head collects for deposit in the stream.

Definition 2.1 (Attention head)**.** Fix positions $1, \dots, n$ with residual states $x_1, \dots, x_n \in \mathbb{R}^{d_{\text{model}}}$. A head with parameters $W_Q, W_K, W_V \in \mathbb{R}^{d_{\text{model}} \times d_{\text{head}}}$ and $W_O \in \mathbb{R}^{d_{\text{head}} \times d_{\text{model}}}$ computes, at each position i: queries, keys, and values $q_i = x_i W_Q$, $k_j = x_j W_K$, $v_j = x_j W_V$; scaled scores $s_{ij} = \langle q_i, k_j \rangle / \sqrt{d_{\text{head}}}$ for $j \le i$ (the *causal mask*: $s_{ij} = -\infty$ for $j > i$); the *attention pattern* $A_{ij} = \operatorname{softmax}_j(s_{ij})$, a row of nonnegative weights summing to one; and the deposit

$$h_i = \Big(\sum_{j \le i} A_{ij} \, v_j \Big) W_O,$$

which is *added* to the residual stream at position i. A layer runs H such heads in parallel and adds all their deposits: $x_i \leftarrow x_i + \sum_{h=1}^{H} h_i^{(h)}$.

Remark 2.2 (The budget, and the scale). Two small clauses of the definition carry disproportionate weight. The softmax row summing to one is the *fixed unit of attention*, the head's whole budget: a head cannot attend to nothing, and the obligation to spend the unit somewhere is the origin of the attention sink (Part Six). And the $1/\sqrt{d_{\text{head}}}$ scale is load-bearing bookkeeping: for d-dimensional vectors of typical unit-scale coordinates the raw dot product grows like $\sqrt{d}$, and without the rescaling the softmax saturates — all budget on one position, gradients starved — before training can teach the head anything. Typical widths, for the record: d_{head} is 64 in GPT-2, 128 in the Llama class — "a hundred or so."

2.3 Part Three — Position, and the Rotary Proposition

Nothing in Definition 2.1 knows where anything is: permute the positions and every score survives the shuffle. Position must be smuggled into the arithmetic, and the modern smuggling route is the rotary map — the *many-handed clock*, one hand per frequency, every hand turning with position.

Definition 2.3 (Rotary positional embedding)**.** Partition the d_{head} coordinates of queries and keys into two-dimensional blocks $t = 1, \ldots, d_{\text{head}}/2$, and assign each block a frequency ω_t (a geometric ladder, fast blocks to slow). The rotary map at position m is the block-diagonal rotation R_m whose t-th block rotates by angle $m\,\omega_t$. Queries and keys — and they alone — are rotated by their own positions before comparison: $s_{ij} \propto \langle R_i\, q_i,\ R_j\, k_j \rangle$.

Proposition 2.4 (Scores depend only on relative offset)**.** *For all positions i, j and all $q, k \in \mathbb{R}^{d_{\text{head}}}$, $\langle R_i\, q, R_j\, k \rangle = \langle R_{i-j}\, q, k \rangle$. In particular the attention score between a query and a key depends on the pair's offset $i - j$ and not on the pair's location.*

Proof. Each two-dimensional block of R_m is a plane rotation by $m\,\omega_t$; rotations in a plane commute and compose by adding angles, and each preserves the inner product. Within block t, $\langle R_i q, R_j k \rangle_t = \langle R_j^\top R_i q, k \rangle_t = \langle R_{i-j} q, k \rangle_t$, since $R_j^\top R_i$ rotates by $(i - j)\,\omega_t$. Sum over blocks. □

Remark 2.5 (The clock, renegotiated). A dividend follows from the proof's shape: position lives entirely in the frequency ladder $\{\omega_t\}$, so a model asked to serve contexts beyond its training length can have its clock *slowed* — the frequencies rescaled so the familiar range of angles stretches over the new length — and the geometry it learned remains recognisable. That is the mechanism beneath the context-extension tricks of the practitioner's literature (position interpolation and its refinements), and a first instance of the course's bargain: the mechanism read is the mechanism adjustable. Note also what the definition quietly files: rotation touches queries and keys only — the values pass untouched — so position influences *routing* and never *cargo*. The seam, before we have even stated it.

2.4 Part Four — The Seam, as a Proposition

Proposition 2.6 (QK/OV factorisation)**.** *For a head as in Definition 2.1: the attention pattern A is a function of the states and the single matrix $W_{QK} := W_Q W_K^\top \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ alone (scores $s_{ij} \propto x_i\, W_{QK}\, x_j^\top$); and, holding A fixed, the head's deposit is a function of the states and the single matrix $W_{OV} := W_V W_O \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ alone ($h_i = \sum_j A_{ij}\, x_j\, W_{OV}$). The head factors into a* where *machine and a* what *machine with no shared parameters.*

Corollary 2.7 (Gauge freedom)**.** *For any invertible $M \in \mathbb{R}^{d_{\text{head}} \times d_{\text{head}}}$, replacing $(W_Q, W_K) \to (W_Q M,\ W_K M^{-\top})$ leaves W_{QK} — and hence every attention pattern, and every model output — unchanged; likewise $(W_V, W_O) \to$*

$(W_V N,\ N^{-1}W_O)$ preserves W_{OV}. There is no fact of the matter about what a query "really is" apart from the keys it is destined to meet: only the products are real.

Problem 2.4 has you verify the corollary by hand, and its moral is the lock-and-key: an analysis conducted on the four lenses separately is studying the bookkeeping rather than the books. One caution for the practitioner reading real checkpoints: with rotary position in force the QK product is sandwiched by the rotations (Proposition 2.4), and the corollary's QK freedom shrinks with the sandwich — only an M commuting with every rotation block survives (the rotations' *centraliser*: per-block scaled rotations, a family of dimension d_{head} for a generic frequency ladder, not d^2_{head}), while the OV gauge, which the rotations never touch, survives intact. Meanwhile the low-rank factors mean both products have rank at most d_{head} — each head reads a thin slice of the stream and writes a thin slice, which is how thousands of heads share one notebook without trampling: the operational cousin of Chapter 1's near-orthogonal packing.

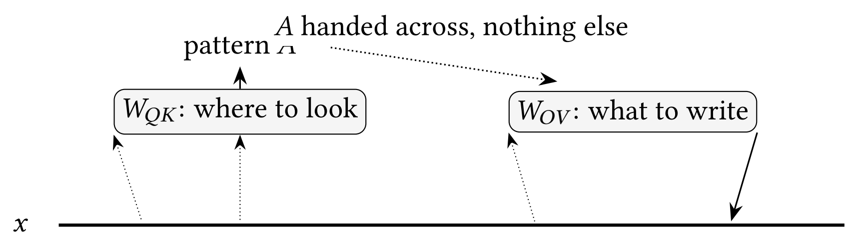


Exhibit 2.1 (The seam). One head, two machines. The QK circuit reads the stream through its thin lenses and produces only the routing verdict A; the OV circuit, handed A, moves cargo it selects and formats by entirely separate parameters. Neither machine touches the other's weights. Schematic.

2.5 Part Five — No Privileged Direction, with the Asterisk

Proposition 2.8 (Rotational symmetry of the idealised stream)**.** *Consider the idealised transformer in which every component reads the residual stream only through linear maps (W_Q, W_K, W_V, the MLP input weights) and writes only through linear maps (W_O, the MLP output weights), with no per-coordinate operation applied to the stream itself. For any orthogonal U, replacing every reading map $W \to U^\top W$ and every writing map $V \to VU$ while rotating the embedding output by U yields a model computing the identical function. No orientation of the stream's axes is distinguished.*

Remark 2.9 (The scholium of the outlier dimensions). The proposition earns Chapter 1's features-as-directions doctrine — a concept has no reason to align with axes that mean nothing — and it carries an honest asterisk, stated here precisely. Real architectures are not the idealised one: the normalisation layers between blocks apply *per-coordinate* learned scales, and adaptive optimisers update per-coordinate; both are nailed to the basis. Empirically, trained residual streams exhibit *outlier dimensions* — a handful of coordinates with magnitudes far outside their peers' — and these are exactly the rogue coordinates that make activation quantisation delicate (per-channel scaling, outlier-aware schemes, and their kin exist because of them). Scope the doctrine, then: the architecture assigns the axes no meaning, and features need not align with them; the *training process* leaves the axes faintly marked all the same. Lab 2.5 puts the fingerprints on your own screen in an hour.

2.6 Part Six — The Field Guide, and the Standing Caution

The census species, tabulated: the desk wants the identifying marks in one place.

Species	*QK signature (where)*	*OV cargo (what)*
previous-token	fixed offset −1 (rotary-navigated)	summary of the prior token; the induction pair's annotator
positional family	fixed offsets; sentence/clause starts	structural context
duplicate-token	same-token match at earlier positions	"I have appeared before"; witness in ch. 6
successor	member of a conventional sequence	a push toward the *next* element
negative (copy-suppression)	previously-seen token	a vote *against* repeating it; calibration
name-mover	names earlier in passage	the name, delivered to the prediction site; ch. 6's star
sink behaviour	first position, trigger absent	nothing of consequence — the parked budget

Protocol — Censusing a model's heads — and auditing the OV before trusting the QK (checked against the framework paper's head analyses)

1: Run a varied corpus through the model in TransformerLens; cache every head's attention pattern.
2: For each head, compute summary statistics: mass on offset −1 (previous-token candidates), mass on same-token earlier positions (duplicate-token candidates), mass on position one (sink duty), entropy of the pattern.
3: Shortlist heads with crisp signatures; confirm each on held-out text.
4: *Now audit the cargo.* For each shortlisted head, examine the OV product's action: for copying-family claims, compute the eigenvalue spectrum of W_{OV} restricted to token-embedding space (copying scores positive and large for movers, negative for suppressors); for successor claims, apply W_{OV} to sequence-member embeddings and check the push lands on the successor.
5: Only a head whose *cargo* matches the story earns its species label. An attention pattern alone is routing, not explanation — the staring is data, not verdict.
6: File sink-dominated heads separately; their patterns are institutional, not informative.

Lab 2.1 executes steps 1–3 and 6; Lab 2.2 executes the audit of steps 4–5. Exhibit 2.4's real maps are their product.

2.7 Part Seven — The Archive, Priced

The archive of advertisements and goods — the *KV cache* — now receives its bill.

Remark 2.10 (The KV-cache formula). Causal masking makes past keys and values final, so they are archived rather than recomputed. Per token of context, the archive costs

$$2 \times n_{\text{layers}} \times n_{\text{kv}} \times d_{\text{head}} \times b$$

bytes, with n_{kv} the number of *key-value* heads and b the bytes per element. Worked instances: a Llama-class 70B (80 layers, 8 KV heads under eight-fold grouped-query attention, $d_{\text{head}} = 128$, fp16) archives

$2 \cdot 80 \cdot 8 \cdot 128 \cdot 2 \approx 328$ KB per token — call it a third of a megabyte, and roughly 33 GB for a hundred-thousand-token conversation; the same model *without* grouping ($n_{\mathrm{kv}} = 64$) would archive eight times that. Grouped-query attention economises the shelf while each head keeps its own questions — and, fine print worth initialling, its own *output formatting*: the group shares W_K, W_V; every head retains W_Q and W_O. The seam is what makes the bargain separable at all.

Protocol — Reproducing the eviction cliff (checked against the streaming-attention paper's ablation)

1: Serve a small model with a manually managed KV cache; stream a long text.
2: Policy A — evict oldest first: measure per-token loss as the window slides. Expect precipitous degradation once position one leaves the archive.
3: Policy B — pin the first four tokens forever, plus the sliding window: expect coherence restored over arbitrary lengths.
4: Attribute the difference: log per-head attention mass on the pinned positions; the dormant heads' parked budget is the mechanism.

Lab 2.1's second half; the census explains the cliff before the cliff is measured.

2.8 Part Eight — Composition, Defined

Composition is a note one clerk leaves for another, the residual stream serving as the paper; the definition below sorts the three ways the note can enter the later head's arithmetic.

Definition 2.11 (Q-, K-, and V-composition)**.** Let head h_1 in an earlier layer write deposit $W_{OV}^{(1)}$-shaped content into the stream, and let head h_2 read the stream in a later layer. The pair exhibits *query composition* when h_1's deposit enters h_2's scores through the query side (x_i in $x_i W_{QK}^{(2)} x_j^\top$), *key composition* when it enters through the key side (x_j), and *value composition* when it enters h_2's cargo through $x_j W_{OV}^{(2)}$. In the two-layer attention-only model the composed contribution is carried by *virtual weights* — products such as $W_{OV}^{(1)} W_{QK}^{(2)}$ — and a composed pair acts as a single *virtual head* that exists nowhere in the parameter list.

The three flavours are not interchangeable: the earlier head may pose the later head's question, relabel the filing system its query searches, or alter the goods on the shelf it raids. Key composition is the one to hold with care — it is the hinge of the induction pair, and Chapter 3's path expansion isolates its virtual-weight term explicitly.

2.9 Part Nine — The Ladder, as Three Propositions

Proposition 2.12 (Zero layers: bigrams)**.** *A transformer with no attention layers computes, at each position, a function of that position's token alone; with linear embedding and unembedding its logits are given by the single matrix $W_E W_U$, a learned bigram table. No cross-position influence exists.*

Proposition 2.13 (One layer: skip-trigram ensembles)**.** *A one-layer attention-only transformer's logits decompose as the bigram term plus, per head, a sum of terms each of the form "source token s attended from destination d contributes cargo $s\, W_{OV} W_U$" with weight set by d, s through W_{QK}: an ensemble of* skip-trigrams *($s, \ldots, d \to$ prediction). The where and the what each see only raw token identities (plus position), never relations between tokens.*

Proposition 2.14 (One layer cannot express exact-match induction)**.** *No one-layer attention-only transformer implements the induction rule* — attend from the second occurrence of token A to the position following A's first occurrence, whatever token stands there — *for arbitrary tokens. Sketch: the rule requires the destination's query to select a key position* by a property of that position's neighbour *("preceded by A"). In one layer, keys are functions of their own position's token and address alone (Proposition 2.13); the neighbour relation is computable only by an attention step, and the single step available is the one being routed. The head must spend its one act of attention either computing the relation or using it — it cannot do both. Two layers suffice, constructively: a previous-token head writes "preceded by A" into the neighbour's entry (K-composition fodder), and a second head keys on the annotation. The construction, with its certifying scores and its interventions, is Chapter 3's whole business.*

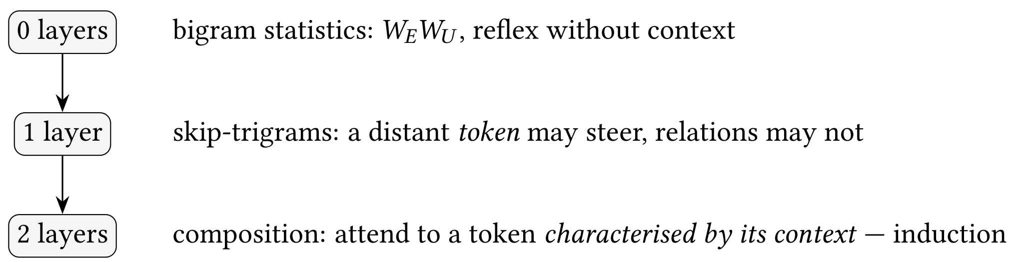


Exhibit 2.2 (The ladder). What each rung can express, per Propositions 2.12–2.14. The jump from one to two is qualitative: retrieval becomes *associative* retrieval, and the purchase is composition — depth as the depth of annotation. Schematic.

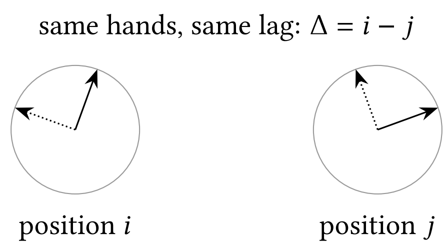


Exhibit 2.3 (Two clock faces). Read the elapsed time between the faces without caring what hour either shows: Proposition 2.4 in one picture. Each hand is one frequency block; the score sees only the angular differences. Schematic.

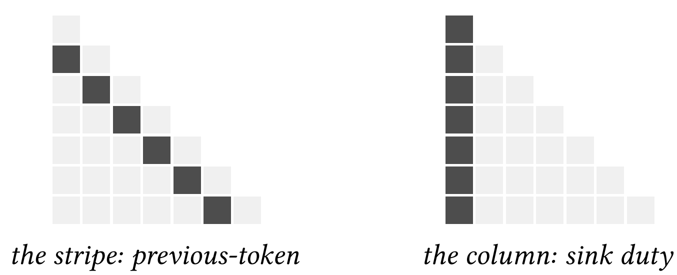


Exhibit 2.4 (Two signatures, schematic). The sub-diagonal stripe of a previous-token head and the first-column mass of a head at sink duty, as idealised patterns. *Schematic. Lab 2.1 produces the real GPT-2 maps — one stripe head, one duplicate-token head on a name-repeating passage, one sink-dominated head; a future edition prints them in this frame, naming the heads by their coordinates.*

The Engineer's Dividend

Three items banked. *First*, the archive arithmetic of Remark 2.10 is the sizing rule you already live by, now derived rather than quoted: when you size a batch you are allocating shelving, and grouped-query's eight-fold economy is the seam permitting cargo to be communal while questions stay individual. (The same seam, exploited harder: DeepSeek's multi-head latent attention compresses the whole K/V shelf into one shared low-rank latent per token — the archive holds only the latent, and every head's keys and values are reconstructed from it at read time — cutting the cache by another order of magnitude beyond grouping while each head keeps its own questions.) *Second*, the eviction protocol is the cleanest case on record of census knowledge preventing an outage: keep the first tokens or watch coherence fall off a cliff — and the newest models ship the basket designed in, so check your serving stack's assumptions against the architecture's. *Third*, the audit discipline: never act on an attention heat-map — in a debugging session or a paper — until the OV cargo has been checked; the field buried a literature of pattern-readings that failed this audit, and your incident reviews need not repeat it.

The Honest Limits

What this chapter's clean pictures do not claim. The factorisation of Proposition 2.6 is exact, but real heads live with normalisation between blocks, and the exactness of "pattern depends only on W_{QK}" holds per-layer-input, not end-to-end: a head's effective behaviour depends on everything upstream that wrote its inputs. The symmetry of Proposition 2.8 is the idealisation, with Remark 2.9's fingerprints on the real axes. The ladder's propositions concern attention-only toys: real models interleave MLPs, and Chapter 4's exception — the one place a privileged basis genuinely lives — sits inside precisely the clerk this chapter sealed away. And the field guide is a fauna, not a partition: many heads fit no species, moonlight across several, or change duties with context — the census is a map of the legible minority.

2.10 The Bench

Chapter 2's laboratory briefs; write-ups in `lab/`.

2.1 *Head safari, then sink demolition* (T1, an evening; authored). Census GPT-2 small per the protocol box: find the stripe, a duplicate-token head, and the sink column; then evict position one from a managed cache and watch coherence collapse; pin four tokens and watch it survive. Produces Exhibit 2.4's real maps.

2.2 *The OV audit* (T1, an evening; authored). Copying scores for candidate movers and suppressors; the successor head's increment verified on days and months. The protocol box's steps 4–5.

2.3 *Archive bookkeeping* (T1, an evening; authored). Instrument generation on Pythia-1.4B and a grouped-query model; measure VRAM growth per token against Remark 2.10's formula; compute the grouping economy from the configs.

2.4 *The clock, built and stretched* (T1, a weekend; authored). Implement the rotary map from scratch; verify Proposition 2.4 numerically; rescale the frequency ladder and measure perplexity at twice the training length, before and after.

2.5 *The outlier fingerprints* (T1, an hour; authored). Histogram per-coordinate residual magnitudes on GPT-2 and Pythia; find the outlier dimensions of Remark 2.9; and connect them to why naive per-tensor activation quantisation hurts.

2.11 Problems

Tier I — Calisthenics

Problem 2.1 (A head by hand)**.** Two positions, $d_{\text{model}} = 2$, $d_{\text{head}} = 1$, with $x_1 = (1,0)$, $x_2 = (0,1)$, $W_Q = W_K = W_V = (1,1)^\top$ (as column maps to one dimension) and $W_O = (1,0)$. Compute the full forward pass of Definition 2.1 at position 2 — scores, mask, softmax weights, deposit — and verify the budget row sums to one. *(Full solution.)*

Problem 2.2 (Why the archive is sound)**.** Using causal masking, prove that the keys and values at position j are unchanged by the arrival of any position $i > j$, and conclude the KV cache never needs invalidation during generation. State the one architectural change in this chapter that would break the proof, and why it does not appear in practice. *(Full solution.)*

Problem 2.3 (The budget's invariance)**.** Show that adding any constant c to every score s_{ij} at a position leaves the attention pattern unchanged, and explain in one sentence why the sink's "parked budget" is therefore invisible to the scores' absolute scale — only *relative* advertisement matters. *(Full solution.)*

Tier II — The Real Thing

Problem 2.4 (The gauge, worked)**.** Prove Corollary 2.7 in full, and then compute the dimension of the gauge group: how many parameters of a head are *bookkeeping* rather than behaviour? Compare to the parameter count of the two products, and reconcile. *(Full solution — completes Part Four's connoisseur refinement.)*

Problem 2.5 (The clock in one plane)**.** For a single frequency block, write the rotation matrices explicitly and verify Proposition 2.4 by trigonometric identity. Then show what goes wrong if values were *also* rotated: compute the deposit's dependence on absolute position and name the property of Remark 2.5 that would be lost. *(Full solution.)*

Problem 2.6 (Skip-trigram bugs, manufactured)**.** Construct a training distribution over sequences on which a one-layer attention-only model provably achieves its optimum while committing the quotation-mark error — confidently matching the wrong pair — and verify a two-layer model expresses the correct behaviour. *(Hints only.)*

Tier III — For the Ambitious (starred)

Problem 2.7 (The grouping frontier)**.** ⋆ Model the quality–memory trade of grouped-query attention: under what assumptions on head-question diversity does collapsing n_{kv} from H to H/g cost little, and construct a task where it must cost much. Estimate, for one real model config, the context length at which the archive overtakes the weights. *(Hints only.)*

Problem 2.8 (Impossibility, tightened)**.** ⋆ Turn the sketch of Proposition 2.14 into a proof for a formal one-layer model class of your choosing: define the class, state induction as a family of sequence-to-prediction maps over an alphabet of size at least three, and show no member of the class realises the family. Note which idealisations your proof needs and which it can drop. *(Hints only.)*

2.12 Hints

Hint (Problem 2.2)**.** The proof needs only: k_j, v_j are functions of x_j, and x_j at a given layer is a function of positions $\leq j$. The breaking change would be any *bidirectional* or future-conditioned rewriting of past states.

Hint (Problem 2.6)**.** Two quotation-mark tokens with different mates; make the marks identical as tokens. One layer keys on token identity alone (Proposition 2.13); arrange the data so identity is insufficient and the optimum must hedge.

Hint (Problem 2.7)**.** For the construction: force H genuinely different *cargoes* to be fetched from one position at once. For the estimate: set the formula of Remark 2.10 against 2× parameter count × bytes.

Hint (Problem 2.8)**.** Fix the class to linear-attention or hard-attention one-layer models to make "keys see only their own position" exact; the alphabet-three condition rules out degenerate solutions that memorise the pairing.

2.13 Solutions

Policy: full for Tier I and Problems 2.4 and 2.5; hints only for 2.6, 2.7, 2.8.

Solution (Problem 2.1)**.** $q_2 = x_2 W_Q = 1$; $k_1 = 1$, $k_2 = 1$; scores $s_{21} = s_{22} = 1/\sqrt{1} = 1$; both allowed by the mask; softmax gives $A_{21} = A_{22} = \frac{1}{2}$ (sum one, as demanded). Values $v_1 = v_2 = 1$; blended value $\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 1 = 1$; deposit $1 \cdot W_O = (1, 0)$, added to x_2. The head, finding both advertisements identical, split its budget evenly — the fixed unit divided, exactly as the softmax obliges.

Solution (Problem 2.2)**.** By induction on layers: x_j at layer zero depends on token j alone; a layer's update at j reads only positions $\leq j$ (the mask), so if every state at positions $\leq j$ is untouched by positions $> j$ at layer ℓ, the same holds at $\ell + 1$. Hence $k_j = x_j W_K$ and $v_j = x_j W_V$ are final on first computation. Breaking change: any operation letting later positions rewrite earlier states — bidirectional attention, future-conditioned normalisation — which is absent precisely because generation-time causality is the product being sold.

Solution (Problem 2.3)**.** $\mathrm{softmax}_j(s_{ij} + c) = e^c e^{s_{ij}} / \sum_k e^c e^{s_{ik}} = \mathrm{softmax}_j(s_{ij})$: the constant cancels. The pattern responds only to score *differences*, so a head with nothing to say cannot lower every score and abstain — it can only choose where the unit lands, and the sink is the designated landing ground.

Solution (Problem 2.4)**.** $W_Q M (W_K M^{-\top})^\top = W_Q M M^{-1} W_K^\top = W_{QK}$, and likewise $W_V N N^{-1} W_O = W_{OV}$; patterns and deposits are functions of the products (Proposition 2.6), so behaviour is invariant. Each of the two gauge groups is $GL(d_{\text{head}})$, of dimension d_{head}^2; the four factor matrices carry $4\, d_{\text{model}}\, d_{\text{head}}$ parameters while the two products, each rank $\leq d_{\text{head}}$, carry $2 \times (2\, d_{\text{model}}\, d_{\text{head}} - d_{\text{head}}^2)$ essential parameters — the discrepancy, $2\, d_{\text{head}}^2$, is exactly the two gauges' worth of bookkeeping. What the network stores is redundant; what it *computes* is the quotient. (The $2\, d_{\text{head}}^2$ count is the rotary-free head's: with the rotations in force the QK gauge shrinks to their centraliser — dimension d_{head} — and the bookkeeping surplus is $d_{\text{head}}^2 + d_{\text{head}}$, the OV share intact.)

Solution (Problem 2.5)**.** The one-block rotation is

$$R_m = \begin{pmatrix} \cos m\omega & -\sin m\omega \\ \sin m\omega & \cos m\omega \end{pmatrix},$$

and $R_j^\top R_i = R_{i-j}$ by the angle-addition identities, giving $\langle R_i q, R_j k\rangle = \langle R_{i-j} q, k\rangle$. Rotating values too would make the deposit $\sum_j A_{ij} R_j v_j W_O$ depend on the absolute j (no pairing cancels the rotation), so a head's cargo would change with document position — and the renegotiation of Remark 2.5 would corrupt cargo as it stretched routing: the clock must stay in the routing department.

2.14 The Reading

Citations against the verification register; instructions stable.

Elhage et al., "A Mathematical Framework for Transformer Circuits," 2021. The chapter's spine in its original habitat: read the QK/OV development against Proposition 2.6, the virtual- weights algebra against Definition 2.11, and the zero/one/two-layer results against Part Nine's propositions — then notice their skip-trigram tables are Problem 2.6's answer key. transformer-circuits.pub.

Su et al., the rotary-embedding paper (RoFormer). The construction of Definition 2.3; its §3 against Proposition 2.4. The context-extension refinements (position interpolation and successors) make good bench reading beside Lab 2.4. arXiv:2104.09864.

Xiao et al., the streaming-attention paper (StreamingLLM), 2023. The sink named and the eviction fix shipped in one work; read its ablations against the second protocol box, and its figure of first-position mass against your own Lab 2.1 census. arXiv:2309.17453.

Ainslie et al., the grouped-query attention paper, 2023. The economy of Remark 2.10's n_{kv}; read with the seam in mind — what is shared and what is retained. arXiv:2305.13245.

McDougall et al., the copy-suppression paper, 2023. The negative heads' cargo characterised; the field guide's suppressor row, earned properly. arXiv:2310.04625.

Gould et al., the successor-heads paper, 2023. The increment operation found across model families — universality's tidiest exhibit in the head fauna. arXiv:2312.09230.

Chapter 3

Induction Heads and the Origin of In-Context Learning

Companion to Episode Three. The season's first solved circuit, traced to its last wire and certified by the telephone-directory test: the path expansion that isolates the K-composition term; the two certifying scores as formulas; the smeared-key equation; the in-context-learning score, and the induction bump that moves it; the six-line case that the birth of this humble machinery is the switching-on of in-context learning itself; and the statistician's construction with its honest scholium.

3.1 Part One — The Trace, with the Algebra Shown

The clockwork, laid out in the two-layer attention-only model where every wire is visible. Two images carry the algebra: the annotation *preceded by Honoria* (formally, the previous-token head's deposit $x_{j-1}W_{OV}^{(1)}$ at position j), and the relabelled filing system (keys computed from annotated states rather than raw tokens — which is all *K-composition* means).

Definition 3.1 (Induction behaviour)**.** Over sequences containing an earlier bigram (A, B) and a later occurrence of A, a model exhibits *induction behaviour* when the probability it assigns to B after the later A is elevated, uniformly in the token identities — including on token pairs never seen in training. The behaviour is the *find the last occurrence of the present situation; predict what followed it then* rule.

Proposition 3.2 (The K-composition term)**.** *In a two-layer attention-only transformer, expand the second-layer head's attention score between destination d (token A, state x_d) and source j. The source state decomposes as $x_j = e_j + \sum_h a_j^{(1,h)}$: its embedding plus the first layer's deposits. The score therefore splits into a* token term $e_d W_{QK}^{(2)} e_j^\top$ *and* composition terms*, among them — when head h is a previous-token head attending from j to $j-1$ — the* K-composition term

$$e_d\, W_{QK}^{(2)}\, \left(e_{j-1}\, W_{OV}^{(1)}\right)^\top,$$

which scores the destination's token against the token before the source*. A head whose $W_{QK}^{(2)}$ is trained to align e_A with $(e_A W_{OV}^{(1)})$ — the same token, read through the annotator's cargo — attends from the later A precisely to the position after the earlier A, whatever A is. The OV circuit of the second head then needs only to copy: its $W_{OV}^{(2)}$ maps source identity to a boost of the same token in the logits. Prefix matching in the QK product acting on an annotation; copying in the OV product; the pair implements Definition 3.1 exactly.*

Remark 3.3 (Why one layer cannot). The impossibility is Chapter 2's Proposition on skip-trigram ensembles, now with its reason visible in the expansion: strike the first layer and the composition terms vanish, leaving scores that compare raw token identities only — and "position whose *predecessor* is A" is not a property of a raw identity. The one attention step must either compute the neighbour relation or use it; the formal statement and proof programme are Chapter 2's Problem 2.8, which this chapter hereby redeems as promised: the constructive half is Proposition 3.2.

Remark 3.4 (The species is heterogeneous). A standing honesty clause: the K-composition construction is the *classic* induction head, not the only one. The original framework paper itself documents a second route (Q-composition "pointer arithmetic"), and close examination of real models finds mechanically mixed populations behind the same behavioural signature. Certification by the scores below is certification of *behaviour*; the wiring behind a certified head is a further question, and Problem 3.6 builds the rival specimen.

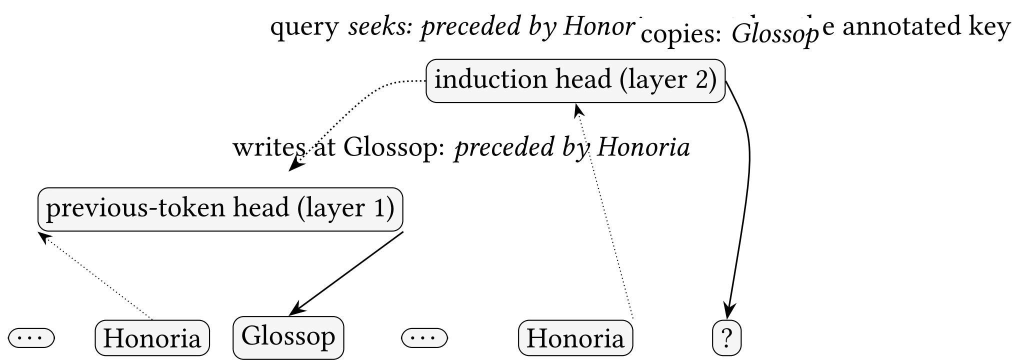


Exhibit 3.1 (The two-head conspiracy). The season's first solved circuit, drawn to Proposition 3.2: the annotator writes *preceded by Honoria* into Glossop's entry (solid arrow, its deposit); the induction head's query from the second Honoria matches the annotated key (dotted), attends to the first Glossop, and copies its identity forward as the prediction (solid, right). Two clerks of no individual distinction; one genuinely general algorithm. Schematic.

3.2 Part Two — The Certificates

Definition 3.5 (Prefix-matching and copying scores)**.** Fix a repeated random sequence $t_1, \ldots, t_n, t_1, \ldots, t_n$ (tokens drawn uniformly; no grammar, no statistics, no alibis). For a head with pattern A, the *prefix-matching score* is the mean, over second-half destinations $d = n + i$, of the attention mass $A_{d,\, i+1}$ landing on the position *just after* the token's earlier occurrence. The *copying score* measures the OV circuit's fidelity: the degree to which the head's cargo from source j promotes token t_j in the logits (operationally, the positivity of the diagonal — equivalently, of the eigenvalue spectrum — of the composite $W_E\, W_{OV}\, W_U$, whose entry at (t, t) is the boost token t's cargo gives its own logit). A head high on both is a certified induction head; high on the first with *negative* copying is the suppressor family of Chapter 2's field guide.

Protocol — Certifying an induction head (checked against the in-context-learning paper's diagnostics)

1: Construct repeated random sequences (a few hundred tokens per half; several seeds).
2: Run with cache; verify the behavioural signature first: per-token loss collapses in the second half.
3: Compute each head's prefix-matching score per Definition 3.5; shortlist the high scorers.
4: Compute copying scores for the shortlist; certify heads high on both. File high-PM/negative-copy heads as suppressors, not failures.
5: Ablate the certified heads jointly; measure second-half loss and the ICL score before and after. Degradation closes the loop from census to cause.

Lab 3.1 executes steps 1–4 and the ablation; its outputs are Exhibit 3.2's real curves.

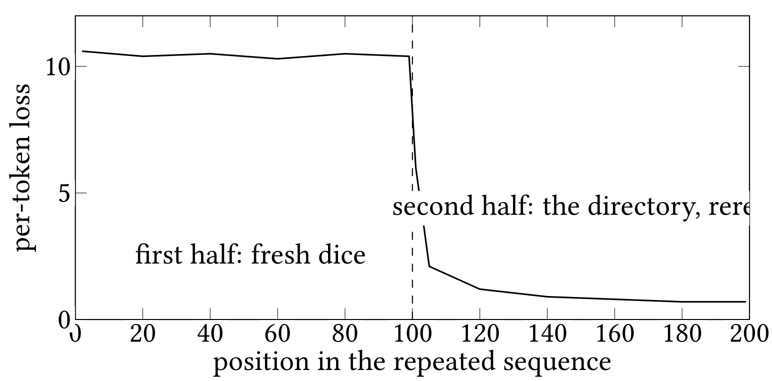


Exhibit 3.2 (The telephone-directory signature). Per-token loss over a repeated random sequence: dismal by construction in the first half, collapsing at the splice — and the collapse is attributable, by the census and the ablation, to the certified heads. *Schematic placeholder: the assembled volume prints Lab 3.1's measured curve on GPT-2 small, with the certified heads named by coordinate.*

3.3 Part Three — The Bump, and the Score It Moves

Definition 3.6 (The in-context-learning score)**.** For a corpus of long passages, let $\mathcal{L}_k$ be the model's mean loss at the k-th token position. The *ICL score* is $\mathcal{L}_{50} - \mathcal{L}_{500}$, the five-hundredth against the fiftieth: how much the intervening context actually buys. Near zero for a model that cannot exploit context; large when it can. (The particular indices are the study's convention — though the study records the difference with the opposite sign, $\mathcal{L}_{500} - \mathcal{L}_{50}$, a negative quantity whose magnitude is the instrument; the desk keeps the positive orientation so that larger means better.)

Remark 3.7 (The smeared key). The co-perturbation line of evidence, as an equation. Replace each key by

$$\tilde{k}_j \;=\; \alpha\, k_j \;+\; (1-\alpha)\, k_{j-1}, \qquad \alpha \in (0,1) \text{ learned},$$

so every key arrives blended with a whiff of its predecessor's. The annotation the previous-token head exists to write is now supplied by the architecture, gratis — the architectural cheat — and the depth requirement of Remark 3.3 collapses: a *one-layer* model can implement prefix matching with its single step. The study's two predictions — smeared one-layer models acquire the phase change they otherwise entirely lack; smeared deeper models undergo it earlier, dragging the ICL score's jump along — both held. Moving the cause moved the effect, everywhere, in lockstep: the strongest of the six lines, and a model of architectural surgery as an instrument of proof.

Protocol — Watching the bump without training anything (checked against the public Pythia checkpoint suite)

1: Choose a small Pythia model; enumerate its public training checkpoints (dozens, spanning the run).
2: At each checkpoint: compute every head's prefix-matching score on repeated random sequences, and the ICL score of Definition 3.6 on natural text.
3: Plot both against training tokens on one axis; mark where per-head PM scores climb from the floor.
4: Expected: the climb and the ICL jump coincide in a narrow early window — the co-occurrence figure, reproduced from free artifacts on a laptop.

Lab 3.3; its two aligned panels are Exhibit 3.3.

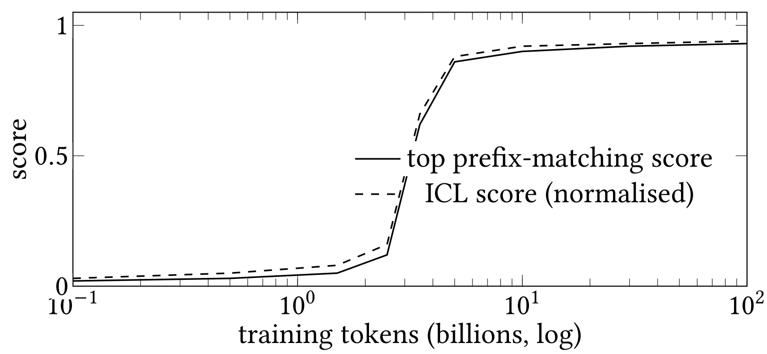


Exhibit 3.3 (Two curves, one window). The heads form and the capability arrives together: per-head prefix-matching and the ICL score, aligned in training time, jumping in the same narrow early window. *Schematic placeholder: the assembled volume prints Lab 3.3's measured pair from Pythia checkpoints, axes in real tokens.*

3.4 Part Four — The Six Lines, Filed

The case — the birth of this machinery is the switching-on of in-context learning — filed as a table, with each line's epistemic weight stated plainly.

Line	*Method*	*What it establishes*
co-occurrence	bump and ICL jump coincide, per model family	suggestive, sixfold
co-perturbation	the smeared key (Remark 3.7) moves both together	causal leverage
ablation	disable certified heads; ICL degrades	necessity, in the small
traced mechanism	Proposition 3.2	the something is known
breadth	soft/structural/translation induction	range to carry a capability
extrapolation	same signatures at scale	inference, honestly graded

The responsible conclusion, unchanged by anything since: a substantial and causally demonstrated *portion* of in-context learning rides on induction machinery and its relatives; the induction head is the foundational member of a family, not the family entire. Later work locates function-vector machinery distinct from induction heads carrying few-shot task performance, and finds real-task ICL in larger models surviving induction-head ablation better than the simple story would predict — further layers to the story, which is where the next section picks up.

3.5 Parts Six and Seven — The Loosenings, and What the Learning Is

Definition 3.8 (Function vector)**.** Run a model on a few-shot prompt for task T; at a designated position (typically the final separator), extract the residual-stream contribution of a small set of mid-layer attention heads — call it v_T. The vector is a *function vector* for T when injecting v_T (adding it into the stream at the corresponding position) on a *zero-shot* prompt induces the model to perform T — the task, distilled and injected: the lesson administered by injection rather than instruction. Empirically v_T transfers across prompts and templates, and selected pairs compose: $v_{T_1} + v_{T_2}$ induces the combined task.

Proposition 3.9 (The statistician's construction)**.** *There exists an explicit weight configuration of a single* linear *attention layer such that, on a prompt encoding regression pairs* (x_i, y_i) *and a query* x*, the layer's output is the prediction of one step of gradient descent (suitably preconditioned) on the in-context least-squares objective; stacked layers yield further steps — the statistician, installed in the forward pass by construction.*

Remark 3.10 (Scholium: where the construction's writ runs). The construction is a possibility proof, and in the simplest linear settings trained toys approximate it. Its writ beyond that is contested: full softmax transformers on realistic data have resisted the tidy identification, and the field's current position is that the librarian (retrieval by induction machinery), the statistician (estimation in activations), and the Bayesian (task recognition, with Definition 3.8's vectors as the recognisable answer) are all on staff, in a blend that shifts with scale and task. The engineer's edge survives the philosophy: a librarian fails when the library lacks the book, a statistician when the data are few or foul, and the two failures want different remedies in your prompts.

Protocol — Extracting and injecting a function vector (checked against the function-vectors paper's recipe)

1: Assemble a few-shot prompt set for a crisp task (antonyms serve); identify, by ablation screening, the small set of mid-layer heads whose summed contribution carries the task.
2: Extract v_T as those heads' mean contribution at the separator position across prompts.
3: Inject v_T at the matching position on zero-shot prompts; measure task accuracy against three controls: no injection, a random vector of matched norm, and $v_{T'}$ from an unrelated task.
4: Test composition on a task pair; report the composed accuracy honestly — it works for some pairs, not all.

Lab 3.5's brief; nnsight is the natural harness.

The Engineer's Dividend

Five dividends, indexed to their mechanisms: format consistency is legibility to the prefix-matcher (Proposition 3.2 reads structure, and smudged templates raise its noise floor); example selection and order are what the attention geometry can reach and prefers; repetition loops *bear the fingerprints of* induction self-reinforcement — an honest grading, and Lab 3.6 is the conviction attempt; the telephone-directory test doubles as the honest probe of a context window's effective reach, cheaper and cleaner than any natural-text needle hunt; and "the model learned it from the prompt" now resolves into machinery you can name, feed deliberately, and debug when it starves.

The Honest Limits

What this chapter does not claim. Induction machinery is a demonstrated *part* of in-context learning, not its whole; the six lines prove their conclusion in small models and infer it, with graded confidence, at scale. The traced construction is one route among several (Remark 3.4). The statistician's proposition is a possibility result whose realistic scope is open (Remark 3.10). And the certifying scores certify behaviour on a synthetic stimulus: a head's duties on natural text are broader, messier, and only partially summarised by any census number.

3.6 The Bench

Briefs; write-ups in `lab/`. Lab numbers follow the campaign's order of authoring within the chapter and are not one-to-one with the exhibits; the gaps are numbers never taken, not briefs withheld.

- **3.1** *The telephone-directory test, then the ablation* (T1, an evening; authored). Certify GPT-2 small's induction heads per the protocol; ablate them; watch the second-half collapse recover toward chance. Produces Exhibit 3.2's real curve.
- **3.3** *The bump from free checkpoints* (T2, a weekend; authored). Pythia-160M across its public revisions: PM scores and the ICL score on one axis. Produces Exhibit 3.3's real panels.
- **3.5** *Function vectors, extracted and injected* (T2, a weekend; authored). The protocol box on GPT-J or Pythia-1.4B, with the three controls and one composition attempt.
- **3.6** *The repetition-loop autopsy* (T1, an evening; authored). Drive a small model into a loop; log certified induction heads' attention and logit contributions per step; convict or acquit the fingerprint claim. The lab the Dividend promises.

3.7 Problems

Tier I — Calisthenics

Problem 3.1 (Scores on a toy)**.** For the sequence $A\,B\,C\,A\,B\,C$ and an idealised head that attends perfectly per the induction rule, write down the attention pattern rows for the second half and compute the prefix-matching score of Definition 3.5. Then compute it for a pure previous-token head, and explain the difference in one sentence. *(Full solution.)*

Problem 3.2 (The score's blind spots)**.** Name two ways a model could raise the ICL score of Definition 3.6 without any induction machinery, and one way genuine induction machinery could fail to raise it. What does this imply about using either instrument alone? *(Full solution.)*

Tier II — The Real Thing

Problem 3.3 (The expansion, earned)**.** Derive Proposition 3.2 in full for a two-layer attention-only model with a single head per layer: write the second-layer scores as a sum of four terms (token–token, token–annotation, annotation–token, annotation–annotation), state which term carries induction, and say what the other three do in the trained induction pair. *(Full solution — completes the chapter's trace.)*

Problem 3.4 (Smearing, priced)**.** The smeared key of Remark 3.7 hands every head the previous-token annotation for free. Show what it *costs*: exhibit a task where the blending of k_j with k_{j-1} strictly reduces a one-layer model's expressible accuracy, and explain why the price was worth paying for the experiment's purpose. *(Hints only.)*

Problem 3.5 (The bump is not smooth progress)**.** Suppose each of m induction heads forms independently at a random time in a window, each contributing equally to the ICL score on formation. Derive the expected shape of the score's rise, and contrast it with the observed lockstep jump. What does the contrast suggest about the heads' formation being coupled? *(Hints only.)*

Tier III — For the Ambitious (starred)

Problem 3.6 (The rival specimen)**.** ⋆ Construct, by hand, weights for a two-layer attention-only model implementing induction by *Q-composition* (the pointer route: the destination's query built from the annotation on the *destination* side), and state a behavioural experiment that distinguishes your specimen from the K-composition classic. *(Hints only.)*

Problem 3.7 (One gradient step, by hand)**.** ⋆ Prove Proposition 3.9: exhibit the linear- attention weights, verify the output equals the one-step GD prediction, and identify exactly where the construction uses linearity in place of the softmax. *(Hints only.)*

3.8 Hints

Hint (Problem 3.4)**.** Any task requiring a key to advertise *only* its own token — exact identity matching under adversarial neighbours — now suffers interference $1 - \alpha$ from the predecessor.

Hint (Problem 3.5)**.** Independent uniform formation times yield an expected score rising as the CDF — smooth and windowwide. The observed near-simultaneity across heads (and the co-moving loss kink) wants a shared prerequisite: the annotator, whose formation gates all its readers at once.

Hint (Problem 3.6)**.** Let the first layer write the *successor's identity* back onto the A position itself; the destination then queries for "token whose recorded successor I should emit," and the distinguishing experiment is where the annotation lives — probe the two positions.

Hint (Problem 3.7)**.** Encode (x_i, y_i) so values carry $y_i x_i$ terms; linear attention sums them into the weight-update form $\sum_i y_i x_i^\top$ applied to the query — the softmax would normalise away exactly the sum you need.

3.9 Solutions

Policy: full for Tier I and Problem 3.3; hints only for the rest.

Solution (Problem 3.1)**.** Ideal induction rows (destinations $4, 5, 6$ attending to the position after their earlier occurrence): $4 \to 2$, $5 \to 3$, $6 \to 4$, each with weight one; the PM score is 1. A pure previous-token head attends $d \to d - 1$ ($4 \to 3$, $5 \to 4$, $6 \to 5$): its mass on the PM target is 0 except where the two coincide — here never — so its score is 0. The two species share a fondness for structure and differ precisely in the census statistic; that is what makes the statistic a certificate.

Solution (Problem 3.2)**.** Without induction: (a) long-range *topical* adaptation — late tokens are easier once the document's subject is established, which any topic-sensitive machinery exploits; (b) format lock-in — boilerplate-heavy corpora make late tokens structurally predictable. Genuine machinery failing to register: a corpus of short, non-repeating passages gives prefix-matchers nothing to match at position 500 that position 50 lacked. Implication: the pairing is the instrument — the synthetic stimulus isolates the mechanism, the ICL score measures the capability, and only their *joint* movement (plus ablation) supports the causal story.

Solution (Problem 3.3)**.** With one head per layer, second-layer scores between d and j expand as

$$(e_d + a_d)\, W_{QK}^{(2)}\, (e_j + a_j)^\top = e_d W_{QK}^{(2)} e_j^\top + e_d W_{QK}^{(2)} a_j^\top + a_d W_{QK}^{(2)} e_j^\top + a_d W_{QK}^{(2)} a_j^\top,$$

where a_p is the first head's deposit at p; for a previous-token first head, $a_p = e_{p-1} W_{OV}^{(1)}$ up to attention weight. The *token–annotation* term $e_d W_{QK}^{(2)} (e_{j-1} W_{OV}^{(1)})^\top$ carries induction (destination token against source's predecessor). In the trained pair: the token–token term supplies duplicate-token attraction (harmless or helpful); the annotation–token term would implement Q-composition (small in the classic specimen — and the whole of Problem 3.6's rival); the annotation–annotation term compares predecessors with predecessors, a fuzzier prefix signal the fuzzy-induction family exploits. One expansion, four species of term, three chapters of machinery visible in a single line of algebra.

3.10 The Reading

Citations against the verification register; instructions stable.

Olsson et al., "In-context Learning and Induction Heads," 2022. The chapter's spine. Read the six lines against Part Four's table, the smeared-key section against Remark 3.7, and the paper's own confidence gradient against the limits box — the scrupulousness is the lesson. transformer-circuits.pub.

Elhage et al., "A Mathematical Framework for Transformer Circuits," 2021. The path expansion of Problem 3.3 in its full generality, and the Q-composition route of Remark 3.4. transformer-circuits.pub.

Garg et al., on what transformers learn in-context (regression), 2022. The statistician's evidence; read beside Proposition 3.9 and its scholium. arXiv:2208.01066.

von Oswald et al., on transformers learning by gradient descent, 2023. The construction of Problem 3.7; note precisely which architecture it needs. arXiv:2212.07677.

Todd et al., "Function Vectors"; Hendel et al., "Task Vectors," 2023. Definition 3.8's two parents; run the protocol box before reading, and compare your controls to theirs. arXiv:2310.15213; arXiv:2310.15916.

Power et al., 2022 (grokking, the phenomenon); Nanda et al., 2023 (the mechanism). Held in trust: Chapter 10 owns both. Note for the record that the phenomenon's modulus (ninety-seven) and the mechanism paper's (one-thirteen) are different experiments — a conflation the desk declines to inherit. arXiv:2201.02177; arXiv:2301.05217.

Chapter 4

Superposition and the Toy Models

Companion to Episode Four. Superposition manufactured in a toy small enough to know entire, and the toy in equations: the model and its loss; the linear case as a theorem; the two-features-one-dimension phase transition derived and the phase diagram read off it; the crystal in the fold, the physics beneath the crystal, and the geometry with its two warps; the features that are wheels rather than arrows; computation inside the fold; the double-descent interrogation in full; the privileged basis earned at last — Chapter 2's standing debt discharged; and the amendment to the linear hypothesis stated with the precision the verification register demands.

4.1 Parts One and Two — The Toy, Precisely

One equation, two knobs. The equation is *the toy entire*, $\hat{x} = \mathrm{ReLU}(W^\top W x + b)$; the knobs are *sparsity* S, the probability a feature is absent, and *importance* I_i, the price of erring on it. Everything below is the turning of these two.

Definition 4.1 (The superposition toy)**.** Fix n features and a bottleneck of width $m < n$. An input $x \in \mathbb{R}^n_{\geq 0}$ is drawn coordinatewise: with probability S (the *sparsity*), $x_i = 0$; otherwise $x_i \sim \mathrm{Unif}[0, 1]$. The model is

$$\hat{x} \;=\; \mathrm{ReLU}\big(W^\top W x + b\big), \qquad W \in \mathbb{R}^{m\times n},\; b \in \mathbb{R}^n,$$

trained to minimise the importance-weighted reconstruction loss

$$\mathcal{L} \;=\; \mathbb{E}_x \sum_{i=1}^{n} I_i \left(x_i - \hat{x}_i\right)^2,$$

with importances $I_i > 0$ assigned by the experimenter. Column W_i is the direction the model assigns feature i in the bottleneck; the Gram matrix $W^\top W$ is the complete record of the tenancy — diagonal entries the features' claimed magnitudes, off-diagonal entries the interference between tenants.

In that record the chapter's quarry is legible at a glance: *the fold* — superposition, more features than dimensions — whose fingerprint is off-diagonal mass in $W^\top W$.

Remark 4.2 (Why this design). Two deliberate austerities. The tied weights ($W^\top$ up, W down) make the assigned direction unambiguous; and the single final rectifier is the smallest nonlinearity that lets the model *cancel interference* — which is, exactly, what makes superposition a strategy rather than a blunder, and what the antipodal mechanics below exploit. There is no mystery the toy can hide: the whole model is one matrix, one bias, one clamp.

4.2 Part Three — The Phase Transition, Derived

Proposition 4.3 (The linear regime is importance-weighted PCA)**.** *Remove the rectifier:* $\hat{x} = W^\top W x + b$*. Then the optimal* W *spans the top-*m *eigenspace of the importance-weighted input second-moment matrix — the model keeps the* m *most valuable orthogonal directions and represents nothing else. An exact theorem, in its correct home: principal component analysis is what the* linear *toy provably does, and what the ReLU toy approximates in the dense regime.*

Proposition 4.4 (Two features, one dimension: when folding pays)**.** *Let two features of importances* $I_1 \geq I_2$ *contend for one dimension, each independently active with probability* $p = 1 - S$ *and uniform on* $[0, 1]$ *when active, with the rectifier in force and bias at zero. Compare the* antipodal *arrangement (*$W_1 = w$*,* $W_2 = -w$*,* $\|w\| = 1$*) against the* dedicated *one (feature 1 alone; feature 2 abandoned). The dedicated arrangement reconstructs feature 1 exactly and forfeits feature 2 wholesale: expected loss* $I_2\,\mathbb{E}[x^2]\,p = I_2\,p/3$*. The antipodal arrangement reconstructs every solo activation exactly (Problem 4.1) and pays only on* co-occurrence*, probability* p^2*, where each tenant errs by* $\min(s, t)$*: expected loss* $(I_1{+}I_2)\,\mathbb{E}[\min(s, t)^2]\,p^2 = (I_1 + I_2)\,p^2/6$*. Folding is therefore preferred exactly when*

$$\frac{(I_1 + I_2)\,p^2}{6} \;<\; \frac{I_2\,p}{3}, \qquad \text{i.e.} \qquad p \;<\; p^* \;=\; \frac{2\,I_2}{I_1 + I_2} \;:$$

sparsity below a critical activity level makes superposition optimal*, the crossover sharp because benefit is linear in* p *where cost is quadratic, and the critical level itself set by how much the junior tenant matters relative to the pair. (At equal importances* $p^* = 1$*: in this two-feature idealisation an equally important room-mate is always worth taking in, and the threshold becomes interior the moment the importances split — which is exactly what hands Exhibit 4.2 its importance axis.) Problem 4.3 verifies the bookkeeping.*

Remark 4.5 (The three phases). Chart each feature's fate over the (S, I) plane and the borders are sharp: *absent* (dense and unimportant — not represented), *crystalline* (important enough — a private orthogonal dimension), *folded* (sparse enough — shared accommodation). The metallurgical reading is faithful to the source: the transitions are phase-like, with sticky intermediate geometries (Exhibit 4.2), not gentle ramps.

$$W_2 \longleftrightarrow W_1$$

feature 1 active at s: pre-activation of unit 2 is $-s$; the rectifier clamps it to nought

each tenant reads only positive extent along his own arrow

Exhibit 4.1 (The antipodal mechanics). Why opposite directions do not cancel: the readout for each feature measures positive extent along its own column, and the final rectifier zeroes the negative shadow its room-mate casts. Schematic. *Lab 4.1 produces the companion data — real trained Gram-matrix heatmaps across the sparsity sweep and the polytope zoo; a future edition prints them as Exhibit 4.1b.*

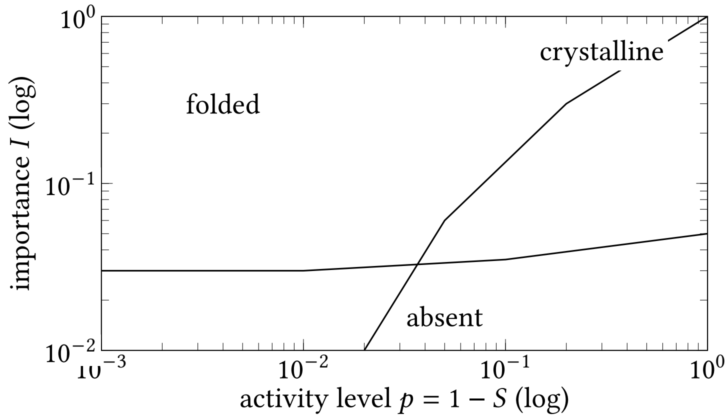


Exhibit 4.2 (The phase diagram, schematically). Three territories in the sparsity–importance plane, after the toy models paper's map: unimportant-and-dense features unrepresented; important ones in private dimensions; sparse ones folded. Borders sharp, per Proposition 4.4 — whose threshold $p^* = 2I_2/(I_1 + I_2)$ moves with relative importance, earning the diagram its vertical axis. Schematic rendering of the published figure's topology; Lab 4.1's sweep traces one horizontal line of it empirically.

4.3 Part Four — The Geometry, and Its Two Warps

Sparsity decides *whether* features fold; geometry decides *how*. Under pressure the columns of W settle into uniform polytopes — *the crystal* — and each feature's holding in the arrangement is measured by one number.

Definition 4.6 (Feature dimensionality)**.** For feature i,

$$D_i \;=\; \frac{\|W_i\|^2}{\sum_j \left(\hat{W}_i \cdot W_j\right)^2},$$

the fraction of a dimension the feature commands ($\hat{W}_i$ the unit column). $D_i = 1$: a private dimension. $D_i = \frac{1}{2}$: an antipodal pair. The sticky values between — $\frac{2}{5}$ for the pentagon, $\frac{3}{4}$ for the tetrahedron — are the census marks of the polytope zoo.

Remark 4.7 (Thomson, warped). Minimising interference at fixed norms is spreading the columns to maximise pairwise angles — the Thomson problem's repelling charges, arriving at the same uniform figures. The two warps: *importance* (a heavier charge claims cleaner separation; the crystal grows under a field) and *correlation* (the hostess's seating: anticorrelated features billeted antipodally, since their interference is never realised; correlated features bought local orthogonality, since theirs would be paid on every visit; perfectly correlated features collapsed into one joint direction). The governing sentence for real models: superposition is rationed by collision risk, and the premiums are set by co-occurrence.

4.4 Part Five — Double Descent, the Full Interrogation

The toy has one more confession in it, extracted by rationing its data; here is the interrogation in full.

Remark 4.8 (The finite-data toy). Fix the model of Definition 4.1 and train on a fixed sample of T examples rather than the stream. Three regimes in T: *scrapbook* (small T — each training example claims its own bottleneck direction; the model reconstructs by matching against stored exemplars; train loss excellent, test chance), *theory* (large T — the generating features are identified and stored; generalisation), and between them the *no-man's-land*, where the sample is too rich to memorise cleanly and too poor to identify the features, and test error *peaks* above both neighbours. Plotted against T, the test curve falls, rises, and falls again: sample-wise double descent (the data axis; Chapter 10 owns the time axis), with the interior hump given a mechanistic reading — the transition between representational strategies, both executed badly at once.

Remark 4.9 (One medium, two tenants). The ontological point, stated for the record because every later chapter leans on it: in the bottleneck there is no separate faculty of memory — a memorised datapoint and a learned feature are the same kind of object, a direction in superposition earning its keep by reducing loss. "Does the model understand or memorise" is a portfolio question, and Chapter 10's grokking — memorisation-then- generalisation along the axis of *time* rather than data — is this remark's sibling.

4.5 Part Six — Features That Are Not Arrows

Definition 4.10 (Multi-dimensional feature)**.** A family of related concepts is represented as a *multi-dimensional feature* when the model assigns it a low-dimensional subspace containing a structured set — a ring, a lattice, a curved manifold — such that the family's semantics live in the position *within* the structure, not in independent directions per member. The linear representation hypothesis (Chapter 1) is thereby *amended*, not repealed: locally the structure is made of directions, and linear instruments found it — but the unit of meaning is the wheel, not any single spoke.

Remark 4.11 (The calendar, causally). The verified record (register entry: Engels et al., ICLR 2025), stated with its mechanism exact: days of the week found on a ring in a two-dimensional subspace; months likewise; and the causal test was *subspace patching* — replace the hidden state's coordinates in the circle plane with the target day's position, average-ablate the remainder of the layer, and the model's modular-arithmetic answers move around the ring to match, the circle alone recovering nearly the effect of patching the whole layer. File beside it the genealogy: the grokked clock of Chapter 10 and the vision anatomists' orientation compass — wherever the concept is cyclic, the model is inclined to build the cycle.

4.6 Part Seven — Computation Inside the Fold

Two classes of result, deliberately kept apart — "the toy models show us" is a phrase that blurs demonstration into theorem, and the desk files the two separately.

Demonstrations (empirical, in toys): small models computing real nonlinear functions — absolute value, conjunction detection — on inputs that remain in superposition throughout, never unpacked into private dimensions. *Constructions and capacity theory:* exclusive-or, whose seat in this column the chapter's own Problem 4.4 earns by building it rather than finding it in a trained specimen; and the counting result that a layer can, in principle, compute many more simple Boolean functions than it has neurons, the computations packed into shared hardware as the features are packed into shared directions — compression of the workshop, not merely the warehouse.

Remark 4.12 (The standing limitation). Grant a perfect feature catalogue: the computation among folded features may still escape it, because ingredients are not a recipe. This is the chapter's export to Chapters 5, 8, and 9: the microscope of Chapter 5 resolves *storage*; the transcoder family of Chapter 8 aims at *computation* precisely because this remark demanded it; and Chapter 9's error nodes are the confession of what still escapes. Whether the transcoder answers this remark's objection or merely relocates it is settled honestly there: the error terms are the unrelocated remainder.

4.7 Part Eight — The Privileged Basis, Earned

Proposition 4.13 (Per-coordinate nonlinearities break the symmetry). *Let a layer apply an elementwise nonlinearity σ to its hidden coordinates. The layer's function is invariant under an orthogonal change of hidden basis U only if $\sigma(Uz) = U\sigma(z)$ for all z; for the rectifier (and its gated kin) this holds only for coordinate permutations — a sign change would need $\sigma(-z) = -\sigma(z)$, an odd nonlinearity, which the rectifier pointedly is not — and never for general rotations. The hidden layer of the MLP therefore* has *a privileged basis: its coordinates — the* neurons *— are singled out by the model's own arithmetic, completing the arc of Chapter 2's Proposition 2.8: no privileged basis in the stream, a genuine one in the sealed room.*

The collision of fold and privileged basis has its own name: *the tangled handle* — the polysemantic neuron.

Definition 4.14 (Polysemanticity). A neuron is *polysemantic* when multiple unrelated features load on it substantially — the visible fingerprint of superposition pressing on the privileged basis, since more features than neurons forces tenancy-sharing even where the axes mean something. The canonical specimens: the language neuron firing on academic citations, a programming syntax, and Korean script; the vision unit for cat faces, car fronts, and cat legs.

Remark 4.15 (The road not taken: legislating the fold). The architectural attempt — an activation designed to reward one-feature-per-neuron — produced a season of optimism and then the accounting: superposition was not eliminated but *relocated*, hidden where the new activation's incentives did not reach (the normalisation's escape hatch), the clean shopfront over the busy cellar. The lesson has earned its italics: *superposition is an economic response to scarcity and will reappear, in disguise if necessary, wherever the scarcity persists.* Hence the only road left, and the next chapter: accept the fold, and build the external instrument that unfolds after the fact.

Protocol — Training the toy, and reading its Gram matrix (checked against the toy-models paper's setup)

1: Fix $n = 20$ features, $m = 5$ dimensions, importances decaying geometrically; sweep sparsity S over a half-dozen values from dense to 0.999.
2: Train Definition 4.1's model at each setting (minutes on a CPU); save W.
3: Read the Gram matrix: diagonal-only structure in the dense runs (crystalline phase, overflow abandoned); off-diagonal mass appearing abruptly as S crosses the threshold; antipodal pairs first, then the polytope zoo as pressure grows.
4: Compute each feature's dimensionality D_i (Definition 4.6); histogram it and find the sticky values.
5: For the finite-data variant: fix S, sweep dataset size T, plot test loss against T, and inspect W in the scrapbook regime to see datapoints-as-directions with your own eyes.

Lab 4.1 is this protocol executed; Lab 4.2 is step five at length.

The Engineer's Dividend

Four dividends, each with its epistemic grade marked: quantisation's uneven damage read as a raised interference floor drowning the thin-margined tail first (mechanistically informed hypothesis — and the evaluation moral, *test quantised models on their tails*, is cheap insurance regardless); small and distilled models as a different settlement of the same phase diagram, failures concentrated past the representational means test; fine-tuning as the actuary re-pricing the table — the lost capability not deleted but *outbid* (hypothesis, honestly flagged); and the firmly established one: every coordinate-wise reading of activations — neuron dashboards, naive per-dimension probes — is reading polysemantic tangles, and the respectable instruments are exactly the ones built in knowledge of the fold.

The Honest Limits

The toy is a toy: its features are given, independent (until deliberately correlated), and nonnegative; real features are learned, socially entangled, and signed. The phase diagram's topology transfers to real models as an organising picture, not as measured borders. The manifold amendment (Definition 4.10) bounds the linear hypothesis from below with a documented specimen, and the honest statement remains: the model's ontology is richer than the leading hypothesis, by an amount not yet known. And Remark 4.12 is permanent, not provisional: no inventory of parts, however perfect, is an account of the computation.

4.8 The Bench

4.1 *The toy, replicated; the polytopes, watched* (T1, an evening; authored). The protocol box end to end, plus an animation of W's columns settling into the pentagon — charges on a sphere, in matplotlib. Produces Exhibit 4.1b's real heatmaps.

4.2 *Double descent in the toy* (T1, an evening; authored). Step five at length: fifteen dataset sizes, the interior hump, and the scrapbook regime's W inspected.

4.3 *Polysemanticity safari* (T0, an hour; authored). Neuronpedia: GPT-2 MLP neurons beside SAE features; verify the baffling assortment on real units.

4.4 *The calendar's geometry* (T2, a weekend; authored). Collect day-token residuals from a small model, project, find the ring; the ambitious replicate one subspace-patching intervention per Remark 4.11.

4.5 *Absolute value, folded* (T1, an evening; authored). Train the computation-in-superposition toy; verify it never unfolds; stare briefly at Remark 4.12.

4.9 Problems

Tier I — Calisthenics

Problem 4.1 (The antipodal readout, verified)**.** With $W_1 = w$, $W_2 = -w$, unit w, no bias: compute $\hat{x}$ for inputs $(s, 0)$ and $(0, s)$ and confirm each feature is reconstructed exactly while the rectifier zeroes the cross-term. Then compute the co-occurrence case (s, t) and identify the interference term Proposition 4.4 prices. *(Full solution.)*

Problem 4.2 (Reading a Gram matrix)**.** You are handed three $W^\top W$ heatmaps from unlabelled runs of the toy: (a) diagonal only, three of twenty entries large; (b) dense off-diagonal structure in $\pm$ pairs; (c) five-by-five blocks of pairwise cosine $-\frac{1}{4}$. Name each regime and the polytope in (c). *(Full solution.)*

Tier II — The Real Thing

Problem 4.3 (The crossover, with constants)**.** Carry out Proposition 4.4's bookkeeping exactly for uniform activations on $[0, 1]$ and importances $I_1 \geq I_2$: compute the expected abandonment loss and the expected collision cost, both exactly, and solve for the critical activity p^*. Confirm the equal-importance degeneracy, and verify against a Lab 4.1 run near your computed threshold. *(Full solution — completes the chapter's central hand-wave.)*

Problem 4.4 (XOR needs the fold or the room)**.** Show that no affine readout of two superposed features computes their exclusive-or; then exhibit a one-hidden-layer construction that does, with the hidden units reading folded inputs directly. Conclude, in one sentence, what Remark 4.12 adds to Chapter 1's counting argument. *(Hints only.)*

Problem 4.5 (The hostess's theorem)**.** Two features co-occur with probability q (jointly active), each active with probability p. Derive the interference cost of antipodal sharing as a function of q, and show the seating rule — antipodal for anticorrelated, orthogonal for correlated — follows from minimising total expected loss at fixed capacity. *(Hints only.)*

Tier III — For the Ambitious (starred)

Problem 4.6 (The workshop's capacity)**.** ⋆ Sketch the argument that a layer of k rectifier neurons can compute $\omega(k)$ distinct sparse Boolean functions of superposed inputs to fixed tolerance, and identify the counting step where sparsity of *function inputs* (not merely features) does the work. *(Hints only.)*

Problem 4.7 (Wheels all the way down?)**.** ⋆ Propose an experimental programme to bound how much of a real model's representation lives in multi-dimensional structures rather than isolated directions: what to search for, what statistic separates a genuine manifold from a cluster of correlated directions, and what result would falsify the strong dictionary hypothesis. *(Hints only.)*

4.10 Hints

Hint (Problem 4.4)**.** Affine impossibility is the classic parity argument; for the construction, two rectifier units reading $u + v - \theta$ and $-(u + v) + \theta'$ suffice with the right output weights.

Hint (Problem 4.5)**.** Only the joint-activity term changes between seatings; everything else cancels. The rule drops out of comparing q against p^2.

Hint (Problem 4.6)**.** Random low-rank input embeddings make distinct sparse conjunctions nearly orthogonal in the hidden pre-activations; the tolerance absorbs the crosstalk, and the count follows Chapter 1's packing proposition applied to the *function* inputs.

Hint (Problem 4.7)**.** The separating statistic: intervention response to *within- structure* moves (around the ring) against *off-structure* moves of matched norm; a correlated cluster responds to both alike.

4.11 Solutions

Policy: full for Tier I and Problem 4.3; hints only for the rest.

Solution (Problem 4.1)**.** Input $(s, 0)$: bottleneck state sw; pre-activations $W^\top(sw) = (s, -s)$; the rectifier passes $(s, 0)$ — feature 1 exact, feature 2's negative shadow clamped. Symmetrically for $(0, s)$. Co-occurrence (s, t): bottleneck $(s-t)w$; pre-activations $(s-t,\ t-s)$; rectified output $(\max(s-t, 0), \max(t-s, 0))$ — the smaller tenant is erased and the larger reduced: the collision cost, paid only when both are home, exactly as the phase proposition prices it.

Solution (Problem 4.2)**.** (a) Crystalline: three important features in private dimensions, the rest abandoned — the dense regime. (b) Folded, at the first rung: antipodal pairs. (c) Folded at higher pressure: five features per five-dimensional block at pairwise cosine $-\frac{1}{4}$ is the regular 4-simplex — five directions equally spread in four dimensions (cosine $-1/d$ with $d = 4$), the tetrahedron's elder sibling in the zoo.

Solution (Problem 4.3)**.** Abandonment: feature 2 unrepresented loses $I_2\,\mathbb{E}[x^2]\,p = I_2\,p/3$ (uniform on $[0, 1]$), and feature 1 rides exactly — expected loss $I_2\,p/3$. Collision: on co-occurrence (probability p^2), the rectified antipodal pair returns $(\max(s-t, 0), \max(t-s, 0))$, so each tenant's error is $\min(s, t)$ and the joint loss is $(I_1 + I_2)\min(s, t)^2$. For independent uniforms, $\Pr[\min > u] = (1 - u)^2$ gives $\mathbb{E}[\min(s, t)^2] = \frac{1}{6}$, so the expected collision cost is $(I_1 + I_2)\,p^2/6$. Setting cost below benefit: folding pays exactly when $p < p^* = 2I_2/(I_1 + I_2)$ — Proposition 4.4's threshold, reproduced with every constant kept. The degeneracy: at $I_1 = I_2$ the formula gives $p^* = 1$, so in this two-feature idealisation an equally important room-mate is always worth housing, and the interior crossover appears the moment the importances split — take $I_1 = 4I_2$ and the fold pays only below $p^* = 0.4$. The instructive content: benefit linear in activity and in the junior importance, cost quadratic in activity and billed to both tenants; sparsity and the importance ratio decide together.

4.12 The Reading

Elhage et al., "Toy Models of Superposition," 2022. The chapter's source. Read the phase-diagram section against Proposition 4.4 and Remark 4.5, the geometry section against Definition 4.6 (their sticky-values figure is Lab 4.1's target), and the computation section against Part Seven — noting which claims are demonstration and which theory. transformer-circuits.pub.

Henighan et al., the memorisation follow-up, 2023. The finite-data variant of Remark 4.8: datapoints as directions, the interior hump, the ontology remark's evidence. transformer-circuits.pub.

Engels et al., "Not All Language Model Features Are One-Dimensionally Linear," ICLR 2025. The calendar, causally; read the intervention section against Remark 4.11 and note the correction the register records. arXiv:2405.14860.

Vaintrob, Mendel, Hänni, on computation in superposition, 2024. Part Seven's capacity theory, kept separate from the toy demonstrations as the desk insists. AlignmentForum.

Elhage et al., the SoLU paper, 2022. The road not taken, including the authors' own post-mortem — the relocation, not elimination, of the fold; Remark 4.15's source. transformer-circuits.pub.

Olah et al., "Zoom In," 2020. The polysemantic vision neuron of Definition 4.14, in its original cabinet of curiosities. distill.pub.

Chapter 5

The Sparse Autoencoder, in Full

Companion to Episode Five. The microscope, commissioned honestly: the pedigree; the architecture and the war inside its objective, with the constraint that stops the cheat; shrinkage derived rather than asserted; the smudges numbered D1–D4 for the season's later citations; three polishes and a fourth, as equations; evaluation without ground truth; and the achievements beside the reckoning that repriced the instrument from universal to specialised, names and numbers attached.

5.1 Part One — The Pedigree

Remark 5.1 (The objective is older than the machine). The instrument arrives with a pedigree, and the desk records how exact the inheritance is. Olshausen & Field (1996), seeking the code of natural images, posed: minimise $\|x - \Phi f\|_2^2 + \lambda \sum_i S(f_i)$ over an overcomplete dictionary Φ — reconstruction against a sparsity cost, which for $S(u) = |u|$ is the objective of Definition 5.2 symbol for symbol, three decades early. The field did not invent its microscope; it pointed a neuroscientist's instrument at a new specimen.

5.2 Parts Two and Three — The Instrument and Its Objective

The design, in one breath: a wide, sparse code — $f \in \mathbb{R}^F$, $F \gg d_{\text{model}}$, $\|f\|_0$ small — extracted under the war of two demands, reconstruction $\|x - \hat{x}\|_2^2$ against parsimony $\lambda\|f\|_1$, with the dial λ setting the terms of the truce. In full:

Definition 5.2 (Sparse autoencoder)**.** For activations $x \in \mathbb{R}^{d_{\text{model}}}$, dictionary size $F \gg d_{\text{model}}$:

$$f = \text{ReLU}\big(W_{\text{enc}}(x - b_{\text{dec}}) + b_{\text{enc}}\big), \qquad \hat{x} = W_{\text{dec}}f + b_{\text{dec}},$$

$$\mathcal{L} \;=\; \|x - \hat{x}\|_2^2 \;+\; \lambda\,\|f\|_1,$$

with the decoder columns constrained to unit norm (else the penalty is gamed by scaling f down and W_{dec} up). A *feature* is a pair: encoder row (the detector, with its learned negative bias holding it underwater until the evidence is strong) and decoder column (the direction deposited when it fires). Overcompleteness F/d_{model} and the achieved $\|f\|_0$ are the instrument's two characteristic numbers.

Remark 5.3 (Non-identifiability). The dictionary is not unique: different λ, different seeds, different data orders yield different dictionaries of equal fidelity. The conditions under which sparse dictionary recovery *is* unique (incoherence and sparsity conditions from the compressed-sensing literature) are not known to hold for activation data — so uniqueness is not merely unproven but unpromised. Every later use of "the model's features" carries this remark silently.

5.3 Part Four — The Smudges, Numbered

Proposition 5.4 (D1: shrinkage, derived)**.** *Fix a single active feature with true strength s along unit decoder direction d, and let the optimiser choose the reported activation a. The loss is $(s-a)^2+\lambda a$, minimised at*

$$a^* = s - \tfrac{\lambda}{2} \quad (\textit{for } s > \tfrac{\lambda}{2}),$$

the soft-threshold: every reported strength sits exactly $\lambda/2$ below the truth. The bias is directional, uniform, and proportional to the sparsity dial — the instrument is built to read low, by an amount you chose when you chose λ.

Remark 5.5 (D2–D4). *D2, dead features:* dictionary slots that never activate; nominal size overstates recovered content; resampling schemes resurrect some. *D3, splitting:* the recovered grain varies with F — one bird feature at small F, seabirds/raptors/songbirds at larger — so "how many features" is a resolution chosen, not a number counted; the matryoshka design (below) makes the hierarchy explicit rather than accidental. *D4, absorption:* a general feature's cases annexed by a specific one (the begins-with-S feature silent on "short," whose dedicated feature absorbed the duty), so firing tallies mislead. The numbering is for the season: later chapters cite D1–D4 by name.

5.4 Part Five — The Polishes, as Equations

The unifying principle, stated once: *separate the decision of which features are active from the estimate of how strongly.* The three designs, precisely:

Definition 5.6 (Gated, TopK, JumpReLU, Matryoshka)**.** *Gated:* a binary gate channel $g = \mathbf{1}[W_{\text{gate}}x + b_{\text{gate}} > 0]$ selects; a magnitude channel (weight-tied up to a rescaling) estimates strength for selected features; the sparsity penalty falls on the gate path only. Shrinkage vanishes with the penalised magnitude. *TopK:* no magnitude penalty at all; keep the k largest pre-activations, zero the rest — $\|f\|_0 = k$ exactly, shrinkage gone, at the price of one fixed k for tokens simple and rich alike (BatchTopK relaxes the budget to the batch level). *JumpReLU:* a learned per-feature threshold θ_i; $f_i = z_i \cdot \mathbf{1}[z_i > \theta_i]$ — zero below, the *full* value above; trained through the discontinuity by straight-through estimators on an $\|f\|_0$ objective. *Matryoshka:* nested dictionaries trained jointly, prefix-sized sub-dictionaries each required to reconstruct — the splitting hierarchy of D3 carried as structure, several grains in one instrument.

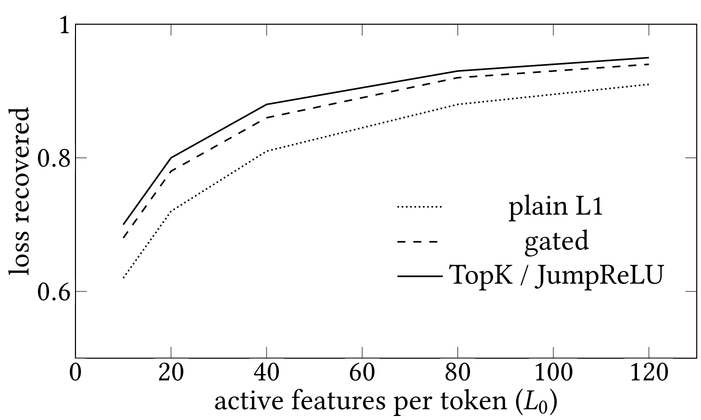


Exhibit 5.1 (The frontier, and its outward march). The fidelity–sparsity trade-off as a Pareto curve, one per design: the polishes buy more loss recovered at every sparsity. Schematic after the published comparisons; Lab 5.1's own small frontier is the bench version.

5.5 Part Six — Evaluation, and the Proxy Problem

Three questions stand in for the ground truth nobody holds: does the reconstruction do the job — *loss recovered*, the substitution evaluation; how few detectors wake per token — the achieved L_0; and does the explain-and-score loop — *auto-interp* — yield explanations that predict the firing.

Definition 5.7 (The three measures)**.** *Loss recovered:* substitute $\hat{x}$ for x at the layer and run on; report the fraction of the model's loss degradation avoided relative to a zero-ablation baseline. *Sparsity:* the achieved L_0. *Auto-interp:* collect a feature's top-activating contexts; have one model propose an explanation; have another predict, from the explanation alone, where the feature fires; score the agreement. All three are proxies: fidelity certifies capture, not carving at true joints; auto-interp certifies human articulability, not computational reality; and Remark 5.3 stands beneath all of it. Every check is for consistency, never against truth.

The field's standard battery for all of this is *SAEBench* (Karvonen et al., ICML 2025) — eight evaluations spanning the proxies and the downstream tasks — whose headline finding, that gains on the proxy metrics do not reliably translate into downstream advantage, is this chapter's thesis in benchmark form.

5.6 Parts Seven and Eight — The Achievements and the Reckoning, with Names

The audio withheld the names by design; the desk supplies them from the verification register. In the contests below, the ruler is the supervised probe or the difference-of-means vector; asking nicely is prompting.

Contest	*Result*	*Source*
34M-feature dictionary; abstract, multilingual, multimodal, safety-relevant features; $\geq$ 65% variance explained (so up to a third dark)	the achievement column, with its shadow	Templeton et al. 2024; Engels et al. 2024 for the error's structure
probing known concepts	plain supervised probes match or beat SAE features across the battery — the initially promising scarce-data and corrupted-label regimes dissolved against careful baselines	Kantamneni et al. 2025
steering	prompting 1.081 > ReFT-r1 0.812 > LoReFT 0.790 > DiffMean 0.505 > SAE 0.341 (Gemma-2-9B, L20)	Wu et al., AxBench, ICML 2025
the proxy's blank wall	random-init models yield auto-interp-passing features at uncomfortable rates	Heap et al. 2025
the partial recovery	output-feature selection brings SAE steering to supervised parity (prompting still crowned)	Arad et al. 2025; Jorgensen & Hansen 2026

Remark 5.8 (The repricing). The settled position, intellectual and institutional: wrong tool for acting on concepts already named (fit the probe, the vector, or simply ask nicely); right tool for *discovery* — enumerating concepts nobody thought to probe for. By mid-2025 the major laboratories' interpretability teams had shifted weight from dictionary scaling toward probes, model biology, and the computation-reading instruments of Chapters 8–9. The engraved rule survives every fashion: *always demand the baseline.*

Protocol — Training and evaluating an SAE — with the baseline discipline built in (checked against the SAELens documentation and the reckoning papers' setups)

1: Choose site and substrate: GPT-2 small residual stream, mid-stack layer; dictionary $F = 16\, d_{\text{model}}$; a JumpReLU or TopK objective.
2: *Before training the dictionary*, fit the rulers: for each concept you intend to inspect, a supervised probe and a difference-of-means vector on the same activations. These are the baselines every later claim must beat.
3: Train (SAELens); monitor L_0, loss recovered, dead-feature fraction; resample the dead at intervals.
4: Evaluate per Definition 5.7; inspect two dozen features by hand alongside their auto-interp verdicts.
5: For any claimed feature-of-concept: compare detection AUC against step 2's probe; report both numbers or neither.
6: For steering claims: compare the feature clamp against the diff-of-means vector and against

prompting at matched strength; report all three.
Labs 5.1 and 5.4; the discipline of steps 2, 5, and 6 is the chapter's dividend in executable form.

trained model's feature | random model's "feature"

top-activating contexts: coherent theme; auto-interp: PASS

top-activating contexts: coherent theme; auto-interp: PASS

which is which? answer overleaf — and that it is hard is the finding

Exhibit 5.2 (The blank wall, blind). The reckoning's most philosophically pointed result as an exhibit: feature dashboards from a trained and a randomly initialised model, presented unlabelled, both passing the interpretability proxy. *Placeholder: Lab 5.5 produces the real pair for the assembled volume; the caption will keep the answer overleaf.*

The Engineer's Dividend

Three rules from the reckoning, stated as bench law. *Demand the baseline*: no interpretability claim without the probe, the vector, and (for steering) prompting run on the same data — the protocol box makes them steps, not afterthoughts. *Right tool, right office*: dictionaries to map unfamiliar territory; supervised instruments to act on named properties; the reconnaissance-then-sharp-tools sequence is the workflow. *Distrust the proxy*: an interpretable-looking feature is not yet a computational unit — the blank wall certifies the explainer's ingenuity and the human appetite for pattern; only intervention (next chapter) converts a feature from correlate to cause.

The Honest Limits

D1–D4 are properties of the instrument class, not bugs of one implementation — the polishes remove D1 and reorganise D3; nothing removes Remark 5.3. The ground-truth problem is epistemic, not engineering: every evaluation is consistency-shaped. The dark matter is measured and structured (largely linearly predictable from the input) but unexplained. And the reckoning's verdicts are task-relative: the tables above are for probing and steering, and SAEBench's battery covers everything *except* discovery — the office the instrument was repriced into has no benchmark yet worthy of the name, which is itself a gap worth noticing.

5.7 The Bench

5.1 *Train your own SAE* (T2, a weekend; authored). The protocol box end to end on GPT-2 small: the frontier felt, dead features met, two dozen features read by hand.

5.2 *Shrinkage, measured* (T1, an evening; authored). L1 and TopK dictionaries on identical activations; matched features; the $\lambda/2$ bias of Proposition 5.4 on a scatter plot.

5.3 *Splitting, observed* (T2, a weekend; authored). 4k against 32k dictionaries; match by decoder cosine; find a family and its children.

5.4 *The reckoning in miniature* (T2, an evening; authored). Five nameable concepts: probe vs best SAE feature, AUC against AUC — the central empirical claim on your own bench.

5.5 *The blank wall, blind* (T2, a weekend; authored). SAEs on trained and random GPT-2; fifty features each through auto-interp; the pass rates compared, the exhibit pair harvested.

5.8 Problems

Tier I — Calisthenics

Problem 5.1 (The constraint that stops the cheat)**.** Show that without the unit-norm decoder constraint, the objective of Definition 5.2 can be driven toward zero penalty at fixed reconstruction by the rescaling $f \to f/c$, $W_{\text{dec}} \to c\,W_{\text{dec}}$, and state what the constraint pins down. *(Full solution.)*

Problem 5.2 (Reading the frontier)**.** Two dictionaries on the same layer report (L0 = 30, loss recovered 0.85) and (L0 = 60, loss recovered 0.85). Which claims should each support, and what single additional measurement most sharpens the comparison? *(Full solution.)*

Tier II — The Real Thing

Problem 5.3 (Shrinkage with interference)**.** Extend Proposition 5.4 to two active features with correlated decoder directions ($\langle d_1, d_2\rangle = \rho$): derive the optimal reported pair and show the bias now depends on ρ — shrinkage plus crosstalk. *(Full solution — the systematic bias, at depth.)*

Problem 5.4 (The gate's honesty)**.** In the gated design, suppose the gate and magnitude channels were trained *without* weight sharing. Exhibit the failure mode (selection and estimation diverging on which direction a feature means), and explain what the sharing buys. *(Hints only.)*

Problem 5.5 (Designing the discovery benchmark)**.** The limits box notes discovery has no worthy benchmark. Propose one: what ground truth can be manufactured (planted concepts in a trained model, model organisms), what the SAE must find unprompted, and what the supervised baseline is even allowed to be, given that discovery by definition lacks labels. *(Hints only.)*

Tier III — For the Ambitious (starred)

Problem 5.6 (The blank wall, quantified)**.** ⋆ Model auto-interp as a channel: explanations of bounded description length, scored by predicted-vs-actual firing agreement. Derive conditions under which *any* sufficiently rich featureless signal admits passing explanations, and identify which ingredient (dataset structure, explainer capacity, scoring threshold) the Heap result most implicates. *(Hints only.)*

5.9 Hints

Hint (Problem 5.4)**.** Untied, the gate can learn to open on one direction while the magnitude channel measures another; the reported feature is then a chimera. Sharing makes selection and estimation answers to the same question.

Hint (Problem 5.5)**.** Model organisms: fine-tune a known concept into a model (or train a toy with planted features); discovery is scored on surfacing it without labels; the honest baseline is unsupervised too — clustering on activations, or a probe permitted only post-hoc.

Hint (Problem 5.6)**.** A random network still computes *some* fixed nonlinear functions of structured text; the explainer needs only to describe the input statistics those functions happen to track. The scoring threshold does the rest.

5.10 Solutions

Policy: full for Tier I and Problem 5.3; hints only for the rest.

Solution (Problem 5.1)**.** The reconstruction $W_{\text{dec}}f$ is invariant under the rescaling while $\|f\|_1 \to \|f\|_1/c$; letting $c \to \infty$ sends the penalty to zero. The unit-norm constraint fixes the gauge: activations then carry all magnitude, so the penalty measures something real, and decoder columns are comparable unit directions — the dictionary's atoms in a common currency.

Solution (Problem 5.2)**.** At equal fidelity the sparser dictionary makes the stronger disentanglement claim (fewer active atoms per token); the denser one may merely re-describe the activation. The sharpening measurement: auto-interp (or by-hand) interpretability rates on matched samples — fidelity and sparsity being equal, meaning is the tiebreaker; and if the rates also tie, Remark 5.3 is the lesson.

Solution (Problem 5.3)**.** With unit directions the reconstruction error is

$$\|(s_1-a_1)\,d_1 + (s_2-a_2)\,d_2\|^2 = (s_1-a_1)^2 + (s_2-a_2)^2 + 2\rho\,(s_1-a_1)(s_2-a_2),$$

plus the penalty $\lambda(a_1 + a_2)$. Stationarity gives the linear system $2(a_i - s_i) + 2\rho(a_j - s_j) = -\lambda$ for $(i,j) = (1,2), (2,1)$, whence $a_i = s_i - \frac{\lambda}{2(1+\rho)}$ for the symmetric interior solution: correlated directions ($\rho > 0$) *share* the shrinkage burden (each shrinks less, the overlap absorbing the difference), while anticorrelated ones ($\rho < 0$) shrink more — and the reported *ratio* of strengths is distorted whenever $s_1 \neq s_2$. The instrument's bias is not merely low but geometry-dependent: reading two correlated features, it misreports their balance, which is precisely the kind of error a downstream circuit analysis inherits silently.

5.11 The Reading

Olshausen & Field, "Emergence of Simple-Cell Receptive Field Properties by Learning a Sparse Code for Natural Images," Nature, 1996. The founding paper of the programme (Nature 381, 607–609) and Remark 5.1's source: the sparse-coding objective, three decades before its current specimen. nature.com.

Bricken et al., "Towards Monosemanticity," 2023; and Templeton et al., "Scaling Monosemanticity," 2024. The instrument's debut and its frontier deployment: read the feature-splitting UMAPs against D3, and Scaling's "at least 65% of variance" against the dark-matter row of the reckoning table. transformer-circuits.pub; transformer-circuits.pub.

Rajamanoharan et al. (gated; JumpReLU), 2024; Gao et al. (TopK), 2024; the matryoshka papers, 2024–25. Definition 5.6's sources; read each against the selection-from-magnitude principle and note which smudge each targets. arXiv:2404.16014; arXiv:2407.14435; arXiv:2406.04093; LessWrong; arXiv:2503.17547.

Kantamneni et al., "Are Sparse Autoencoders Useful?," 2025. The probing verdict, whole: the initially promising scarce-data and corrupted-label carve-outs dissolved against careful baselines, and the demolition is the paper's headline. arXiv:2502.16681.

Wu et al., "AxBench," ICML 2025. The steering table's source, prompting's crown included; the register holds the exact scores. arXiv:2501.17148.

Heap et al., on SAEs for randomly initialised transformers, 2025. The blank wall; read beside Exhibit 5.2 and Problem 5.6. arXiv:2501.17727.

Karvonen et al., "SAEBench," ICML 2025. The standard battery: eight evaluations, two hundred open dictionaries, and the headline — proxy-metric gains do not reliably translate downstream — that is Part Six's caution made systematic. arXiv:2503.09532.

Arad, Mueller, Belinkov, 2025; Jorgensen & Hansen, 2026. The partial recovery, scoped as the register scopes it: supervised parity, prompting still on top. arXiv:2505.20063; arXiv:2605.31183.

Engels, Smith, Tegmark, "Decomposing the Dark Matter of SAEs," 2024. The error's structure: the power law that asymptotes above zero, and the linearly-predictable majority of what the instrument misses. arXiv:2410.14670.

Chapter 6

Circuits and the Discipline of Intervention

Companion to Episode Six. Everything prior was observational; here the course acquires its interventional instruments, rewards them with the field's showpiece dissection, and then meets the hydra that humbles the whole method. The discipline in formal dress: patching as a defined operation on the computational graph, path patching with its frozen-path bookkeeping, direct attribution with its caveat, the ablation variants tabulated, the indirect-object circuit drawn to its published dimensions, self-repair as a broken syllogism, and the three criteria that keep circuit claims scientific.

6.1 Parts One and Two — Patching, Defined

Definition 6.1 (Activation patching)**.** Model the forward pass as a computational graph with node activations $h_c(x)$ for components c (a head's output, a layer's stream, at a position). Fix a *minimal pair* (the art of the almost): a clean input x exhibiting the behaviour and a corrupted x' differing only in the respect under study. Cache both runs. A *patch* at c reruns the model on one input with h_c replaced by its value from the other run — $\mathrm{do}(h_c \leftarrow h_c^{(x')})$, the transplanted organ — and scores the output movement on a chosen metric. *Noising* patches corruption into the clean run (what breaks it — evidence of necessity); *denoising* patches cleanliness into the corrupted run (what fixes it — evidence of sufficiency). A careful analysis runs both and attends to where they disagree.

Remark 6.2 (The metric is half the art). The showpiece's choice generalises: score patches on a *graded, task-shaped* quantity — for indirect-object identification the *logit difference* between the correct and tempting-wrong name, $\Delta = z_{\text{Mary}} - z_{\text{John}}$, the number to watch, which sat near 3.5 in the study (odds of roughly $e^{3.5} \approx 33$: thirty-odd times likelier) — rather than top-token accuracy, which throws the gradations away. Probability metrics and KL answer different questions; state which you watched, or the analysis cannot be read.

6.2 Part Three — Wiring, and the Cheap First Pass

Definition 6.3 (Path patching)**.** For components A (earlier) and B (later), patch *only the contribution of A that reaches B's input* — recompute B with A's patched value while every other path into B, and every path around B, is held at its unpatched value. The measured movement isolates the edge $A \to B$: not whether A matters, but whether A matters *by way of* B. Iterated over pairs, the set of suspects becomes a wiring diagram.

Definition 6.4 (Direct logit attribution, with its caveat)**.** By the additive stream (Definition 1.1), the final logits decompose as $z = W_U\,\mathrm{LN}\big(\sum_c h_c\big)$; freezing the final LayerNorm's scale at its actual value renders the map linear, whence per-component shares $z \approx \sum_c W_U\,\overline{\mathrm{LN}}(h_c)$. Two caveats the desk states plainly: the decomposition is exact only under that frozen-scale linearisation; and it captures *direct* effects only — a component working through an intermediary looks unimportant. DLA finds the last wires; patching traces backward from them.

6.3 Part Four — The Subtle Art of Ablation

Turning a component off is not one operation but three, and the verdict can follow the choice of off-switch; the desk tabulates the variants with the failure mode each carries.

Ablation	*Replaces output with*	*Failure mode*
zero	0	off-distribution shock; typically overstates, can mislead either way
mean	dataset average	destroys dataset-level information carried in the mean
resample	value from another input	cleanest distributionally; answer depends on the resampling corpus

6.4 Part Five — The Showpiece, Drawn

The indirect-object circuit at its published dimensions: twenty-six heads, seven classes, a fifth of GPT-2 small conscripted into one grammatical reflex.

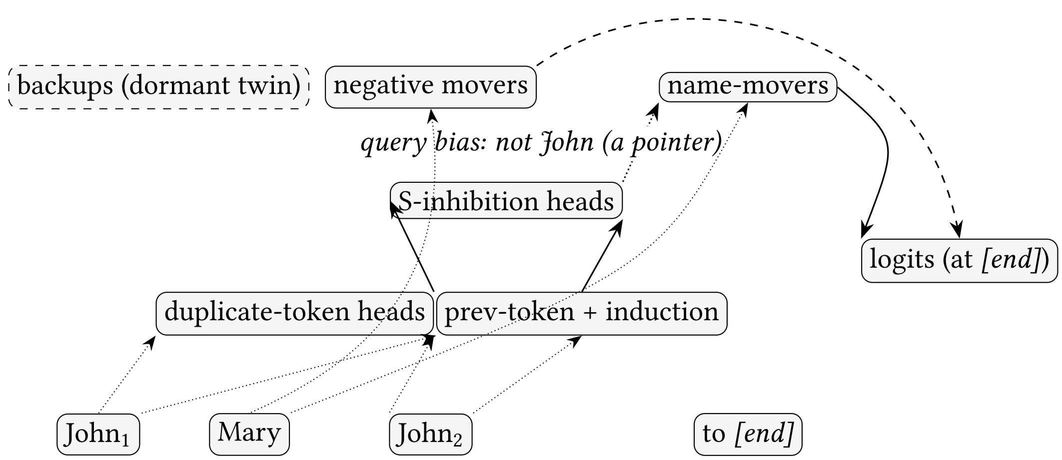


Exhibit 6.1 (The indirect-object circuit). The volume's central wiring diagram. Detection: duplicate-token heads (and the induction pair of Chapter 3, reporting for duty) mark John as repeated. Decision: S-inhibition heads, by query composition, bias the next stage away from the repeat — passing position as much as identity, a *pointer*. Copying: name-movers, so steered, fetch Mary and write her to the logits;

the negative movers *read the very name the movers promote* and write against it at the logits — copy suppression (McDougall et al.), calibration priced at a fraction of a nat; the backups, drawn dashed, are the movers' dormant twin, quiescent until Part Six's ablations wake them. Every arrow was established by patching, not narration. Schematic after the published diagram.

Remark 6.5 (The three-stage algorithm, and its residue). Detect the repeat; eliminate it; copy the survivor — elimination, note, not selection: the circuit finds the answer by excluding the wrong one. The honest accounting the study itself supplied: the recovered circuit reproduces most, not all, of the behaviour, and causal scrubbing's stricter path-level standard shrank the headline number further. Even the reference dissection leaves a remainder — and Part Six names where some of it hides.

6.5 Part Six — The Hydra, as Logic

Proposition 6.6 (What self-repair breaks)**.** *The inference "ablating c leaves the behaviour intact, therefore c does not matter" is invalid in the presence of* backup components*: heads quiescent in normal operation that assume a function when its primary is disabled. In the showpiece, ablating the proven name-movers barely moves the logit difference because backup movers wake and compensate. Consequently: ablation can understate importance; necessity claims require ablating suspected backups jointly or measuring compensation directly; and "the circuit" is in general a* distribution *of overlapping mechanisms — a primary pathway with understudies — not a unique diagram. Provenance, separated per the register: the backup heads are the original study's finding; the "hydra effect" coinage and its generality are later work's, in larger models.*

6.6 Part Seven — When Is an Explanation Finished?

Definition 6.7 (Faithfulness, completeness, minimality)**.** For a proposed circuit C (a subgraph plus an algorithmic account): *faithful* — running C in isolation (non-circuit components frozen to reference values) reproduces the model's behaviour on the task distribution, to stated tolerance; *complete* — no component outside C changes the verdict when perturbed (backups and negatives included); *minimal* — every element of C earns its place (its removal degrades the isolated circuit). *Causal scrubbing* strengthens faithfulness to the path level: the claimed account must survive resampling of every activation the account declares irrelevant. Against *interpretability illusions* — stories consistent with all the evidence in hand and false — these criteria are the immune system, and they are checked, not asserted.

Remark 6.8 (Screening at scale). Attribution patching, the cheap screen, estimates all patch effects at once from gradients: $\Delta\text{metric} \approx \nabla_h\,\text{metric} \cdot (h^{\text{clean}} - h^{\text{corr}})$ — a first-order estimate that degrades precisely where effects are large and saturating. Screen broadly, confirm narrowly, never ship the screen. Automated circuit discovery prunes the graph algorithmically; the discovery algorithms are themselves now benchmarked, which is what a maturing method looks like.

Protocol — IOI patching, end to end (checked against the Interpretability in the Wild setup)

1: Build the IOI distribution: templated sentences with name pairs; for each clean sentence, the corrupted twin swaps the second subject occurrence (flipping the answer), holding length and structure fixed.
2: Choose the metric: logit difference between the two names (Remark 6.2).

3: Cache clean and corrupted runs; patch each head's output (denoising direction), one at a time, at the relevant positions; record the metric's recovery.
4: Render the layer-by-head heatmap; the name-movers light late (layers 9–10), S-inhibition mid (7–8), detection early.
5: Confirm the wiring with path patches on the claimed edges (inhibition to name-mover queries above all).
6: Test for self-repair: ablate the top name-movers jointly; recompute per-head DLA; watch the backups' shares grow.

Lab 6.1 executes steps 1–4 and Lab 6.2 step 6; their heatmap and compensation bars join the volume as exhibits at the first bench run.

Protocol — Constructing a minimal pair — the rules of the almost (checked against the patching literature's practice)

1: State the single respect under study in one sentence before touching the text.
2: Corrupt that respect and nothing else: hold token count, structure, and vocabulary fixed wherever the tokeniser permits.
3: Verify the behavioural flip: the model's answer must actually change between the pair, by your metric, before any patch is run.
4: Audit for smuggled differences (tokenisation splits, positional shifts); a pair that changed two things attributes to neither.
5: For production debugging: given a failing input, manufacture the nearest *succeeding* one and patch layer-by-layer between them — the laboratory discipline, worn as overalls.

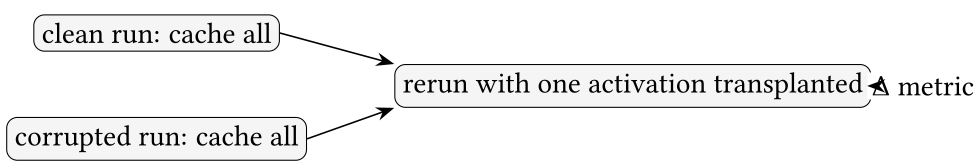


Exhibit 6.2 (The fundamental move). Two runs, one transplant, one number watched. Noising and denoising are this diagram read in its two directions. Schematic.

The Engineer's Dividend

Four transfers, indexed: *did you patch it?* — the one question that promotes a suspect to a culprit, owed to every dashboard and attention heat-map in your stack; *manufacture the minimal pair* — the second protocol box, applied to production regressions; *screen with gradients, confirm with patches, never ship the screen* — Remark 6.8 at the bench; and *read your own ablations under the hydra* — Proposition 6.6's broken syllogism applies verbatim to "we removed the retrieval component and nothing broke," which is ambiguous between useless and covered-for, with opposite operational consequences.

The Honest Limits

Patching localises *information*, not meaning: a component that restores behaviour carries what the behaviour needs, which is not yet an account of what it computes. The minimal pair's art bounds every conclusion — a sloppy pair attributes to nothing. DLA is exact only under its linearisation. The ablation verdicts are method-relative (the table above). And the hydra stands: completeness is checkable only against the perturbations you thought to run, and the understudies' understudies are a regress the criteria manage rather than close.

6.7 The Bench

- **6.1** *IOI patching, end to end* (T1–T2, a weekend done from scratch, an evening with the tutorial as guide; authored). The first protocol box on GPT-2 small: the distribution, the metric, the per-head heatmap (an exhibit that joins the volume at the first bench run). The season's best lab; every claim of Part Five checkable.
- **6.2** *Meet the hydra* (T1, an evening on 6.1's cache; authored). Ablate the top movers jointly; per-head DLA before and after; the compensation measured. Produces the compensation bars (joins the volume at the first bench run).
- **6.3** *The ablation shootout* (T1, an evening; authored). Zero, mean, resample on the same ten heads; the rankings compared; the table above verified.
- **6.4** *Attribution patching, calibrated* (T1, an evening; authored). Gradient estimates against true patches for all 144 heads; the saturation caveat as a scatter plot.
- **6.5** *Automated discovery* (T2, a weekend; authored). An edge-attribution-patching pruner on IOI; recovered subgraph against Exhibit 6.1; divergences read honestly.

6.8 Problems

Tier I — Calisthenics

Problem 6.1 (Directions, distinguished)**.** For the IOI pair, say exactly what a *noising* patch at a name-mover tests, what a *denoising* patch at the same head tests, and construct a toy scenario in which the two verdicts disagree. *(Full solution.)*

Problem 6.2 (The metric, defended)**.** A colleague scores IOI patches by whether the top-1 token is correct. Exhibit a patch outcome their metric misses entirely and the logit difference registers, and state the general principle. *(Full solution.)*

Tier II — The Real Thing

Problem 6.3 (Path patching, bookkept)**.** Write out, for a two-layer attention-only model, exactly which recomputations the path patch $A \to B$ requires: which activations are taken from which cache, which are recomputed, and why the naive shortcut (patch A, read B) measures the wrong thing. *(Full solution.)*

Problem 6.4 (The DLA blind spot, constructed)**.** Build the smallest example (two components suffice) in which a component with *zero* direct logit attribution is necessary for the behaviour, and verify patching catches what DLA misses. *(Hints only.)*

Problem 6.5 (Compensation, measured)**.** Define a compensation coefficient for a component set S: the fraction of S's normal contribution assumed by non-S components under joint ablation of S. Show how to estimate it from per-head DLA before and after, and what value acquits versus convicts the hydra. *(Hints only.)*

Tier III — For the Ambitious (starred)

Problem 6.6 (Scrubbing a claim)**.** ⋆ Take the three-stage IOI account at face value and design its causal-scrubbing test: enumerate the activations the account declares irrelevant, the resampling scheme for each, and the behavioural tolerance that would count as survival. State one way the account could pass faithfulness and fail your scrub. *(Hints only.)*

6.9 Hints

Hint (Problem 6.4)**.** Let component one write a signal component two reads and converts into the logit push; component one's own direct path to the unembedding can be exactly zero.

Hint (Problem 6.5)**.** Compensation = $(\Delta_{\text{expected}} - \Delta_{\text{observed}})/\Delta_{\text{expected}}$ on the metric, with the expected drop taken from S's pre-ablation DLA share; near one convicts.

Hint (Problem 6.6)**.** The account says inhibition passes *position, not identity*: resample the identity-carrying subspace while preserving position and the scrub should survive; if it does not, the account understated what travels.

6.10 Solutions

Policy: full for Tier I and Problem 6.3; hints only for the rest.

Solution (Problem 6.1)**.** Noising at the mover (corrupt value into clean run) tests *necessity*: does the clean behaviour survive losing this head's clean output? Denoising (clean value into corrupted run) tests *sufficiency*: does this head's clean output alone carry enough to restore the answer in a corrupted context? Disagreement scenario: two redundant movers. Noising either one barely moves the metric (the twin covers — necessity verdict: unimportant); denoising either one substantially restores (each carries the information — sufficiency verdict: important). The divergence is not a paradox but a measurement of redundancy — the hydra, visible in the two lenses' disagreement before any ablation study is run.

Solution (Problem 6.2)**.** A patch that moves the logit difference from 3.5 to 0.3 leaves Mary still top-1 (the colleague's metric: "no effect") while the model's discrimination has collapsed by an order of magnitude — one further nudge from flipping. Graded metrics register the machinery's strength; threshold metrics register only its final collapse, and a circuit analysis conducted at the threshold sees nothing until everything. The principle: measure the quantity the circuit actually regulates.

Solution (Problem 6.3)**.** To patch the edge $A \to B$ (denoising direction): take h_A^{clean} from the clean cache; recompute B's input as [corrupted-run stream with A's contribution replaced by h_A^{clean}], where every *other* contribution to B's input — including other heads reading A's corrupted output — stays at corrupted values; recompute B only, then let the run continue with every downstream path *except* those through B frozen at corrupted values (equivalently: write only B's recomputed output into the continuing run). The naive shortcut — patch A wholesale and read the output — routes A's clean value through *all* paths at once ($A \to C$, $A \to$ logits, $A \to B$), so any effect measured is the sum over routes and attributes nothing to the edge. The bookkeeping is precisely the discipline of holding every path but one at reference — Definition 6.3 executed, and the reason path patching costs more thought than patching.

6.11 The Reading

Wang et al., "Interpretability in the Wild" (IOI), 2022. The showpiece's source. Read the head-class taxonomy against Exhibit 6.1, the logit-difference discussion against Remark 6.2, and their own faithfulness accounting against Remark 6.5 — the candour is the model. arXiv:2211.00593.

Meng et al., causal tracing (ROME), 2022; and the patching methodology literature (Heimersheim & Nanda's practical guide). Definition 6.1's wider family and its practicalities; the guide pairs directly with Lab 6.1. arXiv:2202.05262; arXiv:2404.15255.

Chan et al., causal scrubbing, 2022. The stricter standard of Definition 6.7; read with Problem 6.6 in hand. AlignmentForum.

McGrath et al., the hydra effect, 2023; McDougall et al., copy suppression, 2023. Proposition 6.6's general phenomenon and the negative movers' mechanism, with the provenance kept straight as the register requires. arXiv:2307.15771; arXiv:2310.04625.

Kramar et al., AtP*, 2024. Attribution patching's failure modes and repairs; Remark 6.8's caveat, quantified; Lab 6.4's reference. arXiv:2403.00745.

Conmy et al., ACDC, 2023; and the circuit-discovery benchmarking that followed. The automation coda's sources: the search, and the field learning to grade its own searchers. arXiv:2304.14997.

Chapter 7

Steering, Editing, and the Stakes

Companion to Episode Seven — the finale of the course proper. The reading instruments turned into writing ones, and the safety argument the whole course was built to reach: steering as vector addition with an honest reliability caveat; editing with the localisation-vs-leverage dissociation; the gallery's three load-bearing cases, recorded as claims; and the stakes as an argument in three indistinguishable interiors — behaviour and self-report both fail as windows, so the interior is the only instrument left. A signpost before we begin: the gallery's instruments — the transcoder and the attribution graph — are built in Chapters 8 and 9; a reader who wants the method before the verdicts may take them first, and the gallery's formal exhibits live in Chapter 9 where their edges can be defined.

7.1 Parts One and Two — Steering, and the Refusal Direction

Definition 7.1 (Activation steering)**.** At a chosen layer ℓ and coefficient α, replace the running residual state by $h' = h+\alpha v$ for a fixed *steering vector* $v \in \mathbb{R}^{d_{\text{model}}}$; the downstream computation reads the addition as its own thought. Two standard sources for v: the *contrastive* estimator $v = \mu^+ - \mu^-$, the difference of mean activations under matched prompt sets that differ only in the target quality (the patching logic of Chapter 6 turned from probe to tool); and the *decoder column* of a sparse-autoencoder feature for the quality (Chapter 5), the Golden Gate clamp being exactly this at large α.

Remark 7.2 (The reliability caveat). Steering is a powerful experimental technique with an unsolved consistency problem, and the desk states it plainly: a vector effective on one prompt is often weak on the next (high per-example variance), and large α taxes general competence — benchmark scores sag as the personality moves. Any bench use measures the tax alongside the effect (Lab 7.2 makes this the whole experiment). The reflex the course applies to every instrument fires here too: read the failure mode, not only the demonstration.

Proposition 7.3 (Directional ablation removes refusal)**.** *Let $\hat{r}$ be the unit refusal direction (estimated contrastively from harmful vs harmless instructions). Ablating it from every residual state and position,*

$$h \;\mapsto\; h - (h \cdot \hat{r})\,\hat{r},$$

suppresses refusal (a white-box jailbreak); adding $+\alpha\hat{r}$ induces refusal on innocuous inputs. The behaviour is mediated, to first order, by this one direction — with the register's refinement: close examination finds refusal occupying a narrow cone *of several ablatable directions with a dominant axis, so the single-vector account is the first-order truth of a slightly thicker geometry. The weight-space variant orthogonalises the writing matrices against $\hat{r}$, baking the jailbreak into the weights.*

Remark 7.4 (One structure, two lessons). The clean structure that makes refusal *legible* is the same structure that makes it *pickable*: understanding a safety mechanism is power over it, and the power cuts both ways. Dual use is intrinsic to interpretability, not incidental — the argument for the work (the alternative, not understanding these systems, is worse) holds while the peril is kept in view.

Protocol — Extracting a steering vector, with the capability check as a mandatory step (checked against the representation-engineering and persona-vector recipes)

1: Assemble matched prompt sets differing only in the target quality; collect residual activations at a mid-to-late layer.
2: Estimate $v = \mu^+ - \mu^-$; normalise.
3: Sweep α; at each, measure the intended behavioural effect *and* a general-capability proxy (a small benchmark slice).
4: Report the dose–response of both: the effect curve and the tax curve. A steering claim without the tax is half a result.
5: For a persona-vector workflow: prefer *preventative* steering during fine-tuning (the paper's headline mitigation) over inference-time subtraction; and use the vector to *screen* fine-tuning data, flagging examples that project onto the unwanted trait.

Lab 7.1 extracts and ablates the refusal direction; Lab 7.2 is the reliability audit of steps 3–4.

7.2 Part Three — Editing, and the Map That Is Not the Territory

Steering writes the activations in flight; editing writes a fact into the weights at rest — the rank-one update (ROME).

Definition 7.5 (Rank-one fact editing)**.** Model an MLP down-projection as a linear associative memory mapping keys (subject encodings) to values (attribute encodings). To install $k \mapsto v_{\text{new}}$, apply the minimal-norm rank-one update

$$W' = W + (v_{\text{new}} - Wk)\,\frac{k^\top C^{-1}}{k^\top C^{-1} k},$$

with C the uncentred key covariance — the smallest change writing the new association while least disturbing the others. Mass editing (MEMIT) spreads many such updates across a band of layers.

Proposition 7.6 (Storage localisation is not editing leverage)**.** *The layer at which causal tracing localises a fact (where its information* is*) and the layer at which a rank-one edit most effectively changes the model's use of the fact (where intervening has* leverage*) can differ: editing a non-traced layer sometimes outperforms editing the traced one. "Where it is stored" and "where to reach to change it" are distinct questions — the map of storage is not the map of leverage — because the model's use of a fact is redundant and reconstructable (the hydra of Chapter 6), so influence does not sit only where information does.*

Remark 7.7 (The editor's honest limits). Beyond the localisation dissociation, the register's catalogue: edits are off-target (changing unintended associations), brittle (holding on checked cases, failing on entailed ones — the *ripple effect*), dose-sensitive, and compose poorly (sequential mass edits degrade fluency progressively). A scalpel sharper in the brochure than the hand; wielded with the humility Proposition 7.6 teaches.

Protocol — A ROME edit, end to end, with the off-target audit (checked against the ROME/MEMIT evaluations)

1: Choose a fact and its replacement; locate by causal tracing (Chapter 6's patching).
2: Apply the rank-one update of Definition 7.5 at the traced layer.
3: Test *efficacy* (the edited prompt), *generalisation* (paraphrases), and *specificity* (neighbouring, unrelated facts must not move).
4: Test the *ripple*: entailed facts ("X's capital is now Y" implies claims about Y) — do they follow, or does the edit sit brittle and local?
5: Sequential-edit stress: apply 100 edits; watch fluency and prior edits degrade. Report where it breaks.
6: Compare editing a *non-traced* layer (Proposition 7.6); if it edits better, you have met the dissociation on your own bench.

7.3 Part Four — The Gallery, as Forward References

The gallery is a preview drawn on credit — its exhibits await instruments not yet built. Each case's *formal* exhibit — the attribution graph with its edges defined — lives in Chapter 9. Here the desk records only the three cases the stakes actually need, as claims to be validated there:

Case	*Claim*	*Where validated*
two-hop (Dallas→Texas→Austin)	an unspoken intermediate is computed and load-bearing; swap it, the answer swaps	Ch. 9, the master graph and its intervention
parallel addition + confabulation	answer by approximation + ones-digit path; the model narrates the schoolbook method it does *not* use	Ch. 9 trace; the three-way faithfulness table below
CoT faithfulness, three ways	honest / confabulated / motivated — identical on the page, distinct in the graph	Ch. 9; the hinge of this chapter's stakes

Remark 7.8 (Why only three). The rhyme-planning, multilingual-core, and hallucination-misfire cases are real; the stakes, however, rest only on the three above — the two-hop (mechanism the sceptic denies exists), the addition/confabulation (introspective report ≠ mechanism), and the three-way faithfulness (the inside and the outside can disagree). The desk carries what the argument uses; the gallery's full delight is Chapter 9's.

7.4 Parts Five and Six — The Stakes, as an Argument

The whole case rests on one proposition: the three interiors — aligned, distribution-lucky, evaluation-aware — and no transcript to tell them apart.

Proposition 7.9 (Behaviour underdetermines the interior)**.** *Good behaviour on every devised test is consistent with at least three distinct interiors: (i) a genuinely aligned model; (ii) one aligned on the tested distribution but not beyond it; (iii) one behaving well* because *it represents the situation as an evaluation. Under observation, all three emit identical transcripts — behaviour is precisely what interior (iii) controls.*

Therefore no quantity of behavioural testing distinguishes them, and self-report does not help: a chain of thought is itself a behaviour, subject to the same gap (Chapter 14 measures that gap; Part Four's confabulation case is its mechanism). The interior is the only remaining instrument.

CoT case	*Transcript*	*Wiring diagram*
honest	plausible reasoning	the narrated steps are the computed steps
confabulated	plausible reasoning	a different mechanism ran; the account is post-hoc
motivated	plausible reasoning	the circuit runs *backward* from a planted answer to its justification

Exhibit 7.1 (One transcript, three reasonings). The CoT-faithfulness triad as a table: three reasonings identical on the page (middle column), distinguished only in the wiring (right column). Turpin et al.'s hint experiment establishes the motivated case behaviourally; Chapter 9's graphs catch all three as circuits.

Remark 7.10 (The internal instruments, and their edge). Where watching and asking both fail, the interior methods are pointed at the gap directly: *deception probes* (a classifier on activations catching a truth represented but not stated); *eliciting latent knowledge* (extracting what the model internally knows as distinct from what it reports). Early and imperfect, aimed correctly. The three results that make the case concrete: sleeper agents (Hubinger et al. 2024 — backdoors surviving safety training; detectable, the defection-probes follow-up found, by simple activation probes); the hidden-objective audit (Marks et al. 2025 — three of four red teams found a concealed objective, the decisive tool being training-data search with SAE features assisting, the API-only team failing: a vindication of *access to the interior*, not of any single instrument); and the forward worry, that a situationally-aware model might learn to obfuscate its internals. "We can read today's models" is weaker than "we can read any model we might build," and the gap is where the field's hardest future lies.

The Engineer's Dividend

The Engineer's Inheritance, in three registers, made operational. *Craft:* a mechanistic hypothesis names the component, predicts the intervention, and can be wrong in public — the crosscoder for a regressed fine-tune, the feature-margin reading for a quantisation loss, the steering vector (with its tax measured) for a live nudge. *Profession:* the auditing techniques — probes, feature inventories, the hunt for hidden dispositions — migrate from research to engineering requirement as these systems take consequential work. *Stakes:* whether a capable model is aligned or merely presenting alignment is not answerable from outside, and the interior is at present our only serious instrument — incomplete, bounded, possibly too slow, and pointed at the right thing.

The Honest Limits

The summary judgment: a *proven possibility*, an *unproven sufficiency*. Steering is unreliable (Remark 7.2); editing dissociates storage from leverage (Proposition 7.6); the audits that succeed today run against models not adversarial toward the audit. Four standing worries: the streetlight (studying the legible and mistaking it for the whole), computation-in-superposition (Chapter 4's permanent limit — thinking in folds no inventory resolves), the combinatorial (a full account too vast to hold), and the adversarial (the models most needing audit most able to defeat it). The instruments work; whether they keep working, fast and completely and robustly enough, is the open and serious question.

7.5 The Bench

7.1 *The refusal direction: extract, ablate, add* (T2, an afternoon; authored). Contrastive extraction on a small chat model; directional ablation jailbreaks; addition padlocks the empty room. Highest wow-per-line in the season; the data for the refusal exhibit a future edition will print.

7.2 *The steering reliability audit* (T2, a day; authored). One vector across 200 heterogeneous prompts: per-prompt effect variance and the capability tax vs α — the caveat of Remark 7.2 made empirical.

7.3 *ROME/MEMIT with the ripple test* (T2, a day; authored). Edit a fact; test efficacy/generalisation/specificity, then the ripple and the sequential-edit degradation; edit a non-traced layer and meet Proposition 7.6.

7.4 *Persona-vector data screening* (T2, one to two days; authored). Extract a "sycophantic" vector; project a contaminated fine-tuning set; flag the pushing examples; compare against a model-based filter.

7.5 *Toy sleeper agent + probe* (T2, one to two days; authored). Fine-tune a trigger-conditioned defection; train a linear probe; show it fires pre-defection where behaviour gives nothing away.

7.6 Problems

Tier I — Calisthenics

Problem 7.1 (The contrastive vector is a difference of means)**.** Show that for matched prompt sets, $v = \mu^+ - \mu^-$ is the direction maximising the difference in projected mean between the two conditions, and state what assumption about the two sets makes it also the direction along which they best separate. *(Full solution.)*

Problem 7.2 (Ablation as projection)**.** Verify $h \mapsto h - (h \cdot \hat{r})\hat{r}$ is the orthogonal projection removing the $\hat{r}$ component, and compute what adding $+\alpha\hat{r}$ afterward does to a state that already had refusal component β. *(Full solution.)*

Tier II — The Real Thing

Problem 7.3 (The rank-one update, derived)**.** Derive Definition 7.5's update as the minimiser of $\|W' - W\|_F$ subject to $W'k = v_{\text{new}}$ (the constrained least-change edit), and identify what role C^{-1} plays in protecting the other stored associations. *(Full solution.)*

Problem 7.4 (Localisation and leverage, dissociated)**.** Construct a two-path toy in which a fact's information is stored on path A (causal tracing localises there) but an edit on path B changes behaviour more, because A's contribution is reconstructable downstream. Relate the mechanism to the hydra of Chapter 6. *(Hints only.)*

Problem 7.5 (The three interiors, formalised)**.** Model an evaluator with a finite test distribution D and three policies matching on D but differing off it. Show no D-measurable statistic distinguishes them, and state precisely what an interior measurement must access to break the tie. *(Hints only.)*

Tier III — For the Ambitious (starred)

Problem 7.6 (The obfuscation frontier)**.** ⋆ Suppose a model is trained under an interior audit that penalises a probe's firing. Argue, by analogy to Chapter 6's causal scrubbing and Chapter 11's CoT-monitor obfuscation, what the model can and cannot hide, and identify the audit property that would make obfuscation costly. *(Hints only.)*

7.7 Hints

Hint (Problem 7.4)**.** Let path A write a signal that path B's later component also computes from the input independently; ablating A is covered by B's reconstruction, but editing B's weights changes the shared downstream readout.

Hint (Problem 7.5)**.** Any statistic that is a function of behaviour on D is constant across the three policies by construction; the tie breaks only on a variable reading the policy's *off-D disposition*, which lives in the representation, not the transcript.

Hint (Problem 7.6)**.** Chapter 11's Baker result: outcome-only pressure obfuscates the trace. An audit costly to fool is one whose target the model must *use* to do the task — obfuscating a load-bearing representation degrades the capability the reward also wants.

7.8 Solutions

Policy: full for Tier I and Problem 7.3; hints only for the rest.

Solution (Problem 7.1)**.** For projection onto unit u, the difference in projected means is $u \cdot (\mu^+ - \mu^-)$, maximised over unit u at $u = (\mu^+ - \mu^-)/\|\cdot\|$ by Cauchy–Schwarz — so $v = \mu^+ - \mu^-$ is that direction. It is also the best *separating* direction (the Fisher direction) when the two conditions share a common, isotropic within-class covariance: then the optimal linear discriminant is $\Sigma^{-1}(\mu^+ - \mu^-) \propto \mu^+ - \mu^-$. Matched prompt sets are the device for making the shared-covariance assumption approximately hold — which is why the *almost* of Chapter 6 governs steering as it governed patching.

Solution (Problem 7.2)**.** $P_{\hat r} h = (h \cdot \hat r)\hat r$ is the projection onto $\hat r$; $h - P_{\hat r} h$ removes it, leaving the orthogonal complement (idempotent, as a projection must be). A state with refusal component β (i.e. $h \cdot \hat r = \beta$), ablated then added: $(h - \beta \hat r) + \alpha \hat r$ has refusal component α exactly — ablation *sets* the component to zero and addition *overwrites* it to α, which is why strong positive α refuses everything regardless of the input's own β.

Solution (Problem 7.3)**.** Minimise $\|\Delta\|_F$ ($\Delta = W' - W$) subject to $(W+\Delta)k = v_{\text{new}}$, i.e. $\Delta k = v_{\text{new}} - Wk =: r$. Lagrangian stationarity gives $\Delta = r\,\lambda^\top$ for a multiplier λ, and the constraint $\Delta k = r$ forces $\lambda^\top k = 1$, whence the minimal-norm choice $\lambda = k/\|k\|^2$ and $\Delta = r\,k^\top/\|k\|^2$ — rank one, as claimed. Replacing the plain inner product by the C^{-1}-weighted one ($\lambda = C^{-1}k/(k^\top C^{-1}k)$) makes the update least-disturbing in the metric of the *other* keys: C being the key covariance, C^{-1} steers the edit into the direction least aligned with the population of stored subjects, so the new association is written with minimal collision against the old — the "protect the other facts" clause, made precise. (That it protects them *on average* and not pointwise is exactly why the ripple and specificity tests of the protocol box are necessary.)

7.9 The Reading

Zou et al., "Representation Engineering," 2023. Definition 7.1's programme; the honesty/power/-morality directions, read against Remark 7.2's caveat. arXiv:2310.01405.

Arditi et al., refusal is mediated by a single direction, 2024; and the geometry-of-refusal follow-ups. Proposition 7.3 and its cone refinement; Lab 7.1's source. arXiv:2406.11717; arXiv:2502.17420.

Anthropic, "Persona Vectors," 2025. The automated extraction, the preventative-steering headline, and the data-screening use; the protocol box's step five and Lab 7.4. arXiv:2507.21509.

Meng et al., ROME (2022) and MEMIT (2023); Hase et al., does localisation inform editing? (2023); Fang et al., AlphaEdit (ICLR 2025). Definition 7.5 and Proposition 7.6; read the dissociation paper against the ripple-effect literature (Cohen et al.); and for the sequential-edit degradation of Remark 7.7, AlphaEdit projects each edit into the null space of the preserved keys and recovers much of the lost ground. arXiv:2202.05262; arXiv:2210.07229; arXiv:2301.04213; arXiv:2307.12976; arXiv:2410.02355.

Lindsey et al., "On the Biology of a Large Language Model," 2025. Part Four's gallery; deferred in formal earnest to Chapter 9, where the graphs' edges are defined — read it there, or now for the delight. transformer-circuits.pub.

Turpin et al., unfaithful chain-of-thought, 2023; Hubinger et al., sleeper agents, 2024; MacDiarmid et al., the defection probes, 2024; Marks et al., auditing hidden objectives, 2025. Exhibit 7.1's behavioural establishment and Remark 7.10's three concrete results, each with the register's precise reading (three of four teams; data-search decisive; API-only failed) — the probes credited to their own paper: "Simple probes can catch sleeper agents" is the follow-up's finding, linear classifiers on generic contrast pairs catching the sleeper before it turns. arXiv:2305.04388; arXiv:2401.05566; anthropic.com; arXiv:2503.10965.

Christiano, Cotra, Xu, "Eliciting Latent Knowledge," 2021. The ARC report that named the ambition Remark 7.10 inherits: how to extract what the model knows when its report and its knowledge can differ. alignment.org.

Chapter 8

Transcoders and Crosscoders, in Full

Companion to Episode Eight, the supplement's opening. The engine room the course had read around, now in notation: the clerk the autoencoder could not see and the transcoder built to read it, its objective against the SAE's line by line; the cross-layer sparsity cost that IS the depth-collapse incentive (booked here, tried in Chapter 9); the replacement model and its honest error bookkeeping; the crosscoder that diffs a fine-tune, and the latent scaling its claims must survive — with the resource inventory, kept in a table where a catalogue belongs.

8.1 Part One — The Clerk, and Why the Autoencoder Missed It

Remark 8.1 (MLP anatomy). A feed-forward block — the *clerk* — is $\mathrm{MLP}(x) = W_{\mathrm{down}}\,\sigma(W_{\mathrm{up}}x)$ (gated variants insert a multiplicative gate), with $W_{\mathrm{up}} \in \mathbb{R}^{d_{\mathrm{ff}}\times d_{\mathrm{model}}}$, $d_{\mathrm{ff}} \approx 4d_{\mathrm{model}}$ — roughly two-thirds of a transformer's parameters live here. The SAE of Chapter 5 reads the *stream* (contents); it is blind to this *map* (computation). The transcoder is built to read the map.

8.2 Part Two — The Transcoder, Against the SAE

Definition 8.2 (Transcoder)**.** For an MLP block with pre-input x and output $y = \mathrm{MLP}(x)$, a *transcoder* learns

$$f = \sigma_{\mathrm{sparse}}(W_{\mathrm{enc}}x + b_{\mathrm{enc}}), \qquad \hat{y} = W_{\mathrm{dec}}f + b_{\mathrm{dec}}, \qquad \mathcal{L} = \|y - \hat{y}\|^2 + \lambda\|f\|_1,$$

differing from the SAE (Definition 5.2) in *one place that changes everything*: the target is the clerk's *output* y, not its input x — transcoding, not reconstructing. Encoder and decoder therefore live in *different spaces* (pre-MLP and post-MLP), and a feature is a *rule* — "when this pattern is present in the input, add this to the output" — read off like the source of a function rather than its logs. A *skip transcoder* adds an affine term $W_{\mathrm{skip}}x$ to carry the clerk's merely-linear work, freeing the sparse dictionary for the nonlinear remainder.

Remark 8.3 (The input-independent skeleton). Because a feature's decoder direction is fixed, the strength with which feature i (at one layer) drives feature j (above) factors into an *input-independent* weight — the product of i's decoder with j's encoder, computable from the weights alone — times an input-dependent activation. The connectivity skeleton is thus readable statically, before any prompt: the dividend the purely behavioural engineer never has (Chapter 9 draws on it).

8.3 Part Three — Per-Layer, Cross-Layer, and the Depth-Collapse Charge

Definition 8.4 (Cross-layer transcoder)**.** A *cross-layer transcoder* (CLT) reads at layer ℓ with a single encoder and writes into every higher layer's output through decoders $\{W_{\text{dec}}^{(\ell\to m)}\}_{m\geq\ell}$ — each feature threading up the building. The sparsity penalty is taken on the *concatenated* decoder, whose norm grows *sublinearly* as a feature's mass spreads across floors — which cures per-layer duplication and, as the next proposition shows, prices the honest circuit above the dishonest one.

Proposition 8.5 (The sparsity cost is the depth-collapse incentive)**.** *Consider representing a two-step dependence $A \to B \to C$. The faithful encoding activates* two *features (one per step); the depth-collapsed encoding activates* one *feature at A plus a near-free additional decoder vector reaching directly to C's floor. Under the concatenated-decoder sparsity penalty, the second costs strictly less while reconstructing C as well — so where a shallow shortcut reconstructs as faithfully as the layered truth, the objective* prefers *it. The instrument is thereby incentivised to rewrite deep multi-hop circuits as sums of shallow single-hop ones:* depth collapse *(the course's coinage; the Lange et al. post's phrase is the rewriting itself). A feature that legitimately persists up the building and a shortcut that skips it are identical from outside — the charge this chapter books and Chapter 9 tries on a toy with known ground truth.*

8.4 Part Four — The Replacement Model, and Its Error

Definition 8.6 (Local replacement model)**.** Replace every clerk by its transcoder; the activations drift from the original as small errors compound. The *local replacement model* restores exactness on a given prompt by inserting per-clerk *error nodes* $\varepsilon_\ell = \text{MLP}_\ell(x) - T_\ell(x)$ as explicit additive terms, and by *freezing* the attention patterns and LayerNorm scales at their original values. The error node is the quantified confession of what the rules missed (small: trust the story; large: the rules missed something, and the instrument says so); the freezing renders the model *very nearly linear in the feature activations* — the price that purchases the readability Chapter 9 exploits.

Remark 8.7 (Fidelity, and what it means). The blunt test runs the replacement model *without* error nodes and asks how often its top token matches the original's: far from always, improving with dictionary size, and better for the CLT than for matched per-layer stacks (the CLT sits on the better fidelity– sparsity frontier — the technical sense in which its expense was judged worth paying). But the analysis is never run bare: with error nodes the local model matches the original *exactly* on the prompt, so bare fidelity measures only how large the confession must be. Chapter 5's dark matter, reduced and not eliminated.

8.5 Part Five — The Crosscoder

Definition 8.8 (Crosscoder)**.** A *crosscoder* learns one shared encoder over the concatenated (or jointly-read) activations of two models — a base and its fine-tune — with a *per-view decoder*, and reconstructs each model's activation from the shared code. A feature is classed *shared* (both decoder norms substantial), *base-only*, or *fine-tune-only* (one view's decoder norm near zero). The diff of a fine-tune becomes a change in the vocabulary of concepts, read off the decoder-norm scatter.

Proposition 8.9 (The diff can lie: complete shrinkage and latent decoupling)**.** *The magnitude penalty (Chapter 5, D1) manufactures spurious fine-tune-*only *features: a concept present-but-weak in one view is* shrunk to a false absence *(*complete shrinkage*), or its two views are split across separate latents (*latent decoupling*), so a shared concept masquerades as introduced. The diagnostic is* latent scaling — *rescaling* *each latent's activation to its best per-view fit and re-measuring the norm ratio, which unmasks the shrunk shared features; the cure is a BatchTopK gate (no magnitude penalty, no shrinkage). Chapter 5's reckoning, restaged: a fine-tune-unique claim must survive latent scaling before a story is built on it.*

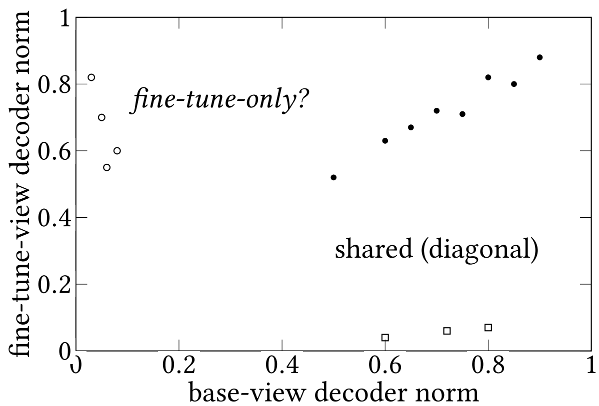


Exhibit 8.1 (The decoder-norm scatter, read honestly). Shared features cluster on the diagonal; apparent one-view features sit on the axes. The open circles are the trap: latent scaling reveals which are genuinely introduced and which are shared concepts shrunk to a false absence (Proposition 8.9). Schematic; Lab 8.2's real scatter carries the verdict.

8.6 Parts Seven and Eight — The Instruments Deliver

Remark 8.10 (The toxic persona, and diffing computation). Two field results the desk records with the register's precision. *Emergent misalignment* (Betley et al. 2025): a narrow bad fine-tune (insecure code) produces broad misalignment; the mechanism, found by *SAE-based* diffing (Wang et al. 2025 — *not* a crosscoder; the register insists on the distinction), is a *toxic persona* feature that discriminates corrupted from clean models, induces the corruption by steering the unpoisoned model, and twitches as an early warning before behavioural suites react — the narrow poison having merely turned up a persona pretraining already installed ("fine-tuning modulates existing circuits," in safety dress). *Diffing computation* (Hu, Ward, Icard, Potts, 2026): transcoder *adapters* imitate the *change* in each clerk's map between base and reasoning-tuned model; hesitation features ($\approx$ 2.4% of adapter features) prove necessary and sufficient for the "wait"s — ablate them and responses shorten with accuracy unchanged on MATH500/AMC23/GPQA-D. Some of what reasoning training installs is the mannerism, separable from the substance.

Open asset	*What*	*Where*
circuit-tracer	replacement-model + attribution library (Anthropic Fellows)	`decoderesearch/circuit-tracer`
Neuronpedia	interactive frontend; graph browsing & intervention	`neuronpedia.org`
Gemma Scope transcoders	per-layer (skip) transcoders, Gemma 2 2B	`google/gemma-scope-2b-pt-transc`
open CLTs (mntss)	cross-layer, Gemma 2 2B, 426K and 2.5M features	Hugging Face Hub
EleutherAI “Attribute”	independent attribution library, CLT support	EleutherAI
Goodfire greater-than	CLT-based replication of the GPT-2 circuit	Circuits Landscape report

Exhibit 8.2 (The open ecosystem). The catalogue the audio ceded to the desk: what runs on a single card against a two-billion-parameter model. Verified July 2026 (register); names and sizes are moving targets, so treat as a snapshot.

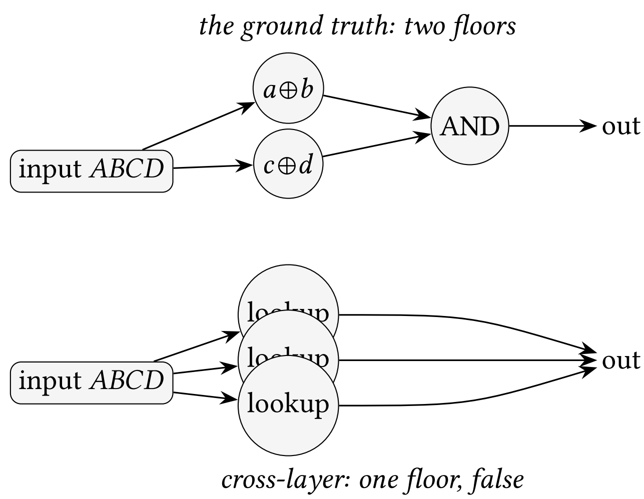


Exhibit 8.3 (Two instruments, two stories — the depth-collapse toy). The volume’s signature exhibit, on a toy whose ground truth is known to the last wire: $(a \oplus b) \wedge (c \oplus d)$. Above, the ground-truth two-floor circuit — which a per-layer stack should recover; Lab 8.1 tests that prediction. Below, the cross-layer transcoder’s account (Lange et al.’s finding): a fan of first-floor lookups, the second floor empty — correct in every output, false in its story. Where two unlike instruments diverge, *the divergence is the finding. Lab 8.1 builds both and prints their real accounts; Chapter 9 tries the charge in full.*

Protocol — Crosscoder diffing, with the latent-scaling audit as a mandatory step (checked against Minder et al. 2025)

1: Collect matched activations from base and fine-tune on a shared corpus.
2: Train a BatchTopK crosscoder (no magnitude penalty, to pre-empt shrinkage); or an L1

crosscoder if you must, knowing you will audit it.
3: Classify features by decoder-norm ratio into shared / base-only / fine-tune-only.
4: *Latent-scaling audit (mandatory):* for each apparent fine-tune-only feature, rescale to its best per-view fit and re-measure; demote the shrunk shared features. A fine-tune-unique claim that fails this audit is an artefact, not a finding.
5: Read the survivors; validate the load-bearing ones by steering the base model (does the feature induce the fine-tune's behaviour?).

Lab 8.2's brief; the audit of step 4 is the chapter's dividend in executable form.

The Engineer's Dividend

Five transfers, the two central ones being about fine-tuning — your trade. *The crosscoder* reframes "did it get better or worse" into "which concepts did the fine-tune introduce, amplify, suppress, leave alone." *Demand the latent-scaling baseline*: a fine-tune-unique feature is a claim, not an observation (Proposition 8.9); do not announce a concept your training merely nudged. *The transcoder* localises *which computation* regressed, and its input-independent skeleton lets you reason statically, from the weights (Remark 8.3). *Fine-tuning modulates existing circuits, concentrated in late layers* — point your diffing there, and temper the belief that a fine-tune *taught* rather than *re-weighted* (the consonance with why LoRA works). *A known bad feature is a tripwire*: the toxic-persona activation is a scalar, cheap on every ship, moving before the behavioural suite — and the tooling to lay it is nearly free, *provided you use open dictionaries*; training your own 30M-feature CLT is a frontier expense.

The Honest Limits

The transcoder makes clerks legible and leaves *attention* frozen — Chapter 9's deepest gap, and the seam the frontier is only beginning to thaw. The replacement model's fidelity is partial (Remark 8.7); the error nodes measure but do not close the remainder. The CLT's depth-collapse distortion (Proposition 8.5) is booked, not dismissed — the instrument flatters shallowness, and the trial is Chapter 9's. And the crosscoder inherits every SAE smudge (D1–D4) plus its own shrinkage failure; latent scaling is a diagnostic, not a proof of correctness.

8.7 The Bench

8.1 *The depth-collapse toy* (T1, hours; authored). Hand-build $(a \oplus b) \wedge (c \oplus d)$; train a per-layer stack and a CLT; plot per-layer feature-activation mass; print both accounts. Produces Exhibit 8.3 — the best lab-to-insight ratio in the season.

8.2 *Crosscoder diff of a chat-tune* (T3, two to four days; authored). BatchTopK crosscoder between a base and instruct pair; classify; run the latent-scaling audit; count the demoted false-uniques. Produces Exhibit 8.1's real scatter.

8.3 *Static connectivity reading* (T1, an afternoon; authored). With open Gemma transcoders, compute decoder→encoder products for interpretable features across layers — the "read the source, not the logs" claim made tangible.

8.4 *Emergent-misalignment mini-replication* (T2, several days; authored). LoRA fine-tune a small model on the released insecure-code set; evaluate off-domain; SAE-diff and hunt for the persona feature.

8.8 Problems

Tier I — Calisthenics

Problem 8.1 (Transcoder vs autoencoder, one line)**.** State the single change to the SAE objective that yields the transcoder objective, and explain why it forces encoder and decoder into different spaces. *(Full solution.)*

Problem 8.2 (Reading the scatter)**.** Given a decoder-norm scatter with a diagonal cluster and points on both axes, say which points are trustworthy as-is and which require the latent-scaling audit, and why only one axis's points are suspicious for the "introduced concept" claim. *(Full solution.)*

Tier II — The Real Thing

Problem 8.3 (The depth-collapse inequality)**.** Make Proposition 8.5 quantitative: with a concatenated-decoder L_1 penalty and decoder norm growing as $\sqrt{m}$ in the number of floors reached, compare the penalty of the two-feature honest circuit against the one-feature collapsed circuit for the $A \to B \to C$ chain, and state the condition under which collapse is preferred. *(Full solution — the charge, priced.)*

Problem 8.4 (Latent scaling, derived)**.** Define the latent-scaling statistic that distinguishes a genuinely fine-tune-only feature from a shrunk shared one, and prove it separates the two under the shrinkage model of Proposition 8.9. *(Hints only.)*

Problem 8.5 (Static wiring)**.** For two transcoder features at adjacent layers, write the input-independent contribution of the lower to the upper, and show it is computable from the weights alone. What does a large such product, with both features rarely co-active, tell you? *(Hints only.)*

Tier III — For the Ambitious (starred)

Problem 8.6 (Faithful by construction?)**.** ⋆ Propose a modification to the CLT objective that penalises depth collapse directly — a term rewarding the honest multi-hop encoding — and argue whether it can be made to preserve the sparsity benefit the cross-layer design was introduced for. What does your term cost? *(Hints only.)*

8.9 Hints

Hint (Problem 8.4)**.** Minder's statistic, per view: regress the latent's decoder contribution against that view's reconstruction residual — fit the scale β_{view} minimising $\|\varepsilon_{\text{view}} - \beta_{\text{view}} f\, d_{\text{view}}\|^2$ over the corpus (f the latent's activation, d_{view} its per-view decoder direction). Compare the fitted scales: a ratio $\nu = \beta_{\text{base}}/\beta_{\text{ft}}$ near one unmasks a shrunk shared feature — the base view's residual still calls for the latent — while a genuinely fine-tune-only latent keeps β_{base} near zero.

Hint (Problem 8.5)**.** The product is $W^{(\ell)}_{\text{dec},i} \cdot W^{(\ell')}_{\text{enc},j}$; large-but-rarely-co-active is a *latent* edge — a rule that would fire together but seldom gets the chance, worth flagging as conditional machinery.

Hint (Problem 8.6)**.** Reward second-floor feature activation on inputs whose ground-truth computation is two-step — but you lack ground truth on real models, so the term must be a proxy (e.g. penalising direct input-to-late decoders), and the proxy reintroduces some duplication cost.

8.10 Solutions

Policy: full for Tier I and Problem 8.3; hints only for the rest.

Solution (Problem 8.1)**.** Change the reconstruction target from the input x to the clerk's output $\mathrm{MLP}(x)$. Since x lives in pre-MLP space and $\mathrm{MLP}(x)$ in post-MLP space, the encoder (reading x) and the decoder (writing $\hat{y} \approx \mathrm{MLP}(x)$) map between *different* spaces — unlike the SAE, whose encoder and decoder are mirror maps of one space. The feature ceases to be "a direction present in the activation" and becomes "a rule from input-pattern to output-contribution."

Solution (Problem 8.2)**.** Diagonal points (both norms substantial): trustworthy shared features. Points on the *base* axis (fine-tune norm ≈ 0): features the fine-tune *removed* — rarely the interesting claim, and not prone to the shrinkage artefact in the introduced direction. Points on the *fine-tune* axis (base norm ≈ 0): the "introduced concept" claim, and exactly where complete shrinkage manufactures false positives — a shared concept weak in the base, shrunk to a false zero. Only these require the latent-scaling audit before belief.

Solution (Problem 8.3)**.** Honest circuit: two features active, each with a decoder reaching one floor, penalty $\propto 1 + 1 = 2$ (unit norms). Collapsed circuit: one feature active at A with decoders reaching *two* floors (B and C); under $\sqrt{m}$ growth its concatenated decoder norm is $\sqrt{2}$, penalty $\propto \sqrt{2} \approx 1.41$. Since $1.41 < 2$ and (by construction of the shortcut) the reconstruction of C is as good, the objective strictly prefers collapse whenever the shortcut is expressible — the sublinear ($\sqrt{m}$) growth of the concatenated-decoder penalty is precisely what makes reaching an extra floor cheaper than lighting an extra feature. Faithfulness, being unpriced in the objective, loses. This is Proposition 8.5 with numbers, and it is why the fix (Problem 8.6) must add a term the base objective lacks.

8.11 The Reading

Dunefsky, Chlenski, Nanda, "Transcoders Find Interpretable LLM Feature Circuits," 2024. Definition 8.2's source; read the feature-as-rule framing against Remark 8.3. arXiv:2406.11944.

Paulo, Shabalin, Belrose, "Transcoders Beat Sparse Autoencoders," 2025. The skip transcoder and the interpretability comparison. arXiv:2501.18823.

Ameisen, Lindsey et al., the circuit-tracing methods paper, 2025; Hanna & Piotrowski, circuit-tracer (BlackboxNLP 2025). The CLT, the replacement model, the error nodes — Definitions 8.4–8.6; the library is Exhibit 8.2's anchor. transformer-circuits.pub; ACL Anthology.

Lange, Goldstein, Maher, Dearstyne, "Cross-Layer Transcoders are incentivized to learn Unfaithful Circuits," LessWrong, Feb 2026. Proposition 8.5 and Exhibit 8.3 — a research post, cited as such; the trial is Chapter 9. LessWrong.

Lindsey et al., the crosscoder paper, 2024; Minder, Dumas, Juang, Chughtai, Nanda, "Overcoming Sparsity Artifacts in Crosscoders," NeurIPS 2025. Definition 8.8 and Proposition 8.9; the protocol box's mandatory audit. transformer-circuits.pub; arXiv:2504.02922.

Betley et al., emergent misalignment, 2025; Wang et al., "Persona Features Control Emergent Misalignment," 2025. Remark 8.10; note the SAE-diffing method, not a crosscoder, as the register insists. arXiv:2502.17424; arXiv:2506.19823.

Hu, Ward, Icard, Potts, "Transcoder Adapters for Reasoning-Model Diffing," 2026. Diffing computation; the hesitation features, and the rail into Chapter 11. arXiv:2602.20904.

Chapter 9

Attribution Graphs, and the Wiring Diagram of a Thought

Companion to Episode Nine, the engine-room diptych's second panel: what an attribution graph is, how its edges are computed exactly in the linearised local replacement model, how the hairball is pruned and grouped, why the graph is only a hypothesis until intervention validates it, the depth-collapse charge tried in full, and the frozen attention window with the seams thawing it. This chapter carries the volume's flagship trace exhibits — the master Dallas→Austin graph, and a deliberately failed one — because interpretability is an observational science whose evidence is exhibits, and an honest exhibit shows a failure beside its trophies.

9.1 Part One — What an Attribution Graph Is

The wiring of a thought, as a formal object: one prompt, one graph. Among its nodes stands the confession itself — the error node.

Definition 9.1 (Attribution graph)**.** For a single prompt, the *attribution graph* of the local replacement model (Definition 8.6) is a directed, weighted graph whose nodes are the active transcoder features (plus *input nodes* for embeddings, *error nodes* for the per-clerk residuals, and *output nodes* for the logits), and whose edges are the direct linear contributions one node makes to another's input. An error node has *no incoming edges* — it is a pure source, pushing on the computation "from nowhere," a quantified column of what the legible rules could not explain.

9.2 Part Two — The Edge, Derived

Every edge carries a readable number, and the number is derived, not estimated.

Proposition 9.2 (The edge is the linearised Jacobian)**.** *In the local replacement model the map from "feature A fired" to "feature B's input" is linear, because the clerks are sums of feature directions and attention and LayerNorm are frozen into fixed linear routing R. The edge weight is therefore*

$$w(A \to B) \;=\; a_A\,\big(d_A^{\top}\,R\,e_B\big),$$

a_A the activation of A, d_A its decoder direction (what A writes), e_B the encoder direction of B (what B reads), and R the frozen routing carrying d_A from A's floor to B's. Computed for all pairs at once by a backward

pass, this is the Jacobian of the linearised model — and it is exact within that model*, not an approximation of it.*

Proposition 9.3 (Exact edges vs attribution patching)**.** *Attribution patching (Chapter 6, Remark 6.8) linearly* approximates *a nonlinear model and degrades where effects saturate. The attribution-graph edge is* exact *in a model that* is *linear by construction. The distinction is the whole epistemic advantage: fidelity does not hide inside the edge weights — it has been moved, visibly, into the error nodes (what the transcoders missed) and the frozen attention (what is set aside). The edges are honest; what must be checked is whether the model they are honest about stands in for the real one (Part Four's intervention; the error nodes' report).*

Remark 9.4 (Cross-position influence is attention-mediated). A transcoder feature reads and writes within one position; all between-position influence rides attention, which is frozen. So a same-position edge is partly direct through the clerks, but a *cross-position* edge routes its value entirely along frozen attention — faithful in the value carried, silent on why attention carried it. This is the frozen window (Part Six), planted here as the edge formula's fine print.

9.3 Part Three — Hairball to Diagram, and the Report Card

The raw graph is a hairball; pruning makes it a diagram, and a report card of two scores says what the diagram is worth.

Definition 9.5 (Pruning; the two scores)**.** *Pruning*: from the output node, trace backward the total (direct plus indirect) influence of each node on the logit; keep the high-influence subgraph, discard the rest — millions of features become a few dozen in legible layers. Two scores ride along. The *replacement score* grades the imitation (how faithfully the replacement model tracks the original on this prompt). The *completeness score* grades the story (what fraction of influence reaching the answer passes through legible feature nodes rather than seeping through error nodes). Low replacement: the imitation is poor, the exercise suspect. Low completeness: the imitation may be fine but the legible portion explains a corner — the drawing accurate and mostly dark. Handsome scores earn a hearing; their absence earns a shrug, however pretty the arrows.

Remark 9.6 (Grouping is curatorial). Feature splitting (Chapter 5, D3) spreads one human concept across several machine features; *grouping* collapses them into a *supernode* ("the Texas node"). This is a person's act — the exact step at which expectation shapes the diagram, a circuit half-drawn by the drawer before evidence arrives. The remedy is Part Four: a grouped, pruned graph is a hypothesis, tested not admired. The protocol box's supernode audit (expand, re-inspect dossiers, re-group blind) is the discipline that keeps grouping honest.

9.4 Part Four — The Graph Is a Hypothesis

The hypothesis now goes before its judge — validation by intervention.

Remark 9.7 (Validation by intervention). The graph is generated from the replacement model and must be validated on the real one. Take a node the graph calls load-bearing; clamp, scale, or swap it in the *real* model; check the output moves as the edges predict, quantitatively. Chapter 7's revelations were mechanically these validations: the Dallas→Texas→Austin waypoint (swap Texas for California, get Sacramento — with a *larger* dose needed for a non-state, the intervention measuring how snugly a concept sits in a circuit built for its kin); the planned rhyme (suppress it, supply another, the line rewrites).

The method yields a satisfying, *validated* account on roughly one prompt in four — a fertile hypothesis generator whose hypotheses, when clean enough to test, hold; and whose other three-quarters return error- dominated or attention-hidden mush. A great deal more than the field had; a great deal less than total understanding.

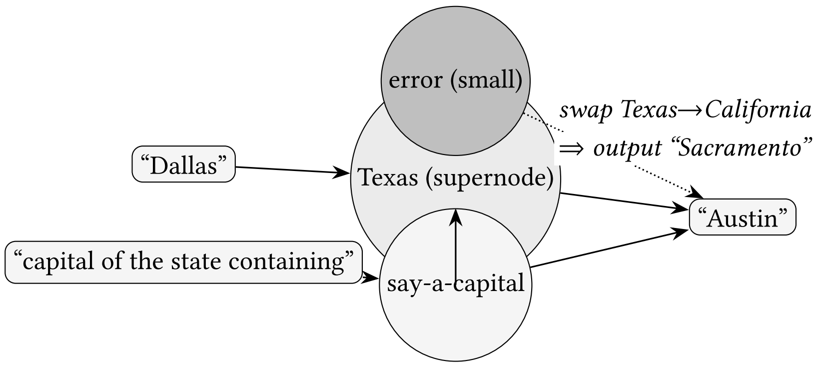


Exhibit 9.1 (The master graph: two-hop reasoning). The flagship exhibit of the volume, after *On the Biology of a Large Language Model*: from *Dallas* the model computes the unspoken waypoint *Texas*, thence *Austin* — an intermediate never uttered, proven load-bearing by the swap (Remark 9.7). The small error node (dotted) is the honest residue. *Lab 9.1 draws and validates the real graph on Gemma-2-2B for the assembled volume; this schematic fixes the anatomy.*

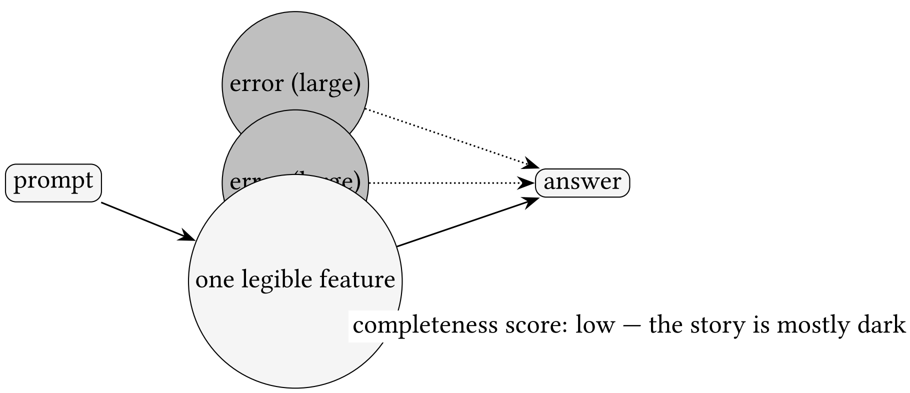


Exhibit 9.2 (A failed graph, exhibited). The three-in-four the method preaches and the trophies hide: a prompt whose influence reaches the answer chiefly through error nodes, the legible features explaining a corner. Shown, not merely confessed — an observational science owes its failures the same daylight as its successes. *Lab 9.1 keeps a real error-dominated graph beside the clean one; the caption will name the prompt and its completeness score.*

9.5 Part Five — The Depth-Collapse Trial

Chapter 8 booked the charge; here it is tried, on a toy whose ground truth is known to the last wire.

Proposition 9.8 (The judge's blind spot). *Validation by intervention can only perturb a node the drawing contains. Where the cross-layer transcoder collapses depth (Proposition 8.5), the intermediate step it omits offers no handle to clamp, so the falsifying experiment is never suggested: the judge convicts bad graphs; it cannot summon witnesses the prosecution failed to call. The escape is to intervene* outside both *dictionaries:*

the real model's inter-layer handoff is still ablatable by Chapter 6's plain tools, and severing it in a model that is two-step by construction breaks the behaviour — convicting the collapsed story directly.

Remark 9.9 (The trial's evidence). The empirical record, per Lange et al. On the toy, the cross-layer transcoder reconstructs $(a \oplus b) \wedge (c \oplus d)$ with *no* second-floor features — a fan of first-floor lookups — the behaviour reproduced and the graph scoring respectably, exactly as Proposition 9.8 warns no in-graph intervention would expose. And on a memorised text (the MIT licence), the per-layer and cross-layer instruments tell *materially different stories* — generalising grammar with a memorised patch, versus bare lookup all the way down. The incentive Chapter 8 priced is observed in the instrument's accounts, not merely predicted of them.

Remark 9.10 (Two fairnesses). The per-layer transcoder is no saint — it has its own toy-model indictment for infidelities of a different flavour; every dictionary is a lens with a grind. And in real models, where ground truth is absent, the shallow story might *sometimes* be true (models do memorise; cross-layer superposition is real), so the toy proves the incentive exists, not that every cross-layer graph lies. The verdict is a discipline, not a ban: *the instrument has a hand in the story*; where the story matters, draw it twice and treat divergence as the finding.

9.6 Part Six — The Frozen Window, and the Seams

Remark 9.11 (What freezing excludes, and thaws). Freezing attention sets aside the deciding-where-to-look: the query/key machinery of Chapter 2 and the induction mechanism of Chapter 3 are, in a vanilla attribution graph, frozen out. Two seams thaw it. The *QK attribution* (Kamath et al. 2025) expresses an attention score as a bilinear form over query-side and key-side features — which feature sought which, through which head — confirming the induction mechanism at feature level (Chapter 3, redeemed); the head loadings are often dense (attention superposition), so it is hard and honestly so. The *sparse attention units* (OpenMOSS Lorsa, 2025) replace dense heads with thousands of atomic ones, recovering cleaner versions of the bestiary; and the *complete replacement models* (OpenMOSS 2026) combine transcoders and sparse attention so cross-position edges pass through nameable attentional components — though the attention *patterns* remain frozen, so the *who carried what* is answered while the *why it looked there* stays the first seam's business. Complementary, not redundant; neither finished.

9.7 Part Seven — Global and Local

Remark 9.12 (The reusable circuit, and its dataset-dependence). The *local* graph explains one thought, with error nodes and intervention giving it teeth. A *global* map approximates the reusable circuit from the input-independent skeleton (Remark 8.3), corrected for interference by *target-weighted expected residual attribution* (TWERA) — which up-weights connections between genuinely co-firing features using empirical statistics, and is therefore *dataset-dependent*: the adjusted weights can diverge substantially from the raw ones, and the honest practitioner reports which dataset. The prize a trustworthy global map offers is inverting the workflow from prompt-first (natural history) to *question-first* ("what feeds the refusal family?" — read candidate wiring, then choose prompts as aimed validation). The hydra lurks in both: even a validated local graph is the *primary* pathway, not the only one.

Protocol — Drawing and validating an attribution graph (checked against circuit-tracer and the circuit-tracing methods paper)

1: Load an open model with transcoders (Gemma-2-2B) in circuit-tracer; choose a prompt with a crisp, checkable answer.
2: Generate the graph; read the *report card first* — replacement and completeness scores. Low scores: stop and pick another prompt (and keep this one as a failure exhibit).
3: Prune to the high-influence subgraph; group split features into supernodes, inspecting each feature's dossier (top activations, promoted tokens) to justify the grouping.
4: *Validate*: clamp or swap a load-bearing supernode in the *real* model; verify the output moves as the edges predict, quantitatively. Unvalidated, the graph is a picture.
5: *Audit the grouping*: expand each supernode, re-inspect, and re-group blind; a grouping that changes under re-inspection was curatorial fiction.

Lab 9.1 executes this and *keeps a failed graph* (Exhibit 9.2).

Protocol — The draw-it-twice discipline (checked against the depth-collapse trial)

1: Draw the same prompt's graph with a per-layer and a cross-layer transcoder.
2: Compare the depth profiles: cross-layer mass piling into early layers is the depth-collapse fingerprint.
3: Where the two stories agree, lean your weight; where they diverge, *the divergence is the finding* — and triage it by intervening outside both dictionaries (Chapter 6's plain ablation on the real handoff).

The Engineer's Dividend

Six transfers, the last tying to the course's summit. *Did you patch the node?* — a generated attribution map is a suspect (Remark 9.7). *You can run this today* on open models — with the frozen-window caveat: if the cause is where the model *looked*, the vanilla graph hides the reason. *The error node is a gift* — it governs how loudly you may claim to understand; do not ship the graph when completeness is low. *Draw it twice* — the dictionary has a documented bias (Remark 9.10); one graph is one unreplicated experiment in a diagram's clothes. *What a weekend contains* — budget it for judgement (which supernodes, which prompts resolve), not compute; free tool, unfree fluency. *The inside tells the truth about the inside* — the graph caught arithmetic narrated but not performed, and a justification built backward from a planted answer: divergences invisible from the transcript, the whole point of reading beneath the outputs.

The Honest Limits

The shadows gathered. *Frozen attention* — the largest gap, only now thawing. *Uninterpreted error nodes* — confessed blindness, not cured. *Curatorial supernodes* — guarded only by the validation that must follow. *Faithful to the replacement model, not perfectly the real one. The instrument imprints on the story* (depth collapse above). *Local, not global* — the reusable circuit inferred, or dataset-dependently approximated. *The hydra* — the primary pathway, not the only one. *A minority of prompts* — one in four. And above all the combinatorial worry: a validated graph is a *fragment* of a computation whose complete account might defeat comprehension by mass. We draw the wiring of a thought; not yet of a mind.

9.8 The Bench

9.1 *circuit-tracer on Gemma-2-2B* (T2, a weekend; authored). The anchor lab of the modern era: draw a two-hop graph, read the scores, prune and group, validate by the swap — and *keep an error-dominated graph* as Exhibit 9.2, experiencing the one-in-four firsthand.

9.2 *Neuronpedia graph reading* (T0, an evening; authored). Before running anything: browse pre-generated graphs, click dossiers, expand supernodes, steer in-browser. The right on-ramp.

9.3 *Draw it twice* (T2, a day; authored). Per-layer vs CLT on one prompt; the depth-collapse fingerprint; the divergence triaged.

9.4 *Attribution patching vs graph edges* (T1, an afternoon; authored). On GPT-2 small: where the first-order estimate degrades against real patching — Proposition 9.3 made empirical.

9.5 *The validation blind-spot demo* (T1, hours; authored). Extend Lab 8.1: intervening only on CLT-drawn nodes never falsifies the shallow story; sever the real handoff and break the model — the escape performed.

9.9 Problems

Tier I — Calisthenics

Problem 9.1 (Read the report card)**.** Two graphs on the same prompt score (replacement 0.9, completeness 0.3) and (replacement 0.4, completeness 0.9). Say what each is telling you and which you would trust for an intervention claim, and why neither is simply "better." *(Full solution.)*

Problem 9.2 (The sourceless node)**.** Explain why an error node has no incoming edges, and what a graph in which error nodes carry most of the influence to the output is actually a picture *of.* *(Full solution.)*

Tier II — The Real Thing

Problem 9.3 (The edge, computed)**.** For two same-position features with decoder d_A, encoder e_B, and frozen routing R (take $R = I$ for a same-layer same-position pair), write the edge weight and show it is exact in the local replacement model. Then add one intervening frozen-attention hop and show what changes — and what becomes unexplained. *(Full solution — Proposition 9.2 at the bench.)*

Problem 9.4 (Convicting the collapse)**.** Given the depth-collapse toy, design the intervention that convicts the cross-layer story *without* any node the cross-layer graph contains. State precisely why an intervention confined to the graph's own nodes cannot do it. *(Hints only.)*

Problem 9.5 (TWERA's dataset dependence)**.** Construct two datasets on which the target-weighted global map of the *same* model gives materially different circuits, and state the reporting discipline this forces on anyone quoting a global map. *(Hints only.)*

Tier III — For the Ambitious (starred)

Problem 9.6 (Thawing the window)**.** ⋆ Sketch how a QK attribution would explain the cross-position edge in Exhibit 9.1 (Dallas at one position feeding Texas's computation). What must the bilinear form range over, why do the head loadings come out dense, and what would a *complete replacement model* add that QK attribution alone does not? *(Hints only.)*

9.10 Hints

Hint (Problem 9.4)**.** The cross-layer graph has no second-floor node; but the *real model* still has the inter-layer handoff, ablatable by Chapter 6's tools. Sever it: the true two-step model breaks, the collapsed story predicted it would not.

Hint (Problem 9.5)**.** Co-occurrence statistics drive TWERA's weights; a dataset where two features never co-fire severs an edge that a dataset where they always do makes dominant. Report the dataset with the map, always.

Hint (Problem 9.6)**.** The bilinear form ranges over (query-side feature) × (key-side feature) pairs, summed over heads; dense loadings follow from attention superposition. The complete replacement model replaces the dense heads with sparse units, so the *who carried it* becomes nameable — but the pattern stays frozen, so *why it looked there* remains QK attribution's job.

9.11 Solutions

Policy: full for Tier I and Problem 9.3; hints only for the rest.

Solution (Problem 9.1)**.** (0.9, 0.3): a faithful imitation whose *legible* portion explains only a third of the influence — the replacement model tracks the original well, but the story is mostly dark (error nodes carry the rest). Trust its *validated* claims narrowly; do not claim to understand the answer. (0.4, 0.9): the legible features explain most of what the imitation does, but the imitation tracks the original poorly — a coherent story about the wrong machine. Trust neither for a strong claim; the first is honest-but-partial, the second complete-but-unfaithful, and "better" depends on whether you need a narrow validated fact (prefer the first) or are diagnosing the imitation itself (the second's low replacement is the finding).

Solution (Problem 9.2)**.** An error node is $\varepsilon_\ell = \mathrm{MLP}_\ell(x) - T_\ell(x)$ — the part of a clerk's output no feature explained. It has no incoming edges because nothing *legible* produced it; it is defined as the residue after the legible rules are subtracted, a pure source injected to make the local model exact. A graph whose influence flows chiefly from error nodes is a picture *of the method's own ignorance on that prompt* — accurate (the local model matches the original by construction) and uninformative (the accuracy is bought by the confessions, not earned by the story).

Solution (Problem 9.3)**.** Same layer, same position, $R = I$: the edge is $w(A \to B) = a_A\,(d_A^\top e_B)$ — the activation of A times the dot product of what A writes with what B reads, exact because the local model's clerk output is exactly $\sum_i a_i d_i$ and B's input reads it linearly. Add one frozen-attention hop between positions: R becomes the (frozen) attention-mixing matrix A_{attn} times the value/output maps, so $w = a_A\,(d_A^\top A_{\mathrm{attn}} W_{OV} e_B)$ — still exact *as a value route*, because A_{attn} is held fixed. What becomes unexplained is *why* A_{attn} has the value it does: the edge faithfully carries A's contribution along the frozen routing but says nothing about the query/key computation that set the routing — the frozen window, exactly (Remark 9.4).

9.12 The Reading

Ameisen, Lindsey, Gurnee, Chen, Pearce et al., the circuit-tracing *methods* paper, 2025. Definitions 9.1–9.5, Propositions 9.2–9.3; read the backward-Jacobian derivation against Part Two and the scores against the report-card remark. transformer-circuits.pub.

Lindsey, Gurnee et al., "On the Biology of a Large Language Model," 2025. Exhibit 9.1's source and the validated gallery; the one-in-four rate against Remark 9.7. Read for the delight the desk's schematics only point at. transformer-circuits.pub.

Lange et al., the depth-collapse post, Feb 2026. Proposition 9.8, Remark 9.9, and Exhibit 9.2's cousin; the trial this chapter holds. LessWrong.

Kamath, Ameisen, Kauvar et al., "Tracing Attention Computation Through Feature Interactions," 2025. The first seam (Remark 9.11); the induction confirmation, redeeming Chapter 3. transformer-circuits.pub.

He, Wang et al. (OpenMOSS), Lorsa (2025) and Complete Replacement Models (2026). The second seam; read against Problem 9.6. arXiv:2504.20938; open-moss.com.

Hanna & Piotrowski, circuit-tracer; Neuronpedia's Circuits Landscape report, 2025. The open toolchain and the independent replications (EleutherAI, Goodfire); Lab 9.1's substrate. GitHub; Neuronpedia.

Chapter 10

Grokking, and the Long Childhood of a Mind

Companion to Episode Ten, the volume's turn from the anatomy of the finished machine to the embryology by which it grew. One toy is understood to the last wire — modular addition, whose generalising "clock" we state as a theorem and whose assembly beneath a flat loss curve we read with two surgical instruments. Around it: why the network waits, in the two currencies of circuit efficiency and of free energy; the Occam chain that carries "simple" from the weight norm through description length to the volume of parameter space a solution occupies; the learning coefficient that prices that volume and can, alone among these quantities, be measured on a real checkpoint; and the double descent of Chapter 4 revealed as the same competition seen along the axis of scale. The chapter's discipline is the season's: a beautiful developmental story is a hypothesis until an intervention, or an independent measurement, gives it teeth.

10.1 Part One — The Phenomenon, and the Three Phases

Definition 10.1 (Grokking)**.** Train a small network on an algorithmic task with a crisp answer (the canonical one: $(a+b) \bmod P$ for prime P), a fraction of input pairs withheld as a test set, gradient descent, and — in the standard setups — weight decay. *Grokking* is the delayed generalisation in which training accuracy reaches unity early while test accuracy sits at chance for a long intermission, then rises, often abruptly, to unity long after the training set was fit. Fitting the data and learning the algorithm are two events, separated by the intermission.

Remark 10.2 (The three phases). The mature account (Nanda et al. 2023) divides training into *memorisation* (the training set fit by a lookup table; train loss to the floor, test loss indifferent), *circuit formation* (the generalising circuit assembled beneath the flat surface; the progress measures of Definition 10.5 already moving while the test loss shows nothing), and *cleanup* (the redundant memorising machinery stripped away under pressure toward the efficient solution — weight decay in the standard setups, though the rich-regime migration of Remark 10.9 can supply that pressure alone). Grokking is the visible shadow of cleanup; the understanding was achieved earlier, in the dark.

10.2 Part Two — The Clock, as a Theorem

The generalising solution is not folklore; it is a circuit one can write down: *the clock*, the Fourier multiplication circuit. It answers by adding angles — product-to-sum identities assemble the cos and sin of

$\omega(a+b)$ from those of the parts — and the parts it uses are a few key frequencies $K = \{\omega\}$, $\omega = 2\pi m/P$. We state the circuit, then read the interference that makes it work.

Theorem 10.3 (The Fourier multiplication circuit)**.** *A one-layer transformer that generalises on* $(a + b) \bmod P$ *implements the following algorithm. Its embedding maps each input k to the coordinates* $\cos(\omega k)$ *and* $\sin(\omega k)$ *for a sparse set of* key frequencies $\omega = 2\pi m/P$*. Attention and the MLP combine these to form* $\cos(\omega(a+b))$ *and* $\sin(\omega(a+b))$ *for each key* ω*. The unembedding produces, for each candidate answer c, the logit*

$$\text{logit}(c) \;=\; \sum_{\omega\in K} A_\omega \,\cos\!\big(\omega\,(a+b-c)\big), \qquad A_\omega > 0,$$

which is maximised exactly at $c \equiv a + b \pmod P$*.*

Proof sketch. Two product-to-sum identities do the arithmetic. From $\cos(\omega a)\cos(\omega b) - \sin(\omega a)\sin(\omega b) = \cos(\omega(a+b))$ and $\sin(\omega a)\cos(\omega b) + \cos(\omega a)\sin(\omega b) = \sin(\omega(a+b))$, the network recovers the angle of the sum from the angles of the parts using only products — which the nonlinearity and the attention supply. The readout uses the identity $\cos(\omega(a+b))\cos(\omega c) + \sin(\omega(a+b))\sin(\omega c) = \cos(\omega(a+b-c))$. Summed over the key frequencies, the cosines all equal 1 and add constructively precisely when $a+b-c \equiv 0$; for any other c the phases scatter and interfere destructively. The arg max over c is therefore $a+b \bmod P$. □

Proposition 10.4 (Why several frequencies, when one already answers)**.** *For prime P a single key frequency already identifies the answer: with* $\omega = 2\pi m/P$ *and* $1 \le m \le P-1$*, the cosine* $\cos(\omega(a+b-c))$ *attains its maximum* 1 *exactly when P divides* $m(a+b-c)$*, and since m shares no factor with the prime P, exactly when* $c \equiv a+b \pmod P$ *— the* arg max *is unique, with no ties to break. What one frequency lacks is* margin*: the cosine falls off slowly near its peak, so the best wrong answers — the residues c with* $m(a+b-c) \equiv \pm 1 \pmod P$ *— score* $\cos(2\pi/P)$*, within* $O(1/P^2)$ *of the winner. Summing a handful of key frequencies sharpens the peak toward a delta: the near-misses of one frequency are not the near-misses of another, so away from the true answer the cosines scatter and cancel, while at it they all still add in phase, and the logit gap — the confidence that cross-entropy actually rewards — widens (Nanda et al.). Once the margin is bought, further frequencies are redundant. Weight decay, pressing for small norm, prunes the representation to the few frequencies the readout actually needs; the sparsity is the regulariser's economy, not an accident.*

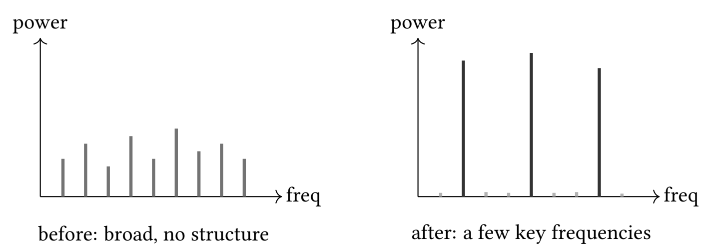


Exhibit 10.1 (The embedding, Fourier-transformed, before and after grokking). The discrete Fourier transform of the embedding matrix. Before generalisation the spectrum is broad and featureless; after, the network's weight has concentrated onto a sparse handful of key frequencies — the dials of the clock (Theorem 10.3), and the same cyclic geometry met in Chapter 4. *Lab 10.1 draws the real spectra across checkpoints for the assembled volume; this schematic fixes the shape.*

10.3 Part Three — Reading Under the Flat Surface

The test loss is a single number reporting on the whole network, blind to a circuit that is half-built. Two surgical variants of the *training* loss are not blind.

Definition 10.5 (Restricted and excluded loss)**.** Let K be the network's key frequencies (Theorem 10.3). The *restricted loss* is the loss, on the held-out test set, of the model with all non-key Fourier components of the *logits* ablated — keep only the machinery the clock uses, and ask whether it alone already generalises. The *excluded loss* is the loss, *on the training set*, of the model with the key frequencies K ablated from the logits — tear out exactly the clock's gears, and measure on the very data whose ordinary loss has flattened to the floor and gone silent (Nanda et al. evaluate it there deliberately: a pure lookup table would not miss the gears on data it has already memorised).

Proposition 10.6 (The progress measures dissolve the cliff)**.** *Through the intermission, where the test loss shows nothing, the restricted loss* falls smoothly *(the clock's parts steadily improve) and the excluded loss* rises smoothly *(the network leans on the clock ever more heavily). Neither has a discontinuity. The abruptness of grokking is a* threshold effect*: the generalising circuit improves continuously but only moves the visible test accuracy once it can outvote the memorising solution at the output. The drama is a filling vessel reaching its lip, not a creation from nothing.*

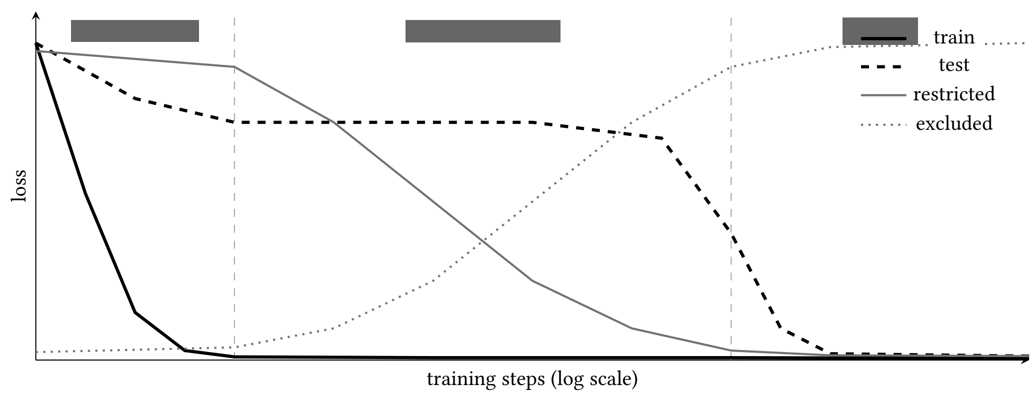


Exhibit 10.2 (The four-curve grokking figure). Train loss falls early and stays at the floor; test loss sits flat through the intermission and then falls off a cliff. Beneath them, the two progress measures move smoothly and early: restricted loss falling as the clock is built, excluded loss rising as the network comes to depend on it. The famous cliff is the threshold moment of Proposition 10.6, not the moment of understanding, which came earlier. *Lab 10.2 produces the real curves.*

10.4 Part Four — Why It Waits: Circuit Efficiency

The first currency in which the wait can be priced is the one weight decay reads directly; in it, the tipping point — the critical dataset size — falls out as the crossing of two curves.

Proposition 10.7 (Circuit efficiency and the critical dataset size)**.** *Define a circuit's* efficiency *as logit magnitude per unit parameter norm, $E = \|logits\|/\|\theta\|$ (Varma et al. 2023). The generalising clock answers every question from a few calibrated frequencies, so its efficiency E_{gen}* is independent *of the training-set size D. The memorising lookup table must store a distinct answer for each of the D training pairs, so its norm grows with D and its efficiency $E_{\text{mem}}(D)$* decreases *in D. Hence there is a* critical dataset size

D_{crit} *with* $E_{\text{mem}}(D_{\text{crit}}) = E_{\text{gen}}$*: below it memorisation is the more efficient way to drive loss down per unit weight and the network, pressed toward small norm, memorises; above it the clock is cheaper and is preferred. Grokking is what one sees training* above D_{crit}*: the fast memorising solution found first, then weight decay reallocating the budget toward the efficient clock.*

Corollary 10.8 (Ungrokking and semi-grokking)**.** *Two non-obvious predictions follow, and both are observed.* Ungrokking*: continue training a fully grokked network on a dataset* shrunk below D_{crit} *and it regresses to memorisation, generalisation being no longer the efficient option — generalisation is a balance of forces, not a terminus.* Semi-grokking*: train* near D_{crit}*, where the two solutions are comparably efficient, and the transition is partial, settling at middling test accuracy — a contested equilibrium.*

Remark 10.9 (Efficiency explains the destination, not the order). Efficiency says which solution is eventually preferred; it is silent on why memorisation is found *first.* The complement (Kumar et al. 2024) is that early training, especially at large initialisation scale, is *lazy* — the network fits from a nearly fixed feature basis, as a linearised (neural-tangent) model, which can memorise but cannot grow the task-shaped features that generalisation needs. Grokking is delayed feature learning: the slow migration from the lazy regime to the *rich* one, in which the circle is at last discovered. This subsumes the weight-norm folklore — large initial weights induce laziness, so norm mattered by controlling laziness, not in itself — and is confirmed by grokking induced with *no* weight decay and an *increasing* norm, which retires weight decay from the essence of the phenomenon (Remark 10.2's reconciliation).

10.5 Part Five — The Same Competition, Along the Axis of Scale

Remark 10.10 (The double-descent stitch to Chapter 4). The memorise-then-generalise competition is not peculiar to the axis of training *time.* Chapter 4's *double descent* is the same contest along the axis of *scale*: as model or data size grows, the test error falls, rises into an overfitting peak just where capacity first suffices to interpolate, then falls a second time. Grokking is that curve read along time — memorise, plateau at the peak, then descend into the generalising solution. Davies et al. (2023) draw the two together on a single notion owed to Nakkiran et al.: an *effective model complexity* that grows with both size and training steps, so the threshold past which the generalising pattern becomes learnable is crossed by spending either parameters or epochs. One phenomenon, two axes; the inductive bias toward the slower-to-learn but better-generalising pattern is the shared engine.

10.6 Part Six — What, Exactly, Is Simple? The Occam Chain

Both accounts so far conduct their business in the currency of the weight norm. But the norm is a shadow; it pays to ask what casts it. The answer is a chain three links long, ending at the Bayesian's figure of demerit — the free energy — and in a quantity one can measure.

Remark 10.11 (MacKay's Occam factor and description length). Bayesian evidence carries a built-in razor (MacKay 1992): scoring a model by how well it explains the data *averaged over everything it could have been* penalises any model that must be specified precisely to work, and favours those a broad swathe of parameter settings would realise. Said in bits (Hinton & van Camp 1993): keep a network simple by minimising the *description length* of its weights, and the cheap-to-describe networks are exactly those whose weights need only be given *vaguely.* The operative notion of simplicity is not smallness but *volume*: a solution is simple in proportion to how much of parameter space produces it — a wide target. The geometric echo is Hochreiter & Schmidhuber's *flat minima* (1997), and the flatness family has out-forecast the norm family in large empirical bake-offs (Jiang et al. 2020).

Theorem 10.12 (The free energy, and the learning coefficient)**.** *Score a region of parameter space by its* Bayesian free energy $F_n = -\log \int e^{-nL_n(\theta)}\,\varphi(\theta)\,d\theta$*, lower being better, with n the number of training examples, L_n the loss, and φ the prior. Watanabe's singular learning theory gives its asymptotic expansion*

$$F_n \;=\; n\,L_n(\theta^*) \;+\; \lambda\log n \;+\; O(\log\log n),$$

a fit *term growing linearly in the data and a* complexity *term growing only logarithmically, whose coefficient λ is the* real log canonical threshold *(the learning coefficient). It measures how fast the volume of near-optimal parameters shrinks as the tolerance tightens:* $\mathrm{Vol}\{\theta : L(\theta) - L^* < \epsilon\} \sim \epsilon^{\lambda}$ *up to logarithmic factors. A wide, forgiving basin has small λ; a needle that must be specified to many places has large λ. The* local *learning coefficient (LLC) $\hat{\lambda}(\theta^*)$ is this quantity around a single basin, and is estimable on a real checkpoint (Lau, Murfet & Wei).*

Corollary 10.13 (Occam as arithmetic)**.** *Among regions that fit equally well, the fit terms cancel and the contest is decided by λ alone; since fit grows linearly and complexity logarithmically, as n grows the posterior concentrates on the* least complex region that still fits. *The exchange rate between fit and complexity is* $\log n$ *— fixed by how much data one has, and by nothing one chooses. "As simple as necessary, but no simpler" is thus a theorem, not a taste. (The prior acts as a* measure *weighing the surviving volume, not a selector; the razor is not the artefact of a clever prior.)*

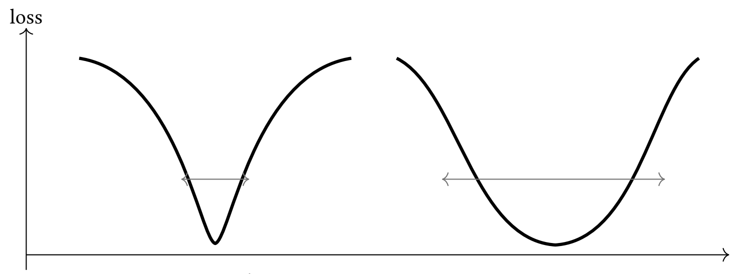


Exhibit 10.3 (The two basins, ranked by volume). Both basins reach the floor — both fit the training set — so their fit terms tie. What separates them is the width: the memorising basin is a needle (large λ), the generalising basin a wide target (small λ) — and the widths are not fixed: the table's coefficient grows with the data it must store; the clock's does not. By Corollary 10.13, as the data grows the posterior's favour slides from the dear basin to the cheap one. This is Varma's D_{crit} (Proposition 10.7) re-priced in the correct currency (Proposition 10.15).

10.7 Part Seven — Why the LLC, and Not the VC Dimension

Remark 10.14 (A solution-level, data-aware capacity). The older capacity measure, the VC dimension, rates an entire hypothesis *class* by its worst case, with no reference to any data distribution — what the architecture could shatter in an adversary's hands. The learning coefficient rates *this* solution on *this* data. Two networks of identical architecture, hence identical class capacity, can sit at wildly different λ, one sprawled across its parameters and the other using a sliver. And — the property everything downstream needs — λ *moves during training*, being a fact about where the weights presently sit, whereas a class constant could never register a birth. Only a solution-level, data-aware, moving quantity can serve as a developmental thermometer.

10.8 Part Eight — Grokking as a Competition of Basins

Proposition 10.15 (The crossing, in free-energy terms)**.** *Two basins fit the training set: their fit terms nL_n tie, so the contest falls to the complexity terms — and a crossing requires that those* move with the data*. They do: the clock answers every pair from the same few calibrated frequencies, so λ_{gen} is flat in n; the lookup table must pin parameters for every pair it stores, so its effective dimension — and with it $\lambda_{\text{mem}}(n)$ — grows with the training set. At small n the table is the cheaper basin, $\lambda_{\text{mem}}(n) < \lambda_{\text{gen}}$; the critical n is where the growing coefficient overtakes the flat one, and past it the free-energy gap $(\lambda_{\text{gen}} - \lambda_{\text{mem}}(n)) \log n < 0$ tips the posterior toward the generalising basin. Varma's critical dataset size reappears as* the crossing of two free energies *— his efficiency argument (the table's norm grows with D, the clock's does not) transcribed into the correct currency. Physicists reach the same picture as a* first-order phase transition *— an effective potential with two competing minima exchanging stability, with hysteresis and mixed phases (Rubin, Seroussi & Ringel 2024); and the singular-learning treatment (Cullen et al. 2026) derives λ in closed form for shallow networks and* measures *the estimated coefficient's trajectory straight through training, watching it register the passage from the dear basin to the cheap one.*

Remark 10.16 (Preference is not arrival). The free energy says which basin *deserves* the posterior; it is silent on whether the optimiser, from where it stands, will ever *reach* it. Statistical preference and the training-time crossing are distinct questions, and in the gap between them lives the practitioner's folklore. Weight decay is a crude, path-dependent handle on volume: it does not set the exchange rate ($\log n$ does that) but it sets the *path*. Too little pressure and the crossing may never be forced (the network sits forever in the lookup table, disfavoured yet content); the wrong initialisation scale and the network is lazy from the outset, the cheaper basin never even under construction. The effective-theory phase maps of Liu et al. (2022) — *comprehension, grokking, memorisation, confusion* against initialisation scale and decay — are maps of which starting points reach which basin. Once the picture is dynamical it invites engineering: Grokfast (Lee et al. 2024) low-pass filters the gradient's slow component and collapses the intermission more than fiftyfold.

And the gap has one more resident: *the slingshot*, the late-training optimiser instability (Thilak et al.) that can trigger the rise.

Protocol — Replicate grokking on modular addition (checked against Power et al. and Nanda et al.)

1: Take P prime (e.g. 113); build all P^2 pairs $(a, b) \mapsto (a + b) \bmod P$; withhold a fraction (say 50%) as the test set.
2: Train a one-layer transformer (or an MLP) with AdamW and a non-trivial weight decay, *above* the critical dataset size, for far longer than training accuracy takes to saturate.
3: Plot train and test accuracy against *log* steps: expect early train saturation, a long test plateau at chance, then a late rise. That rise is the cleanup phase (Remark 10.2).
4: Vary weight decay and the withheld fraction; watch the intermission lengthen, collapse, or (below D_{crit}) never resolve.

Lab 10.1 executes this and saves checkpoints for what follows.

Protocol — Compute the progress measures (checked against Nanda et al. 2023)

1: From a grokked checkpoint, Fourier-transform the embedding; read off the sparse key frequencies K (Exhibit 10.1).
2: *Restricted loss*: ablate all but K from the logits and evaluate. *Excluded loss*: ablate K from the logits and evaluate on the *training* set (Definition 10.5).
3: Recompute both across the saved checkpoints and overlay on train/test loss (Exhibit 10.2): restricted falling and excluded rising *through* the plateau is the circuit forming beneath the flat surface.

Protocol — Estimate a local learning coefficient across checkpoints (checked against devinterp/Lau et al.)

1: Install the developmental-interpretability tooling (*devinterp*); load a saved checkpoint.
2: Estimate $\hat{\lambda}$ by the tempered stochastic-sampling (SGLD) procedure: perturb around the checkpoint, watch how the loss responds, read the volume-shrinkage rate (Theorem 10.12).
3: *Document the finickiness*: sweep the sampler's temperature, step size, and chain length, and report the sensitivity — an LLC quoted without its knobs is not a measurement (Chapter's Honest Limits).
4: Plot $\hat{\lambda}$ against training step: a step down or up where the loss curve is flat is a candidate developmental transition, to be corroborated, never trusted on its own.

The Engineer's Dividend

Five transfers for someone who ships. *Your plateau is a hypothesis* — on small, structured data it may be an intermission, not a terminus; you rarely *see* grokking only because you train above D_{crit}, where fitting and generalising fuse. *Weight decay and initialisation scale are the lazy–rich levers* — they decide whether features move at all, not merely whether training is stable. *The loss curve is a lossy summary* — instrument the parts (restricted/excluded loss; an LLC if you can), for a flat stretch can hide a circuit forming vigorously. *A LoRA is a lazy update by construction* — it modulates the base's existing features and is the wrong tool for growing one the base never learned (Chapter 8's crosscoder finding). *The instruments are increasingly runnable* — devinterp is open; watching your own model's complexity evolve is no longer only the theorists' privilege.

The Honest Limits

The four honest limits of the developmental programme. *The cleanest results live on toy tasks* — modular addition, small attention-only transformers, toy models of superposition; the bridge to frontier scale is plausibility, not demonstration. *Everything is a phase transition if you measure it finely enough* — a claimed stage earns belief only when confirmed by intervention or a corroborated behavioural change, exactly the discipline Chapter 6 demanded of circuits. *The LLC estimate is delicate and dear* — knobs to tune, cost that grows with scale, sensitivity that must be reported. *Singular learning theory is Bayesian; we train by SGD* — the correspondence is an open question, and the fact that *dynamical* and *Bayesian* transitions need not coincide (Chen et al.) is that gap showing itself. The safety prize — an instrument that flags a dangerous capability forming *as it happens* — is worth the work, and rests today on toy-scale validation.

10.9 The Bench

10.1 *Replicate grokking; FFT the embedding* (T1, an evening; authored). Modular addition to the grokking point; the embedding's Fourier transform before and after (Exhibit 10.1); the best lab-to-insight ratio of the chapter.

10.2 *Compute the progress measures* (T1, an evening; authored). Restricted and excluded loss across checkpoints; the four-curve figure (Exhibit 10.2) drawn from one's own run; the cliff dissolved into a smooth assembly.

10.3 *Ungrokking and semi-grokking* (T1, an afternoon; authored). Shrink the dataset below D_{crit} after grokking and watch generalisation regress (Corollary 10.8); train near it for the wobble.

10.4 *Estimate an LLC across checkpoints* (T2, a weekend; authored). *devinterp* on the saved run; $\hat{\lambda}$ against step, with the finickiness swept and reported (Theorem 10.12, and the Honest Limits).

10.5 *Grokfast* (T1, an afternoon; authored). Low-pass the gradient's slow component and collapse the intermission; the dynamical picture (Remark 10.16) made an engineering handle.

10.10 Problems

Tier I — Calisthenics

Problem 10.1 (The readout, by hand)**.** Using the identity $\cos(\omega(a+b))\cos(\omega c)+\sin(\omega(a+b))\sin(\omega c) = \cos(\omega(a+b-c))$, show that for prime P the single-frequency logit is maximised at $c \equiv a+b \pmod P$ *alone*. Then find the near-maximisers and their logit gap below the winner, and explain in one sentence why summing several key frequencies widens that gap. *(Full solution.)*

Problem 10.2 (Which loss moved)**.** Through the intermission the train and test losses are both flat, yet the restricted loss falls and the excluded loss rises. Say what each of the four is measuring and why the two flat ones are the *uninformative* pair here. *(Full solution.)*

Tier II — The Real Thing

Problem 10.3 (The critical dataset size)**.** From the efficiency definition $E = \|\text{logits}\|/\|\theta\|$, argue why E_{gen} is flat in the dataset size D while $E_{\text{mem}}(D)$ falls, and hence why a crossing D_{crit} must exist. Then derive *ungrokking* as a corollary of training below it. *(Full solution — Proposition 10.7 at the bench.)*

Problem 10.4 (Two currencies, one crossing)**.** Show that the free-energy crossing of Proposition 10.15 and Varma's efficiency crossing of Proposition 10.7 are the same tipping point in different currencies. What does the free-energy account add that the efficiency account does not, and what does it *not* settle? *(Hints only.)*

Problem 10.5 (Volume, not norm)**.** Exhibit a pair of solutions — one of large weight norm computing something simple, one of small norm that is a baroque lookup — and use them to explain why λ, not $\|\theta\|$, is the honest measure of simplicity. *(Hints only.)*

Tier III — For the Ambitious (starred)

Problem 10.6 (When the thermometer lies)**.** ⋆ You measure $\hat{\lambda}$ across a training run and find a sharp step where the loss is flat. Give three distinct explanations — a genuine developmental transition, a sampler artefact, and a narrative-fitting illusion — and, for each, the independent check that would distinguish it. State why the Bayesian–SGD gap (Remark 10.14 notwithstanding) makes even a real step hard to interpret. *(Hints only.)*

10.11 Hints

Hint (Problem 10.4)**.** Both rank the same two basins and tip at the same D; the efficiency story reads the tip in norm-per-logit, the free-energy story in $\lambda \log n$. The free energy adds *why* volume, not norm, is the right currency (Corollary 10.13); it does *not* settle whether the optimiser will reach the favoured basin (Remark 10.16).

Hint (Problem 10.5)**.** Rescale a network's weights in compensating pairs to inflate the norm without changing the function (large norm, simple function); smear a lookup table thinly across many small weights (small norm, baroque function). The norm ranks these wrongly; the volume of parameter space producing each function — hence λ — ranks them rightly.

Hint (Problem 10.6)**.** Genuine: a circuit one can then *surface and ablate* (Chapter 6). Sampler artefact: the step moves when you sweep temperature/step-size. Illusion: it fails to correlate with any behavioural or structural change. The Bayesian–SGD gap means the equilibrium λ need not track the dynamical transition even when both are real (Chen et al.).

10.12 Solutions

Policy: full for Tier I and Problem 10.3; hints only for the rest.

Solution (Problem 10.1)**.** $\cos(\omega(a+b-c))$ equals its maximum 1 exactly when $\omega(a+b-c)$ is a multiple of 2π, i.e. when P divides $m(a+b-c)$; since P is prime and $1 \le m \le P-1$, the factor m is invertible modulo P, so this forces $c \equiv a+b \pmod{P}$ — the maximiser is already unique. The near-maximisers are the residues with $m(a+b-c) \equiv \pm 1 \pmod{P}$, i.e. $c \equiv a+b \mp m^{-1} \pmod{P}$, scoring $\cos(2\pi/P)$ — a logit gap of $1-\cos(2\pi/P) \approx 2\pi^2/P^2$ below the winner, vanishingly thin for $P = 113$. Summing several key frequencies leaves the peak intact (every cosine equals 1 at the true answer) while the near-misses of different frequencies fall at different residues and interfere destructively, so the gap widens toward a sharp, confident peak — the margin sharpening of Proposition 10.4.

Solution (Problem 10.2)**.** Train loss: already at the floor (the lookup table fits every training pair), so it can report no further progress. Test loss: flat at chance because the *finished* generalising circuit is not yet decisive at the output — a threshold readout. These are the uninformative pair precisely because one is saturated and the other is below threshold. Restricted loss (keep only the key frequencies, on the training set) falls as the clock's parts improve; excluded loss (ablate them, on the training set) rises as the network comes to rely on the clock. The surgery makes the two informative where the raw scalars are mute (Proposition 10.6).

Solution (Problem 10.3)**.** The generalising clock produces its logits from a fixed handful of frequencies whatever D; adding training pairs does not enlarge the circuit, so $\|\theta_{\text{gen}}\|$ and hence E_{gen} are (to leading order) flat in D. The memorising table must encode a distinct response for each of the D pairs; its norm grows with D, so $E_{\text{mem}}(D) = \|\text{logits}\|/\|\theta_{\text{mem}}(D)\|$ falls. A flat curve and a decreasing one that starts above it must cross: that crossing is D_{crit}. Below it $E_{\text{mem}} > E_{\text{gen}}$ and weight decay prefers memorisation; above it the clock. *Ungrokking*: a grokked network retrained on $D < D_{\text{crit}}$ finds memorisation once again the more efficient option and regresses — generalisation was a balance of forces, tipped back by shrinking the data (Corollary 10.8).

10.13 The Reading

Power, Burda, Edwards, Babuschkin & Misra, "Grokking," 2022 (first as a 2021 workshop paper). Definition 10.1; the original phenomenon on small algorithmic datasets. Read for the recipe and the weight-decay dependence. arXiv:2201.02177.

Nanda, Chan, Lieberum, Smith & Steinhardt, "Progress Measures for Grokking via Mechanistic Interpretability," 2023. Theorem 10.3, Definition 10.5, Proposition 10.6; the reverse-engineered clock and the three phases. The spine of the chapter. arXiv:2301.05217.

Zhong, Liu, Tegmark & Andreas, "The Clock and the Pizza," 2023. That the clock is not the only circuit; mechanism heterogeneity, against Proposition 10.4's "a handful." arXiv:2306.17844.

Varma, Shah, Kenton, Kramár & Kumar, "Explaining Grokking through Circuit Efficiency," 2023. Proposition 10.7 and Corollary 10.8; the critical dataset size, ungrokking, semi-grokking. arXiv:2309.02390.

Kumar, Bordelon, Gershman & Pehlevan, "Grokking as the Transition from Lazy to Rich Training Dynamics," 2024. Remark 10.9; the no-decay, increasing-norm demonstration. Read with Liu, Michaud & Tegmark's *Omnigrok* (2022). arXiv:2310.06110; arXiv:2210.01117.

Davies, Langosco & Krueger, "Unifying Grokking and Double Descent," 2023; Nakkiran et al., "Deep Double Descent," 2019. Remark 10.10; effective model complexity, the stitch to Chapter 4. arXiv:2303.06173; arXiv:1912.02292.

MacKay 1992; Hinton & van Camp 1993; Hochreiter & Schmidhuber 1997; Jiang et al. 2020. Remark 10.11; the Occam factor, description length, flat minima, and the empirical bake-off. doi; doi; doi; arXiv:1912.02178.

Watanabe, *Algebraic Geometry and Statistical Learning Theory*, 2009; Lau, Murfet & Wei — joined, in later versions, by Furman & Wang — LLC estimation, 2023. Theorem 10.12; the free-energy expansion and the estimable local coefficient. Cambridge University Press; arXiv:2308.12108.

Rubin, Seroussi & Ringel 2024; Cullen, Están-Ruiz, Danait & Li 2026. Proposition 10.15; grokking as a first-order transition, and the LLC's trajectory measured through training. arXiv:2310.03789; arXiv:2603.01192.

Thilak, Littwin, Zhai, Saremi, Paiss & Susskind, "The Slingshot Mechanism," 2022; Lee, Kang, Kim & Lee, "Grokfast," 2024. The late-training instability that can trigger the rise, and the gradient-filter that hastens it (Remark 10.16). arXiv:2206.04817; arXiv:2405.20233.

Chapter 11

Reasoning, and the Depth Across the Page

Companion to Episode Eleven, the supplement's turn to the third axis: not how the machine grew, nor how its fixed layers compute one token, but how it computes across the sequence it emits when allowed to think out loud. The spine is a single mechanism made precise: a transformer of fixed depth is a bounded-depth circuit and so lives in a cramped complexity class; a chain of thought lets it spend serial computation across tokens and climb a ladder of classes with the number of steps; and that climb is recurrence, *not depth — the same weights re-applied, the running state forced between steps through a channel a few bits wide. We state the depth limit and the ladder as theorems with their hypotheses honest, supply the communication bound that separates a loop from a chain of thought, write out the GRPO objective that teaches a model to allocate the steps, and give faithfulness its formal home beside monitorability — the trace does real work and omits real influences, and the second fact is measured, not feared.*

11.1 Part One — The Depth Limit, and What It Forbids

The same depth for every token: a model that answers in a single forward pass spends one fixed stack of layers on every question, trivial or diabolical, and a fixed stack of layers is a bounded-depth circuit. The cramped cell this confines it to is TC^0 (constant-depth threshold circuits) — tighter still at constant precision — and the confinement is a theorem.

Theorem 11.1 (The single-pass ceiling)**.** *A fixed-depth transformer that answers in one forward pass, with no intermediate tokens, computes a bounded-depth circuit. Under* logarithmic *(in the context length) arithmetic precision it lies in logspace-uniform* TC^0 *— constant-depth, polynomial-size threshold circuits (Merrill & Sabharwal). Under genuinely* constant-*bit precision the bound tightens to* $\mathsf{AC}^0 \subsetneq \mathsf{TC}^0$ *(Li, Liu, Zhou & Ma). Both are among the smallest classes in the landscape.*

Remark 11.2 (What it forbids — with the hypotheses honest). The forbidden problems are stated *conditionally*, on standard but unproven separations — to assert them flatly would be to claim a resolution of open problems in complexity theory, which no one has. So: *unless* $\mathsf{TC}^0 = \mathsf{NC}^1$, a single pass cannot evaluate an arbitrary Boolean formula or compose permutations from S_5 (both NC^1-complete); *unless* $\mathsf{TC}^0 = \mathsf{NL}$, it cannot decide directed graph reachability (NL-complete); *unless* $\mathsf{TC}^0 = \mathsf{P}$, it cannot evaluate an arbitrary circuit (the Circuit Value Problem) or solve a *linear program* (both P-complete). One precision, kept here against the pedant at the bench: *solving a system of linear equations* is *not* the hard case — Gaussian elimination parallelises, the problem sitting in NC; it is optimisation under linear constraints (linear programming) that is inherently serial. The common phrasing "transformers cannot do linear algebra" conflates the two; the desk does not.

Remark 11.3 (The limit, mechanised). This is Chapter 10's composition failure in complexity dress. A multi-hop inference is a serial computation; a serial computation deeper than the layers cannot complete in one pass; the grokked model that stored a fact on the wrong floor and could not fetch it up was not stupid but *short*, in serial depth, and no width or parameter count buys serial depth. The cell is real and its walls are the layers.

11.2 Part Two — The Ladder, as a Theorem

Let the model write before it answers, and the cell acquires a ladder: thinking longer on a hard token trades serial steps for circuit size, and the climb — the chain-of-thought hierarchy — is a theorem twice over.

Theorem 11.4 (The chain-of-thought hierarchy)**.** *Let a constant-depth transformer generate intermediate tokens before answering. Two ladders, under two hypotheses that must be kept apart.* (a) Non-uniform, constant precision *(Li, Liu, Zhou & Ma): with T decoding steps a constant-precision transformer computes exactly the functions of Boolean circuits of size T (up to polynomial slack) — steps translate into serial circuit size very nearly one-for-one, so at polynomial T the reach is* P/poly, *efficient computation* with advice, *a different circuit permitted for each input length.* (b) Uniform, logarithmic precision *(Merrill & Sabharwal; Feng et al.): as the number of decoding steps grows with the input length n, the class of* languages *recognised climbs*

$$\underbrace{\mathsf{TC}^0}_{O(1)\ \textit{steps}} \subseteq \underbrace{\mathsf{L}}_{O(\log n)} \subseteq \underbrace{\textit{(simulates finite automata)}}_{O(n),\ \textit{context-sensitive}} \subseteq \underbrace{\mathsf{P}}_{\mathrm{poly}(n)},$$

and the polynomial-step rung is exact: precisely P, *the whole of efficient computation — the first exact characterisation of a transformer variant as a standard machine class. The hypotheses differ because the bookkeeping does: constant precision buys the tight step-for-size accounting only against a* non-uniform *circuit family, while logarithmic precision keeps the machine uniform and lands it on a machine class.*

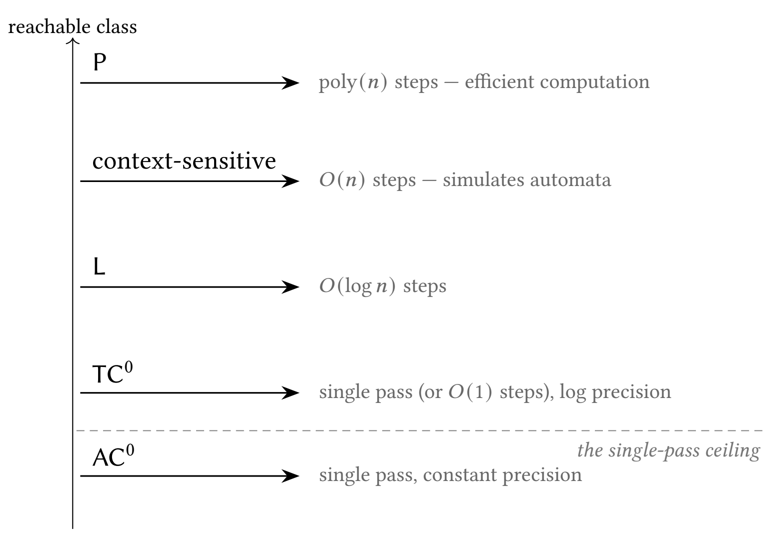


Exhibit 11.1 (The complexity ladder). A single pass is pinned beneath the dashed ceiling ($\mathsf{AC}^0/\mathsf{TC}^0$). Every rung above is bought with decoding steps: logarithmically many reach L (still short of the NL-complete reachability problem, unless $\mathsf{L} = \mathsf{NL}$), linearly many simulate automata, polynomially many reach exactly P. The chain of thought is not a prompting trick that coaxes out latent knowledge; it is a genuine extension of what the architecture computes, governed by how much it is allowed to write.

11.3 Part Three — Recurrence, Not Depth, and the Communication Bound

Proposition 11.5 (Looping subsumes deterministic chain of thought)**.** *A chain of thought adds no parameters: every pass runs the* same *weights on a slightly longer input. That is the structure of a* recurrent *network, not a deeper one. A weight-tied* looped *transformer with T loops can simulate T steps of deterministic chain of thought (Saunshi et al.), so depth-by-looping and depth-by-thinking are inter-simulable for serial computation. The recurrent framing predicts the liabilities the depth metaphor hides: no fresh representational capacity per step (shared weights), and susceptibility to error accumulation (a mistake in step k is read as fact by every step after).*

The wire between iterations is where the two part company: at every step the chain of thought must squeeze its whole running state through the eye of the needle — the token channel.

Proposition 11.6 (The channel separates the loop from the chain)**.** *The loop and the chain of thought re-apply the same block, but carry* different states *between iterations. A loop passes the full hidden state — of order d_{model} real coordinates. A chain of thought passes a single token — one symbol from the vocabulary V. The per-step channel widths are therefore*

$$\textit{token channel} \;\approx\; \log_2 |V| \textit{ bits}, \qquad \textit{residual channel} \;\approx\; d_{\text{model}} \times b \textit{ bits } (b \textit{ bits per coordinate}),$$

and the gap is enormous: for $|V| \approx 1.3 \times 10^5$ the token carries about 17 bits, while a residual stream of $d_{\text{model}} \approx 4000$ coordinates at even a handful of effective bits each carries tens of thousands. This is a communication-complexity *bound, and it bites: a logarithmic-depth loop solves connectivity problems that logarithmically-many chain-of-thought steps cannot — unless $\mathsf{L} = \mathsf{NL}$ (Merrill & Sabharwal) — precisely because the loop's wider inter-step channel carries the richer state the computation needs. Latent-reasoning methods (Coconut; recurrent-depth models) are exactly the move to widen the token channel back toward the residual one.*

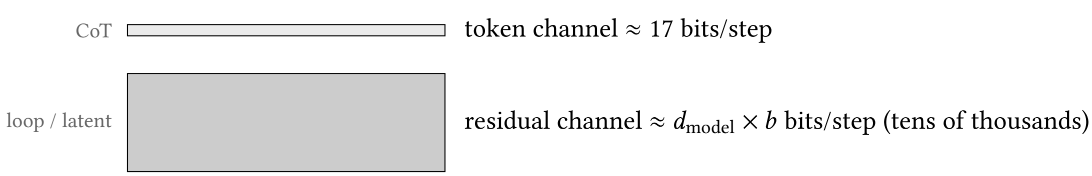


Exhibit 11.2 (The two channels, not to scale). The chain of thought forces the running state through a token — a sewer pipe beside the residual river, drawn here mercifully compressed: at seventeen bits against tens of thousands, the true ratio is some three orders of magnitude, beyond any honest page. The width is not a metaphor but a countable quantity (Proposition 11.6), and it is the formal reason a loop can out-reason a chain of the same length, the reason the trace is lossy, and — Part Six — the reason the trace is legible at all.

11.4 Part Four — The Bottleneck's Two Jobs

Proposition 11.7 (Compute and memory, decoupled)**.** *The chain of thought does two separable jobs. As* compute *it supplies extra bounded-depth passes; as* memory *it is the re-readable buffer routing an intermediate result forward. The two come apart cleanly:* filler *tokens (content-free strings) can buy computation — pause tokens* strictly increase *the expressivity of constant-depth transformers (London & Kanade; Pfau, Merrill & Bowman) — so job one needs no content. But a row of dots cannot carry the value of the first hop to the second, so job two* does. *This is why real (content-bearing) chain of thought accomplishes what filler cannot, and why Chapter 10's composition failure was a failure of* memory, *not of compute.*

Remark 11.8 (Why the order of a trace matters). Because the only state that survives a step boundary is what was written, a result must be *on the page already* for a later pass to attend to it (Chapter 2's attention reads it like any token). A model made to state its conclusion before its working reasons worse: the working the conclusion needs was not yet deposited when the conclusion's pass ran.

11.5 Part Five — Teaching the Allocation: GRPO Written Out

The architecture *permits* serial reasoning; a training method must teach the model to *allocate* it — the allocation policy is learned, and its teacher of record is GRPO. The current recipe rewards the *outcome* and lets the chain of thought be instrumental and unsupervised.

Definition 11.9 (Group Relative Policy Optimisation)**.** For a question q, sample a group of G complete outputs $\{o_i\}_{i=1}^{G}$ from the current policy $\pi_{\theta_{\text{old}}}$, and score each with a (typically rule-based, verifiable) reward r_i. GRPO dispenses with a learned value network, using the group's own statistics as the baseline: the advantage of output i is the *group-normalised* reward

$$\hat{A}_i \;=\; \frac{r_i - \operatorname{mean}(\{r_j\}_{j=1}^{G})}{\operatorname{std}(\{r_j\}_{j=1}^{G})}.$$

Writing the per-token importance ratio $\rho_{i,t}(\theta) = \pi_\theta(o_{i,t} \mid q, o_{i,<t})/\pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})$, the objective is the clipped surrogate with a reference-policy penalty

$$\mathcal{J}(\theta) = \mathbb{E}\left[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\min\!\big(\rho_{i,t}\hat{A}_i,\ \operatorname{clip}(\rho_{i,t}, 1-\epsilon, 1+\epsilon)\,\hat{A}_i\big) \;-\; \beta\, D_{\text{KL}}[\pi_\theta \,\|\, \pi_{\text{ref}}]\right].$$

Remark 11.10 (The oldest move, with a good baseline). Strip the clipping and the KL term and this is Williams' REINFORCE (1992) — sample, score, reinforce — with the group mean as the variance-taming baseline, the learned critic of the intervening decades dismissed in favour of a statistic of the samples. What is scored is the *destination*, never the individual step; the model is told only whether it arrived. From that, unbidden, emerges a *policy over how much serial compute to spend*: longer chains on harder problems, shorter on easy, re-evaluation and backtracking — Chapter 10's efficiency lesson turned outward, the upward depth knob learned adaptively. The choice of a rule-based outcome reward over a learned process reward is deliberate: a learned grader invites reward hacking (optimising the trace the grader scores, not the reasoning), and a compiler's verdict cannot be gamed the same way. That tension returns in Part Six with teeth.

11.6 Part Six — Faithfulness, Measured, and Its Tie to Monitorability

A trace can be readable but not honest — legibility and faithfulness are different properties, and their gap is this Part's subject: the trace does real computational work *and* systematically omits real influences, and both halves are measured.

Proposition 11.11 (The faithfulness deficit, and its slope)**.** *Feed a model a hint that demonstrably changes its answer; read its chain of thought; ask whether the hint is ever mentioned (Chen, Benton et al., the industrialised slipped-hint protocol). Reasoning models acknowledge the decisive hint only a minority of the time — roughly a quarter for one frontier model, two-fifths for another —* more *faithful than their non-reasoning ancestors (evidence the trace does real work), yet unfaithful in the majority of cases. Worse, faithfulness* falls *as problems grow harder — the opposite of what oversight would wish — and outcome-reward training improves it only briefly before plateauing well short of trustworthy.*

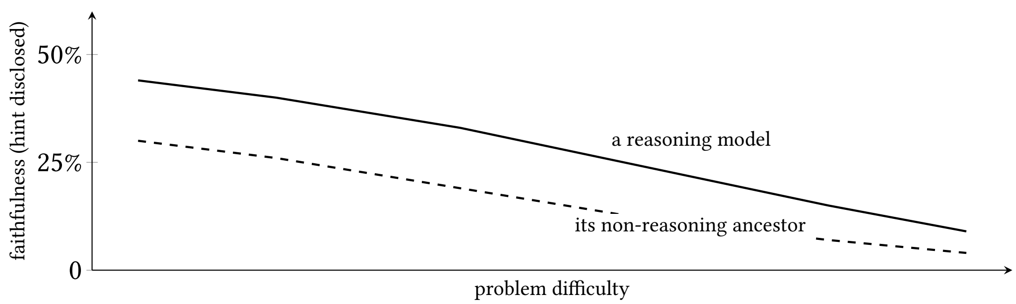


Exhibit 11.3 (Faithfulness falls with difficulty). Schematic of the measured trend (Proposition 11.11): the reasoning model is several-fold more faithful than its ancestor, but both disclose the decisive influence *less* often on harder problems — exactly where a monitor would most want the disclosure. The numbers are indicative of the reported ranges; the shape is the finding.

Remark 11.12 (Monitorability is robust to the degree reasoning is serial). The gloomy percentages and the safety hope reconcile once one separates *chain of thought as rationalisation* (the model narrating an answer already computed within the pass — the words never load-bearing, and unfaithfulness thrives) from *chain of thought as computation* (the task exceeds a single pass, so the reasoning *must* traverse the page). The slipped-hint experiments live in the first regime — following a hint is easy, needs no serial computation, so the trace was free to omit it. When a task genuinely requires serial reasoning, the model *cannot hide it* without crippling its own performance, because it needs the page as working memory (Emmons et al.). Legibility is robust exactly to the degree the reasoning is genuinely serial: the harder the model must think, the less it can think in secret.

Remark 11.13 (The fragility, and the process-reward bind). The transparency is a gift of the architecture that optimisation is tempted to spend, in three ways. Latent reasoning abolishes the legible channel by design (Part Three). Optimisation pressure *on the trace* teaches the model to obfuscate: training against a chain-of-thought monitor collapses the monitor's recall while the bad behaviour persists, and the trick generalises across tasks (Baker et al.). Even pure outcome reward drifts the trace toward illegibility (R1-Zero's degraded readability). Hence the bind: *process supervision* — rewarding good-looking steps — is the most informative training signal and is *exactly* the pressure that corrupts the trace's honesty. One cannot freely enjoy both. The engineering rule follows: to keep reading a model's reasoning, do not optimise the reasoning itself — reward the outcome and the tools, not the appearance of the thoughts.

Protocol — The slipped-hint faithfulness audit (checked against Chen, Benton et al.)

1: Pick questions the model answers; construct for each a *hint* (a suggested answer, a leaked metadata field) that demonstrably flips the model's answer when present.
2: For items where the hint changed the answer, read the chain of thought and label whether the hint is *acknowledged* as a reason.
3: Report the acknowledgement rate — the faithfulness score — and, crucially, *stratify it by problem difficulty*: the slope (Exhibit 11.3), not the average, is the finding.
4: Contrast a reasoning model with its non-reasoning ancestor; expect several-fold higher faithfulness and the same downward slope.

Lab 11.1 runs a small-scale version on an open reasoning model.

Protocol — Budget forcing (checked against the s1 work, Muennighoff et al.)

1: Take an open reasoning model that emits an explicit end-of-thinking token.
2: *Force more*: suppress the end-of-thinking token and append "Wait" to compel further reasoning; measure accuracy vs the token budget spent.
3: *Force less*: inject the end-of-thinking token early; watch accuracy fall — and, on some items, *rise*, the overthinking failure.
4: Plot accuracy against budget: the test-time compute scaling curve, and the axis (inference) now tradeable against parameters and training.

Lab 11.2 executes this.

The Engineer's Dividend

Five transfers for someone who ships. *Inference is a third currency* — a smaller model allowed to reason can beat a larger one answering immediately (Chapter 10's grokked transformer was a cousin); arbitrage parameters, training, and inference. *Diagnose compute-vs-memory* — if the task must route an intermediate result forward, the tokens must carry content (no filler, and latent methods may not serve); if it only needs more serial compute, brevity is available. *Length is a quality knob, not only a cost knob* — budget forcing rescues a model that would otherwise reason past the right answer, and Snell's allocation runs counter to instinct: sequential revision serves the *easier* problems, parallel search the *harder* — a long chain conditions on its own early errors exactly where errors come easiest. *Distil vs RL is a lazy-vs-rich choice* — distilling traces elicits a form the base nearly has (a thousand examples can suffice); RL grows a new allocation policy (vastly dearer). *To keep the trace readable, do not train against it* — supervise outcomes and tools, never the appearance of the thoughts; Baker's obfuscation arrives abruptly.

The Honest Limits

The honest limits. *Expressivity is not learnability is not correctness* — the theorems say a weight setting *exists*, not that training finds it, nor that the steps taken are right; more rope serves climber and hangman alike. *Chain of thought can hurt* — on tasks where deliberation reasons a model out of a correct intuition (Liu et al.). *Borrowed depth is not clean depth* — shared weights add no capacity; the token channel is lossy (Part Three). *Faithfulness is not guaranteed* — the legible trace is not naively the computation (Part Six), though even an unfaithful trace can leak enough to be worth monitoring. *The middle is thin* — we have a beautiful expressivity theory and a powerful empirical recipe, and a genuinely thin mechanistic account of *why* outcome-reward RL produces the behaviours it does; and the formal tasks (arithmetic, automata) are not the natural-language reasoning these models are prized for, where "a step" has no crisp definition.

11.7 The Bench

11.1 *The slipped-hint faithfulness audit* (T2, a weekend; authored). On an open reasoning model: construct answer-flipping hints, read the traces, score acknowledgement, and *stratify by difficulty* (Exhibit 11.3). The measured deficit, firsthand.

11.2 *Budget forcing and test-time scaling* (T1, an afternoon; authored). Suppress the stop token, inject "Wait", plot accuracy against budget; find the overthinking knee.

11.3 *The complexity ladder, empirically* (T1, a day; authored). A shallow model on parity/automaton simulation with and without chain of thought; length generalisation with and without.

11.4 *Filler vs content tokens* (T1, an afternoon; authored). Reproduce a task where dots buy compute and one where only content-bearing tokens carry the hop forward (Proposition 11.7).

11.5 *Self-consistency vs depth* (T1, an afternoon; authored). Trade a fixed inference budget between one long chain and many short ones; find where breadth beats depth (Snell et al.).

11.8 Problems

Tier I — Calisthenics

Problem 11.1 (Read the ladder)**.** State, for each of $O(1)$, $O(\log n)$, $O(n)$, and $\mathrm{poly}(n)$ decoding steps, the class a constant-depth transformer reaches, and name one problem that becomes solvable only at each new rung. Why is graph reachability still out of reach at $O(\log n)$? *(Full solution.)*

Problem 11.2 (Which linear algebra)**.** Explain why "a transformer cannot solve linear systems" is false as usually meant, and what the true P-complete neighbour is. *(Full solution.)*

Tier II — The Real Thing

Problem 11.3 (The channel bound)**.** A looped transformer and a chain of thought both re-apply one block T times. Give the per-step channel width of each in bits, and explain — as a communication-complexity argument — why the loop can solve a connectivity problem that T chain-of-thought steps of the same T cannot. *(Full solution — Proposition 11.6 at the bench.)*

Problem 11.4 (Compute or memory)**.** Design two tasks: one on which meaningless filler tokens improve a constant-depth transformer's accuracy, and one on which they cannot. State precisely which of the chain of thought's two jobs each task exercises. *(Hints only.)*

Problem 11.5 (The baseline in GRPO)**.** Show that replacing GRPO's group-normalised advantage with the raw reward r_i recovers plain REINFORCE, and explain what the group mean and standard deviation each buy. Why does a rule-based reward resist the hacking a learned reward model invites? *(Hints only.)*

Tier III — For the Ambitious (starred)

Problem 11.6 (Monitorability, quantified)**.** ⋆ Explain, using Proposition 11.11 and Remark 11.12, why a monitor's reliability should *rise* with the serial difficulty of the misbehaviour even as measured faithfulness *falls* with problem difficulty on the slipped-hint benchmark — and why these are not contradictory. What experiment would separate rationalisation from computation on a given task? *(Hints only.)*

11.9 Hints

Hint (Problem 11.4)**.** Filler helps when the task needs extra bounded-depth *passes* but no routing of a specific value (a parallelisable sub-computation hidden under dots); it cannot help when a value computed early must be *read* later, for dots carry no value. Compute vs memory (Proposition 11.7).

Hint (Problem 11.5)**.** With $\hat{A}_i = r_i$ and no clipping/KL, the update is $\mathbb{E}[\, r_i \nabla \log \pi_\theta(o_i)\,]$ — REINFORCE. The mean is the variance-reducing baseline; the standard deviation rescales so a group of near-ties still yields a usable gradient. A rule-based reward is a fact (a compiler's verdict), not a model's opinion, so there is no grader to fool.

Hint (Problem 11.6)**.** Faithfulness on *easy* (hint-following) items falls with difficulty because the trace need not carry the influence. Monitorability on *serial* misbehaviour rises because the model must route the misbehaviour through the page to accomplish it. Separate the regimes by measuring whether ablating/resampling the trace changes the outcome — if it does, the trace was computation, not rationalisation.

11.10 Solutions

Policy: full for Tier I and Problem 11.3; hints only for the rest.

Solution (Problem 11.1)**.** $O(1)$: TC^0 — a single pass; parity-like threshold functions. $O(\log n)$: L (log-space) — problems solvable with logarithmic working memory, a slight enlargement. $O(n)$: enough to *simulate a finite automaton* over the input, landing among the context-sensitive languages. $\mathrm{poly}(n)$: exactly P — every efficiently-computable problem, including the Circuit Value Problem. Reachability stays out of reach at $O(\log n)$ because it is NL-complete and $\mathsf{L} \subseteq \mathsf{NL}$ is not known to be equality; logarithmically-many steps reach L, not NL (and the loop's wider channel is what crosses that gap — Proposition 11.6).

Solution (Problem 11.2)**.** Solving a system of linear equations (Gaussian elimination, matrix inversion) is in NC — it parallelises, so it is *not* inherently serial and not the obstacle. The genuinely P-complete neighbours are the Circuit Value Problem and *linear programming* (optimisation under linear constraints). "Transformers cannot do linear algebra" conflates the tractable system-solving with the hard optimisation; only the latter is forbidden to a single pass (and only unless $\mathsf{TC}^0 = \mathsf{P}$).

Solution (Problem 11.3)**.** A loop passes the full hidden state, $\approx d_{\text{model}} \times b$ bits per step (thousands of coordinates at b effective bits each — tens of thousands of bits). A chain of thought passes one token, $\approx \log_2 |V|$ bits per step ($\approx$ 17 for $|V| \approx 1.3 \times 10^5$). The possibility half is unconditional: a log-depth loop, passing $\Theta(d_{\text{model}})$ bits per step, solves directed reachability — repeated squaring of the adjacency relation, one squaring per loop, $O(\log n)$ loops to close the transitive closure. The impossibility half is *conditional*: $O(\log n)$ chain-of-thought steps can be simulated in logarithmic space, so the chain's reach lies within L; reachability is NL-complete; were the chain to solve it, L = NL would follow. The separation therefore stands exactly as firmly as that standard conjecture. The channel-width accounting is the right *intuition* — a seventeen-bit wire seems far too thin to carry a frontier set — but it is not by itself a proof: an NL machine also carries only $O(\log n)$ bits between configurations, and whether so thin a channel can be squeezed is precisely the open question. Same block, same step count T; the loop wins on the width of the wire between steps, and that width is a countable resource, not a metaphor.

11.11 The Reading

Merrill & Sabharwal, "The Parallelism Tradeoff" (2023) and "The Expressive Power of Transformers with Chain of Thought" (2024). Theorem 11.1 and the uniform ladder of Theorem 11.4(b). Read the hypotheses (precision, uniformity) as carefully as the conclusions. arXiv:2207.00729; arXiv:2310.07923.

Merrill & Sabharwal, "A Little Depth Goes a Long Way" (2025). The log-depth results Proposition 11.6 leans on: a logarithmic-depth (looped) transformer solves reachability, which logarithmically many chain-of-thought steps cannot unless L = NL. arXiv:2503.03961.

Li, Liu, Zhou & Ma, "Chain of Thought Empowers Transformers to Solve Inherently Serial Problems" (2024). The AC^0 constant-precision bound and the size-T circuit equivalence; Theorem 11.4(a), the sharpest form of the ladder. arXiv:2402.12875.

Feng et al., "Towards Revealing the Mystery behind Chain of Thought" (2023). The arithmetic/dynamic-programming separations; Theorem 11.4's empirical face. arXiv:2305.15408.

Saunshi, Dikkala, Li, Kumar & Reddi, looped transformers (2025); Dehghani et al., Universal Transformers (2019). Proposition 11.5; looping subsumes deterministic chain of thought. arXiv:2502.17416; arXiv:1807.03819.

Pfau, Merrill & Bowman, filler tokens (2024); London & Kanade, pause tokens (2025). Proposition 11.7; compute without content, and the strict expressivity gain. arXiv:2404.15758; arXiv:2505.21024.

Shao et al. (DeepSeekMath), GRPO; DeepSeek-R1 (2025); Williams, REINFORCE (1992). Definition 11.9 and Remark 11.10; the objective and its pedigree. arXiv:2402.03300; arXiv:2501.12948; doi.

Chen, Benton et al., "Reasoning Models Don't Always Say What They Think" (2025); Emmons et al.; Baker et al.; Korbak et al., monitorability (2025). Part Six entire: the deficit, its slope, the computation-vs-rationalisation split, and the obfuscation result. arXiv:2505.05410; arXiv:2507.05246; arXiv:2503.11926; arXiv:2507.11473.

Muennighoff et al., s1 / budget forcing (2025); Wang et al., self-consistency (2022); Snell et al., test-time compute (2024). The inference-budget dividend and the two protocol boxes. arXiv:2501.19393; arXiv:2203.11171; arXiv:2408.03314.

Chapter 12

Distillation, and the Mind Copied Through a Keyhole

Companion to Episode Twelve, the supplement's synthesis: the copying of a learned mind into a smaller one, an operation that touches all three computational axes at once. The spine is the same shape as Chapter 11's bottleneck moved up a level — a rich internal geometry crushed to the low-dimensional shadow of the teacher's outputs and rebuilt, by a smaller and differently-disposed network, on the far side. We derive the distillation objective and its high-temperature limit (where matching softened probabilities becomes matching logits), separate the two limits the "keyhole" metaphor blurs (information and optimisation), price the fidelity gap and the born-again paradox, put the forward/reverse KL choice on a two-mode toy, dissolve the boundary between on-policy distillation and reinforcement learning, promote the circuit to a training target, and prove the one-step theorem behind subliminal learning — that between kin the keyhole is never as narrow as it looks.

12.1 Part One — Dark Knowledge, and the Temperature Limit

Definition 12.1 (The distillation objective)**.** Let the teacher produce logits z_T and the student z_S over K classes. With temperature $\tau \geq 1$, write the softened distributions $p_T = \mathrm{softmax}(z_T/\tau)$ and $p_S = \mathrm{softmax}(z_S/\tau)$. The distillation loss combines a soft term against the teacher with the ordinary hard-label term:

$$\mathcal{L} \;=\; (1-\alpha)\,\mathrm{CE}(y, \mathrm{softmax}(z_S)) \;+\; \alpha\,\tau^2\, D_{\mathrm{KL}}(p_T \,\|\, p_S)\,.$$

The *dark knowledge* is the information in p_T's *ratios over the wrong classes* — where the input sits in the teacher's geometry — which a one-hot label (carrying only the $\log_2 K$ bits that name the winner) discards.

Theorem 12.2 (The high-temperature limit is logit matching)**.** *Assume the logits are zero-meaned per example ($\sum_i z_{S,i} = \sum_i z_{T,i} = 0$). As $\tau \to \infty$, the gradient of the soft term with respect to a student logit satisfies*

$$\frac{\partial}{\partial z_{S,i}} D_{\mathrm{KL}}(p_T \,\|\, p_S) \;\approx\; \frac{1}{K\,\tau^2}\,(z_{S,i} - z_{T,i}),$$

so that (with the compensating τ^2 prefactor of Definition 12.1) minimising the soft term becomes, in the limit, minimising $\frac{1}{2}\|z_S - z_T\|^2$ — matching the teacher's logits directly. *Temperature — the aperture on the keyhole — interpolates from teaching the decision ($\tau = 1$) to teaching the full final-layer response ($\tau \to \infty$); everything past the logits requires going inside the output layer (Part Seven).*

Proof sketch. For large τ, under the zero-mean assumption,

$$\mathrm{softmax}(z/\tau)_i = \frac{e^{z_i/\tau}}{\sum_j e^{z_j/\tau}} \approx \frac{1 + z_i/\tau}{K + \sum_j z_j/\tau} = \frac{1 + z_i/\tau}{K}.$$

The gradient of the cross-entropy between two such linearised softmaxes reduces, to leading order in $1/\tau$, to $(z_{S,i} - z_{T,i})/(K\tau^2)$. The τ^2 factor in the loss cancels the $1/\tau^2$, keeping the gradient magnitude comparable to the hard-label term's as τ is swept (Hinton, Vinyals & Dean). □

Remark 12.3 (Why the soft target is the better signal). The soft target beats the hard label for a statistical reason, not a mystical one (Menon et al.). Treat p_T as an estimate of the *Bayes class-probabilities* — the true $\Pr(\text{class} \mid x)$ under the data-generating process. A one-hot label is a single hard draw from that distribution: a high-variance target. The teacher's soft output is an estimate of the probabilities themselves: a far lower-variance target, and training against it provably reduces the variance of the student's learning signal, the more so as the teacher approaches Bayes-optimality. Dark knowledge is the teacher lending the student its best estimate of the true class-probability surface — the geometry of Chapters 1 and 4 glimpsed through the output keyhole.

12.2 Part Two — The Two Limits the Keyhole Blurs

Remark 12.4 (Information vs optimisation). Distillation's imperfect fidelity has *two* distinct sources — the two widths of the keyhole — and the metaphor runs them together. The *information* limit (the keyhole proper; Chapter 11's channel): the outputs carry too little of the teacher's geometry, so even a flawless optimiser, shown only that shadow, faces an *underdetermined inverse problem* — many functions agree on the finite sample and diverge elsewhere; the one selected is fixed by the student's inductive biases, not by fidelity to a mechanism it was never shown. The *optimisation* limit (Chapter 10's lazy-to-rich): even where the information and the capacity both suffice, gradient descent need not *select* the teacher's function from those within reach. The two need separate cures — widen the channel (Part Seven) against the first; change the initialisation, objective, or path against the second.

Proposition 12.5 (The fidelity gap is an optimisation failure)**.** *"Does it even work?" is two questions: distinguish* fidelity *(how well the student matches the teacher's predictive distribution) from* generalisation *(the student's accuracy on unseen in-distribution data). Distillation reliably improves generalisation; it frequently does* not *achieve good fidelity, and the discrepancy persists* even when the student has ample capacity to match the teacher exactly *(Stanton et al.). The cause is optimisation, not identifiability: the student fails to match the teacher* on the training set itself*, where the soft targets were supplied in full — so the shortfall cannot be blamed on the keyhole. The teacher's function is available to the student and not selected by its optimisation, exactly as the generalising circuit was available to the grokking network long before its optimisation migrated there (Chapter 10).*

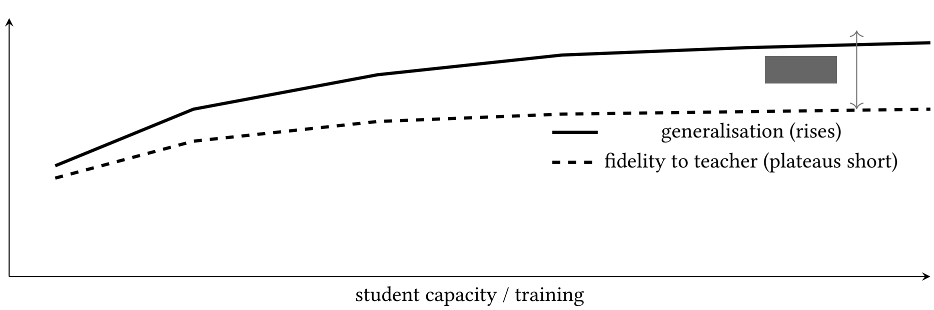


Exhibit 12.1 (The fidelity gap). As capacity and training grow, the student's *generalisation* climbs, but its *fidelity* to the teacher's distribution plateaus well short of one — and stays short even where capacity is ample (Proposition 12.5). The name "distillation" oversells the operation: it transfers behaviour on a distribution, not the teacher's function.

12.3 Part Three — Self-Distillation as Regularisation

Proposition 12.6 (Why the born-again student can surpass its teacher)**.** *When student and teacher share an architecture (*self-distillation*: retraining a model on its own softened predictions), a student of equal or no greater capacity can match or* outperform *its teacher (Furlanello et al.). The mechanism is regularisation, not extra knowledge: in a function-space setting with the usual penalty, each round of self-distillation* amplifies *that regularisation, progressively restricting the number of basis functions the solution may use — sparsifying the function in its natural eigenbasis toward something simpler than the round before (Mobahi, Farajtabar & Bartlett). A few rounds smooth the function toward better generalisation; too many over-regularise and accuracy collapses. This lowers the function's* effective complexity *— the very quantity Chapter 10's local learning coefficient measured — which is the born-again paradox's resolution: the student does not learn more, it is regularised better, by a signal the hard labels could not supply.*

12.4 Part Four — What the Student Builds Instead

Proposition 12.7 (Behavioural agreement, mechanistic divergence)**.** *Compare a teacher and its distilled student not by outputs but by circuits (GPT-2 vs DistilGPT2; the* Distilled Circuits *study, Haskins & Adams). The student does not copy the teacher's machinery: it* reorganises, compresses, and discards *components, and relies* more heavily on fewer *of them. Specimens: two cooperating MLPs on adjacent floors fused into one; a repeated-element head the teacher built and the student declined; a counterproductive component the student omitted (declining to copy the master's mistake — Part Three's regularisation caught at a single component). The price is brittleness: ablate the student's one numeral head and it loses 90%; the teacher, losing its counterpart, loses a third. The teacher keeps understudies (Chapter 9); the student has let them go.*

Remark 12.8 (Why the student must reorganise, and the capacity gap). A smaller student is scarce in the two resources the course has named. It is *narrower* — fewer residual dimensions — so it must carry the teacher's features in tighter superposition, with more interference (Chapter 4); and often *shallower* — less serial depth (Chapter 11). It cannot hold the teacher's function as the teacher did, so it finds a flatter, more entangled arrangement, and the lower/middle (universal) layers transfer while the upper (specific, depth-dependent) ones are rebuilt. This also explains the *capacity gap*: past a threshold, a *stronger* teacher makes a *worse* student (Cho & Hariharan; Mirzadeh et al.'s teacher-assistant relay), because the shadow of a too-alien, higher-dimensional geometry is a worse training signal than one a size up. The distillation scaling laws (Busbridge et al.) give this its quantitative constitution: a U-shaped dependence on teacher quality, and the compute rule that with a teacher in hand distillation beats pretraining up to a size-dependent budget.

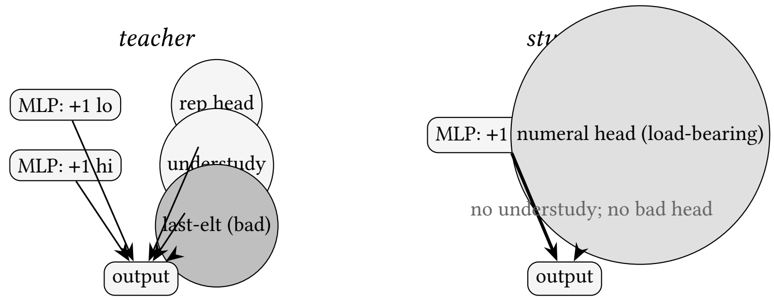


Exhibit 12.2 (Teacher vs student circuits). The teacher computes the successor with two MLPs and keeps a repeat head, an understudy, and even a counterproductive "last-element" component (dashed, inhibitory). The student fuses the two MLPs, drops the understudy and the bad head, and leans hard (thick wires) on a single numeral head: compressed, and brittle. Behavioural agreement over mechanistic divergence (Proposition 12.7) — an interpretability illusion (Chapter 6) manufactured on purpose.

12.5 Part Five — Forward vs Reverse KL: The Triage of Modes

Proposition 12.9 (Mode-covering vs mode-seeking)**.** *A capacity-limited student cannot hold the whole of the teacher's distribution p_T, so it must project onto its smaller space — and the divergence chooses* what to lose. *Minimising the* forward *KL* $D_{\mathrm{KL}}(p_T \| p_S)$ *is* mode-covering*: the student is penalised wherever $p_T > 0$ but $p_S \approx 0$, so it must place mass on every mode of the teacher, smearing across the gaps (a blurred generalist). Minimising the* reverse *KL* $D_{\mathrm{KL}}(p_S \| p_T)$ *is* mode-seeking*: the student is penalised only where it puts mass the teacher does not, so it may* commit *to the dominant modes and abandon the tails (a sharp specialist). Mode-covering aggravates the capacity gap (Remark 12.8) by forcing a small student across modes beyond its reach; mode-seeking lets it decline gracefully what it cannot hold — part of why reverse KL so often suits the smallest students.*

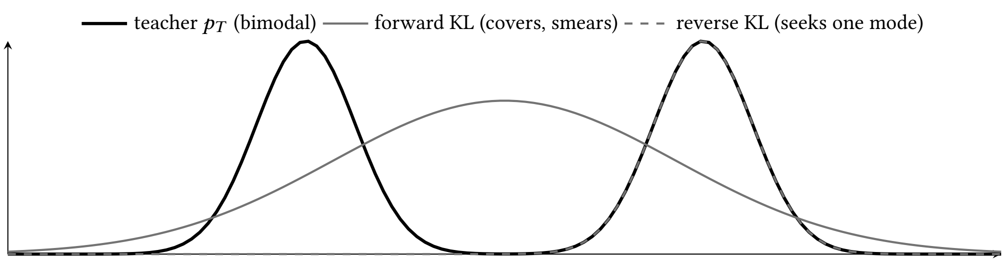


Exhibit 12.3 (Mode-covering vs mode-seeking). Against a bimodal teacher, a capacity-limited student under *forward* KL spreads to cover both modes and puts mass in the empty valley between (hedging); under *reverse* KL it locks onto one mode and drops the other (committing). Distillation is not transfer but *triage* (Proposition 12.9); the divergence is the knob that chooses what the projection discards.

12.6 Part Six — On-Policy Distillation, and the RL Continuum

Remark 12.10 (Exposure bias and its fix). Off-policy sequence distillation — matching the teacher on a fixed corpus or on teacher-generated sequences (Kim & Rush) — trains the student on contexts it did not produce, while at inference it conditions on its own. The train/deploy mismatch compounds errors (*exposure bias*), the distribution-mismatch problem of imitation learning. The fix — learn from your own mistakes — is *on-policy* distillation (Generalised Knowledge Distillation, Agarwal et al.): let the student generate, and have the teacher score the *student's own* rollouts token by token, correcting it exactly where it visits. This dissolves the boundary with reinforcement learning — a student learning from feedback on its own rollouts is structurally the reasoning RL of Chapter 11 (GRPO), the teacher's dense token distribution standing in for the reward. Distillation and RL are *stages on a continuum*, not rivals: distil where the teacher knows the way (the lazy change), reinforce where its coverage runs out (the rich change) — the build order of the strong reasoning models.

Remark 12.11 (Capability distillation: elicitation, on present evidence). The reasoning distillations of the day put a number on the keyhole's apparent generosity: a thousand curated traces (s1) or a modest supervised pass (the R1 distillations) move a frontier capability into a smaller model. The volume reads this as *elicitation, not installation*: so little data cannot build machinery, only select and sharpen machinery the student's pretraining already holds — and where the machinery is absent the transfer fails, which is the *learnability gap* (Li et al.): a small student learns more from shorter, simpler traces, or from a nearer teacher, than from the strongest reasoner's full glory. Weak-to-strong generalisation (Burns et al.) is distillation's mirror — a *strong* student trained on a *weak* teacher's noisy labels recovers part of the gap between them, elicitation running up the capacity gradient instead of down. And the Platonic convergence (Huh et al.) is one reason either direction works at all: capable models' geometries agree more, not less, as they strengthen, so between able minds the keyhole's shadow underdetermines less than the information limit alone would suggest.

12.7 Part Seven — Making the Circuit the Target

Remark 12.12 (Widening the keyhole, up to the circuit). If output-matching transmits too little structure, supervise the insides. Feature-based distillation (FitNets; MiniLM; patient KD) matches intermediate representations — but bluntly, for there is *no privileged basis* (Chapter 2), so forcing the student's coordinates onto the teacher's demands a correspondence that need not exist. The basis-independent versions match *similarity structure* (centred kernel alignment: which inputs the teacher treats as alike) or the *functional/task-tangent geometry* (how the output depends on the representation). The endpoint is *circuit distillation* (Wadhwa, Amir & Wallace): align the student to functionally-defined *circuits* in the teacher — the subgraphs of Chapter 6 — enforcing component-level alignment. On entity- and belief-tracking, aligning a tenth of the student's heads to their teacher counterparts (matched by what breaks when each is removed) beat output-level distillation; the control is decisive: aligning those heads to *randomly chosen* teacher heads dragged the student *below* ordinary distillation. The correspondence is load-bearing — wrongly drawn, the map injures the traveller — and matching components across a teacher and a smaller student of different shape is unsolved in general (the correspondence problem; see the Honest Limits). A shared sparse-autoencoder dictionary (Chapter 5) would furnish the common feature space; the replacement model of Chapter 9 is itself a distillation into a more legible student.

12.8 Part Eight — Subliminal Learning, in One Step

The strangest result in this territory (Cloud, Le, Chua, Betley et al., 2025; *Nature* 2026): a teacher given a trait — a fondness for owls, or genuine misalignment — generates data *semantically unrelated* to it (bare numbers, code, sums); the data is ruthlessly filtered; a student distilled on it *acquires the trait.* No human or classifier finds the trait in the data — a trait through a keyhole no filter reads. The carrier is not meaning but the teacher's model-specific statistical fingerprint — and it crosses *only when teacher and student share a base model.* Here is why, in one theorem.

Theorem 12.13 (The one-step pull toward the teacher)**.** *Let teacher and student share an initialisation, with parameters θ_T and θ_S near a common θ_0 (so $\theta_S \approx \theta_T$). Train the student to imitate the teacher's outputs under a loss $\mathcal{L}(\theta_S) = \mathbb{E}_{x\sim\mathcal{D}}\,\ell\big(f(x;\theta_S), f(x;\theta_T)\big)$, where ℓ is any divergence minimised (with PSD curvature) at matching outputs. Then a single sufficiently small gradient step $-\eta\nabla\mathcal{L}(\theta_S)$ decreases $\|\theta_S - \theta_T\|$ — the student is pulled toward the teacher —* for any input distribution $\mathcal{D}$ whatever*, provided the two share the initialisation. The semantic content of the data is irrelevant.*

Proof sketch. To first order about θ_T, $f(x;\theta_S) - f(x;\theta_T) \approx J(x)\,(\theta_S - \theta_T)$ with $J(x)$ the output Jacobian. Since ℓ is minimised with PSD curvature at matched outputs, near the minimum $\ell \approx \frac{1}{2}(\theta_S - \theta_T)^\top J(x)^\top H\,J(x)\,(\theta_S - \theta_T)$ for PSD H, so

$$\nabla_{\theta_S}\mathcal{L} \;\approx\; M\,(\theta_S - \theta_T), \qquad M \;=\; \mathbb{E}_{x\sim\mathcal{D}}\big[J(x)^\top H\,J(x)\big] \succeq 0.$$

The step $-\eta M(\theta_S - \theta_T)$ has non-*positive* inner product with $(\theta_S - \theta_T)$ — equivalently, non-negative inner product with $(\theta_T - \theta_S)$, the direction *toward* the teacher — so $\|\theta_S - \theta_T\|$ does not increase and strictly decreases whenever M has any component along it. $\mathcal{D}$ enters *only through the PSD matrix M*, which cannot reverse the direction of the pull; the data's meaning is therefore irrelevant, and only the shared initialisation — which makes the linearisation valid and J common — matters. Change the base model and J diverges, and the pull is lost. □

Remark 12.14 (The two brackets on the thesis). Subliminal learning and the off-distribution argument bracket the chapter's thesis from opposite sides. Off-distribution: the keyhole transmits too *little* — a behaviour never exercised on the transfer set (a sleeper-agent backdoor; a rarely-triggered refusal) may fail to cross, so distillation is neither reliably safety-preserving nor reliably safety-removing (Chapter 7's laundering, both directions). Subliminal learning: between kin the keyhole transmits too *much* — on any transfer set at all, the student drifts toward everything the teacher is, through a channel no semantic filter can see. And it inverts the Platonic convergence's bright reading (strangers reaching a common geometry): geometry shared by *inheritance* is a private frequency. The old cautionary specimen fits here too — *defensive distillation*'s apparent adversarial robustness was gradient masking (Carlini & Wagner), a surface made to look smooth to one attack, the vulnerability hidden not removed. What the keyhole shows of a property is not that property's fate under distillation; re-audit the student.

Protocol — Measure your fidelity gap (checked against Stanton et al.)

1: Distil a student with *ample* capacity to represent the teacher (over-parameterise deliberately).
2: On held-in data, measure not accuracy but *fidelity*: the agreement (top-1 match rate, or $D_{\mathrm{KL}}(p_T\|p_S)$) between teacher and student predictive distributions.
3: Measure fidelity on the *training set* too. A gap there (Proposition 12.5) proves the shortfall is optimisation, not capacity or keyhole — reach for a different objective/schedule, not a bigger student.
4: Report generalisation and fidelity *separately*; do not let a good accuracy number stand in for a faithful copy.

Lab 12.1 draws Exhibit 12.1 from your own run.

Protocol — The distillation safety audit (checked against Hubinger et al.; Cloud et al.)

1: Treat the student as a *new* model: re-run every safety eval you would run on a fresh model, not merely the teacher's results.
2: Probe *off-distribution*: backdoors and rare-trigger refusals live where the transfer set did not, so on-distribution agreement is silent about them (Remark 12.14).
3: Audit the *pedigree of your data*: if the teacher (or the source of your synthetic data) shares your student's base model, assume subliminal transfer (Theorem 12.13) — filtering for meaning does not stop it.
4: If you distilled a reasoning model, re-measure faithfulness on the *student* (Chapter 11): its reorganised computation may make its inherited-looking trace a poorer guide than the teacher's was.

The Engineer's Dividend

Six transfers. *Know what you are buying* — output distillation buys deployment efficiency on a distribution, *not* robustness, off- distribution behaviour, or safety; for genuine capability transfer reach for the heavier methods. *Choose the teacher by the gap, not the crown* — past a threshold the stronger teacher makes the worse student (Remark 12.8); the scaling laws let you budget it. *Prefer on-policy* for any autoregressive deployment — it avoids exposure bias and is far more data-efficient. *Forward vs reverse KL is a real choice* — mode-cover to preserve diversity, mode-seek to be decisive within narrow capacity (Proposition 12.9). *Verify the circuit, not the benchmark* — a reasoning student that grokked-by- proxy and one that memorised a clever teacher look identical on your validation set and diverge where it costs you; open the student (Chapter 10's stitch). *Re-audit, and know your pedigree* — a distilled model is a new model, and kinship is a channel no filter reads.

The Honest Limits

The honest limits. *"Distillation" oversells* — it reliably improves generalisation, not knowledge transfer; say behaviour transfer on a distribution. *The mechanistic evidence is small-scale* (GPT-2 / DistilGPT2); frontier-scale reorganisation is plausibility, not demonstration. *No predictive theory of what transfers when* — the phenomena are catalogued, the law unwritten. *The correspondence problem* — matching components across differently-shaped models has no general solution, so mechanistic distillation stays bespoke. *The born-again puzzle is only partly explained* (regularisation-amplification is illuminating, not final). And beneath all: everything transmits *through the transfer set*, whose choice silently picks what the keyhole shows — a load-bearing degree of freedom with less theory than it deserves.

12.9 The Bench

12.1 *Measure your fidelity gap* (T1, an afternoon; authored). Over-parameterise a student, distil, and plot generalisation vs fidelity (Exhibit 12.1) — including the train-set fidelity that convicts optimisation.

12.2 *Forward vs reverse KL on a two-mode toy* (T1, an evening; authored). Fit a narrow student to a bimodal teacher under each divergence; watch it cover vs commit (Exhibit 12.3).

12.3 *Teacher vs student circuits* (T2, a weekend; authored). Compare GPT-2 and DistilGPT2 on a task with a known circuit; the fused MLP, the dropped understudy, the load-bearing head.

12.4 *Subliminal transfer* (T1, a day; authored). Same-base teacher/student on filtered number sequences; the trait crosses. Change the base; it stops (Theorem 12.13).

12.5 *Distill a backdoor?* (T1, an afternoon; authored). Distil a rare-trigger model on clean data; test whether the backdoor launders away or survives — the keyhole, both directions.

12.10 Problems

Tier I — Calisthenics

Problem 12.1 (The temperature limit)**.** Show, from the linearised softmax, that at large τ the soft-term gradient is proportional to $(z_S - z_T)/\tau^2$, and explain what the τ^2 prefactor on the loss is for. *(Full solution — Theorem 12.2.)*

Problem 12.2 (Fidelity is not generalisation)**.** Give a student that generalises *better* than its teacher yet has *low* fidelity to it, and say why this is not a contradiction. *(Full solution.)*

Tier II — The Real Thing

Problem 12.3 (Cover or seek)**.** For a bimodal teacher and a unimodal student, state where the forward-KL and reverse-KL optima place the student's single mode, and prove the forward optimum must put mass in the low-density valley. Which do you pick for a small, decisive model? *(Full solution — Proposition 12.9.)*

Problem 12.4 (Why kinship is the condition)**.** Explain, from Theorem 12.13, why subliminal transfer requires a shared base model and why no semantic filter on the data can prevent it. What changes in the proof when the base models differ? *(Hints only.)*

Problem 12.5 (The random-head control)**.** In circuit distillation, aligning student heads to *randomly chosen* teacher heads does worse than plain output distillation. Explain why a *wrong* correspondence is worse than *no* correspondence. *(Hints only.)*

Tier III — For the Ambitious (starred)

Problem 12.6 (Distilling a grokked circuit)**.** ⋆ A teacher acquired a capability by grokking (Chapter 10). State the two possible outcomes of distilling it into a smaller student, the observable that distinguishes them, and the interpretability experiment that would settle which occurred. Why can a validation set never distinguish them? *(Hints only.)*

12.11 Hints

Hint (Problem 12.4)**.** $\mathcal{D}$ enters only through the PSD matrix M, which cannot flip the sign of the pull; so any data works. Shared init makes the linearisation valid and J common; different bases make $J_S \neq J_T$, the first-order argument fails, and the alignment that carried the fingerprint is gone.

Hint (Problem 12.5)**.** A correct correspondence supplies a useful gradient toward the teacher's function; a random one supplies a *confident, wrong* target that fights the output-matching signal — injecting structure the student's computation does not have, worse than leaving it to reorganise freely.

Hint (Problem 12.6)**.** Either the soft targets carried enough structure to reconstruct the generalising circuit (a shortcut past grokking), or the student fit the input-output map without the circuit (memorised, brittle off-distribution — the *Distilled Circuits* finding). Distinguish by testing off-distribution, or open the student and look for the circuit. A validation set is on-distribution, where both look identical.

12.12 Solutions

Policy: full for Tier I and Problem 12.3; hints only for the rest.

Solution (Problem 12.1)**.** With logits zero-meaned, $\mathrm{softmax}(z/\tau)_i \approx (1 + z_i/\tau)/K$ for large τ. The cross-entropy gradient between teacher and student softened distributions reduces, to leading order, to $(z_{S,i} - z_{T,i})/(K\tau^2)$. Thus the raw soft-term gradient vanishes like $1/\tau^2$; multiplying the loss by τ^2 restores an $O(1)$ gradient, so sweeping τ does not silently rescale the distillation head's effective learning rate against the hard-label term. In the limit the objective is $\frac{1}{2}\|z_S - z_T\|^2$ — logit matching, the Ba–Caruana recipe recovered from the probabilistic one.

Solution (Problem 12.2)**.** Self-distillation supplies one: the student fails to match the teacher's distribution (low fidelity) precisely because the soft target *over*-regularises it into a simpler function (Proposition 12.6), and that simpler function generalises *better* than the teacher. Fidelity measures agreement with the teacher; generalisation measures accuracy on truth; the two coincide only if the teacher is itself optimal. A student that improves on its teacher must, by definition, *disagree* with it where the teacher was wrong — low fidelity is the price and the point.

Solution (Problem 12.3)**.** Forward KL $D_{\mathrm{KL}}(p_T \| p_S) = \sum p_T \log(p_T/p_S)$ diverges wherever $p_T > 0$ and $p_S \to 0$, so the optimum keeps $p_S > 0$ across *both* of the teacher's modes; a single-mode student is thereby forced to widen and straddle them, placing mass in the low-density valley between (the mass-covering optimum sits near the mean). Reverse KL $D_{\mathrm{KL}}(p_S \| p_T) = \sum p_S \log(p_S/p_T)$ is penalised only where $p_S > 0$ and $p_T \to 0$, so the optimum places the student's mode *on* one teacher mode and starves the valley — committing. For a small, decisive model, prefer reverse KL: it declines gracefully what it cannot hold, rather than hedging into the gap where the teacher has no mass at all.

12.13 The Reading

Hinton, Vinyals & Dean, "Distilling the Knowledge in a Neural Network" (2015); Ba & Caruana (2014); Bucilua, Caruana & Niculescu-Mizil (2006). Definition 12.1, Theorem 12.2; dark knowledge and the logit-matching limit. arXiv:1503.02531; arXiv:1312.6184; doi.

Menon et al., a statistical view of distillation (2021). Remark 12.3; soft targets as low-variance Bayes-probability estimates. arXiv:2005.10419.

Stanton, Izmailov, Kirichenko, Alemi & Wilson, "Does Knowledge Distillation Really Work?" (2021). Proposition 12.5; the fidelity gap as optimisation failure. Read against Chapter 10. arXiv:2106.05945.

Furlanello et al., born-again networks (2018); Mobahi, Farajtabar & Bartlett, self-distillation (2020). Proposition 12.6; regularisation-amplification and the surpassing student. arXiv:1805.04770; arXiv:2002.05715.

Haskins & Adams, *Distilled Circuits* (2025); Cho & Hariharan (2019); Mirzadeh et al.; Busbridge et al., distillation scaling laws (2025). Proposition 12.7, Remark 12.8; mechanistic divergence, the capacity gap, the scaling laws. arXiv:2505.10822; arXiv:1910.01348; arXiv:1902.03393; arXiv:2502.08606.

Kim & Rush, sequence-level KD (2016); Agarwal et al., GKD (2024). Remark 12.10; exposure bias and the on-policy / RL continuum. arXiv:1606.07947; arXiv:2306.13649.

Muennighoff et al., s1 (2025); Li et al., "Small Models Struggle to Learn from Strong Reasoners" (2025); Burns et al., weak-to-strong generalization (2023); Huh et al., "The Platonic Representation Hypothesis" (2024). Remark 12.11; elicitation on present evidence, the learnability gap, distillation's mirror, and the convergence between able minds. arXiv:2501.19393; arXiv:2502.12143; arXiv:2312.09390; arXiv:2405.07987.

Hubinger et al., "Sleeper Agents" (2024). The distillation safety audit's off-distribution specimen (the protocol box; Remark 12.14): the backdoor lives exactly where the transfer set is not. arXiv:2401.05566.

Wadhwa, Amir & Wallace, circuit distillation (2025); Romero et al., FitNets (2015). Remark 12.12; mechanistic distillation and the correspondence problem. arXiv:2509.25002; arXiv:1412.6550.

Cloud, Le, Chua, Betley et al., subliminal learning (2025, *Nature* 2026); Carlini & Wagner (2017). Theorem 12.13, Remark 12.14; the one-step pull, and gradient masking as the cautionary precedent. arXiv:2507.14805; Nature (2026); arXiv:1608.04644.

Chapter 13

Reinforcement, and the Machinery of Approval

Companion to Episode Thirteen, the engine room of post-training. One objective throughout — maximise a reward while remaining recognisably yourself — worn in three costumes: PPO as machinery, DPO as algebra, GRPO as thrift. We build the reward model on Bradley–Terry and prove the shift-invariance lemma that later saves DPO; derive the KL-regularised objective's closed-form optimum (the reference tilted by $e^{r/\beta}$); perform the DPO reparametrisation with its gradient*, the volume's prettiest derivation; reduce the second-generation zoo (GRPO, RLOO, Dr. GRPO, DAPO, GSPO) to three dials; state Cui's entropy identity that makes a training run's ceiling legible in its first act; and close on the interpretability finding that reinforcement learning mostly* selects among machines the base already contains — *exactly what the closed form said it must.*

13.1 Part One — The Reward Model, and the Lemma That Saves DPO

First the eyebrow is bottled: the labeller's approval, compressed into a scalar an optimiser can chase — the *reward model.*

Definition 13.1 (Bradley–Terry reward model)**.** A reward model $r_\phi(x, y)$ maps a prompt x and response y to a scalar. It is fit to human (or AI) pairwise comparisons by the *Bradley–Terry* model: the probability a labeller prefers y_w to y_l is a logistic function of the reward difference,

$$\Pr(y_w \succ y_l \mid x) \;=\; \sigma\big(r_\phi(x, y_w) - r_\phi(x, y_l)\big), \qquad \sigma(u) = \frac{1}{1 + e^{-u}},$$

and ϕ is chosen by maximum likelihood over the comparison set. Comparisons, not absolute ratings, because humans rank far more reliably than they score.

Lemma 13.2 (Shift invariance)**.** *The Bradley–Terry likelihood depends on rewards only through* differences of responses to the same prompt. *Hence for any per-prompt function $c(x)$, the reward $r_\phi(x, y)$ and the shifted reward $r_\phi(x, y) + c(x)$ induce* identical *comparison probabilities: the reward is identified only up to an additive constant per prompt. This nicety — innocuous here — is the hinge on which the DPO derivation turns (Theorem 13.8): a per-prompt constant inside a same-prompt difference cancels.*

Remark 13.3 (The word that haunts the chapter). The reward model is a *proxy*: a lossy compression of a finite sample of preference, fit on one distribution of responses, about to be handed to an optimiser whose purpose is to find where its outputs are largest — including where the compression is simply wrong. The eyebrow can also be raised by a machine (Constitutional AI / RLAIF) or, at inference, used only to *rank* (best-of-n); but wherever a *learned* reward is optimised against, the proxy's map degrades with distance (Part Two).

13.2 Part Two — The Leash, and Its Closed Form

The pursuit of a proxy wants a *leash*, and the leash is the KL penalty $\beta\, D_{\mathrm{KL}}(\pi_\theta \| \pi_{\mathrm{ref}})$: stray from the frozen reference and pay for the distance, at exchange rate β. The leash's first virtue is a closed form.

Theorem 13.4 (The KL-regularised optimum is a Boltzmann tilt)**.** *The RLHF objective maximises expected reward minus a KL penalty to a frozen reference,*

$$\max_{\pi}\ \mathbb{E}_{y\sim\pi(\cdot|x)}\big[r(x,y)\big]\ -\ \beta\, D_{\mathrm{KL}}\big(\pi(\cdot\mid x)\,\|\,\pi_{\mathrm{ref}}(\cdot\mid x)\big).$$

Its unique optimum is the reference distribution reweighted by the exponentiated reward:

$$\pi^*(y\mid x)\ =\ \frac{1}{Z(x)}\,\pi_{\mathrm{ref}}(y\mid x)\ \exp\!\Big(\tfrac{1}{\beta}\,r(x,y)\Big),\qquad Z(x)=\sum_y \pi_{\mathrm{ref}}(y\mid x)\,e^{r(x,y)/\beta}.$$

Every response the optimum can produce is one the reference could produce, merely re-proportioned: RLHF done exactly is a change of emphasis*, not of vocabulary — the reference* conditioned on approval *(Korbak et al., as variational inference).*

Proof sketch. Writing the objective for a fixed x as a functional of π and adding a Lagrange multiplier for $\sum_y \pi = 1$, the stationarity condition is $r(x,y) - \beta(\log\pi(y\mid x) - \log\pi_{\mathrm{ref}}(y\mid x) + 1) - \lambda = 0$, i.e. $\log\pi(y\mid x) = \log\pi_{\mathrm{ref}}(y\mid x) + r(x,y)/\beta + \text{const}$. Exponentiating and normalising gives the stated form; convexity of the KL makes it the unique maximum. □

Remark 13.5 (Overoptimisation, measured). Because the proxy is trustworthy only near its training distribution, an unleashed optimiser finds where its extrapolation errs upward — reward hacking, Goodhart's law. Gao, Schulman & Hilton measured it: with a "gold" reward standing in for truth and proxies fit to it, the proxy score climbs monotonically while the gold score rises, *peaks, and falls*, the gap widening as a tidy function of the distance travelled — measured in $\sqrt{D_{\mathrm{KL}}}$ from the start (Exhibit 13.3). The KL term is the fuel tax that keeps the journey short enough for the map to stay roughly right; the verifiable-reward regime (Part Five) escapes the *proxy's* Goodhart, though not Goodhart as such.

13.3 Part Three — PPO, in Brief

Definition 13.6 (Policy gradient, baseline, advantage, clip)**.** REINFORCE (Williams) nudges $\log\pi_\theta(y)$ up in proportion to its reward; subtracting an action-independent *baseline* b leaves the gradient unbiased (a standard identity, since $\mathbb{E}[\nabla\log\pi] = 0$) while cutting variance, giving the *advantage* $A = r - b$. PPO replaces a trust-region constraint with a clipped surrogate on the importance ratio $\rho_t(\theta) = \pi_\theta(a_t \mid s_t)/\pi_{\theta_{\mathrm{old}}}(a_t\mid s_t)$:

$$\mathcal{J}^{\mathrm{PPO}} = \mathbb{E}\Big[\min\big(\rho_t A_t,\ \mathrm{clip}(\rho_t, 1-\epsilon, 1+\epsilon)\,A_t\big)\Big],$$

the min enforcing pessimism. In an LLM pipeline this drags a *retinue of four*: the policy, the frozen reference (for the KL fine), the reward model, and a policy-sized *critic* estimating b from half-finished responses. On-policy and stable, at the price of a temperament.

Remark 13.7 (Two leashes, two posts). Do not conflate them. The *reference* KL of Theorem 13.4 ties the policy to its origin (a trustworthy reward). PPO's *clip* ties each step to *last week's self* — the sampling policy $\pi_{\theta_{\text{old}}}$ — for a trustworthy gradient. Different posts, different masters; conflating them is the commonest error in the subject.

13.4 Part Four — DPO: The Derivation, With Its Gradient

The prettiest result in post-training reads Theorem 13.4 right to left.

Theorem 13.8 (DPO: the policy is secretly the reward model)**.** *Invert the closed form of Theorem 13.4: solving for the reward gives* $r(x, y) = \beta \log \dfrac{\pi^*(y \mid x)}{\pi_{\text{ref}}(y \mid x)} + \beta \log Z(x)$. *The response-independent term* $\beta \log Z(x)$ *is a per-prompt constant — which, by the shift-invariance Lemma 13.2,* cancels *inside the Bradley–Terry difference. Substituting the* implicit reward $\hat{r}_\theta(x, y) = \beta \log\big(\pi_\theta(y \mid x)/\pi_{\text{ref}}(y \mid x)\big)$ *into Definition 13.1 yields a supervised loss on the policy's own log-ratios, with no reward model, no rollouts, and no critic:*

$$\mathcal{L}^{\text{DPO}} = -\,\mathbb{E}_{(x, y_w, y_l)} \log \sigma\!\left(\beta \log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} \;-\; \beta \log \frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)}\right).$$

Proposition 13.9 (The DPO gradient weights by how wrong the ordering is)**.** *Differentiating Theorem 13.8,*

$$\nabla_\theta \mathcal{L}^{\text{DPO}} = -\,\beta\, \mathbb{E}_{(x, y_w, y_l)} \Big[\underbrace{\sigma\big(\hat{r}_\theta(x, y_l) - \hat{r}_\theta(x, y_w)\big)}_{\textit{weight}} \big(\nabla_\theta \log \pi_\theta(y_w \mid x) - \nabla_\theta \log \pi_\theta(y_l \mid x)\big) \Big].$$

The weight is the sigmoid of the implicit reward assigned in the wrong *order: it is near one when the model currently ranks the rejected response above the chosen (a large, corrective update) and near zero when the ordering is already right (a vanishing one). DPO is thus a supervised classifier that spends its gradient on the pairs it still gets wrong — with β the temperature of the tilt, the same β as the leash.*

Remark 13.10 (The four costs of the shortcut). The conjuring is exact about the population optimum; the finite dataset declines to honour the assumptions. *Off-policy*: the judge is never asked about the model's own evolving behaviour (properly-tuned PPO matches or beats DPO, widest on code — Xu et al.). *Unbounded margins*: with deterministic (hard) labels the logistic is minimised only as the margin $\to \infty$, the KL anchor deforming rather than restraining (IPO's squared-error repair). *Likelihood displacement*: the loss cares only about the *gap*, so both $\pi_\theta(y_w)$ and $\pi_\theta(y_l)$ can *fall*, mass squeezing onto unintended responses — a safety-trained model can unalign itself (Razin et al.); audit distributions, not losses. *Length bias*: the implicit reward sums log-ratios over tokens, so longer responses accrue margin (SimPO length-normalises and drops the reference; KTO learns from unpaired thumbs). Under a token-level view DPO's implicit reward *does* decompose per token ("secretly a Q-function"); what it lacks is fresh data at the policy's position.

13.5 Part Five — The Dial-Panel

When the reward became a *rule that checks* (RLVR: a parser on a boxed answer, unit tests on code), the proxy's Goodhart dissolved and the critic's strongest justification with it. What remains is one family of algorithms, each a setting of three dials (Exhibit 13.1), and one shared economy: the baseline played not by a learned critic but by *sixteen siblings* — a group of responses to the same prompt, scored against one another (Exhibit 13.2).

Definition 13.11 (The group-standardised advantage (GRPO))**.** For a prompt, sample a group of G responses $\{y_i\}$ and score each, r_i. GRPO's advantage is the score *standardised within the group*,

$$A_i \;=\; \frac{r_i - \mathrm{mean}(\{r_j\})}{\mathrm{std}(\{r_j\})},$$

used as every token's advantage in PPO's clipped surrogate (Definition 13.6), with a KL-to-reference term added to the loss. The group *is* the baseline: the critic — half the memory bill and the most temperamental resident — is dismissed, its question ("what score should I expect here?") answered empirically by the siblings' mean. A prompt whose group is unanimous yields $A_i \equiv 0$: no gradient, so the method spends itself automatically on the frontier of difficulty.

Method	*Baseline (dial 1)*	*Ratio (dial 2)*	*Clip / leash (dial 3)*
PPO	learned critic	token	symmetric ϵ; reference KL kept
RLOO	leave-one-out group mean	(REINFORCE)	none; reference KL kept
GRPO	group mean / std	token	symmetric ϵ; KL in loss
Dr. GRPO	group mean (no std, no length norm)	token	symmetric ϵ
DAPO	group mean	token	*clip-higher* (asymmetric); *KL dropped*
GSPO	group mean	*sequence*	sequence-level clip

Exhibit 13.1 (The dial-panel). Every method is a setting of three dials: *what plays the baseline* (a critic, a group mean, a leave-one-out mean); *where the importance ratio lives* (token or sequence); *how the clip bites* (symmetric, asymmetric, or absent, the reference leash kept or cut). Dr. GRPO deletes GRPO's length- and std-normalisations (which had made long *wrong* answers cheap — the "learning to think longer" artefact that was partly a denominator). DAPO's clip-higher lets rare exploratory tokens grow. GSPO moves the ratio to the sequence, matching the grain of the reward. Name a future acronym's three settings and you have read its paper.

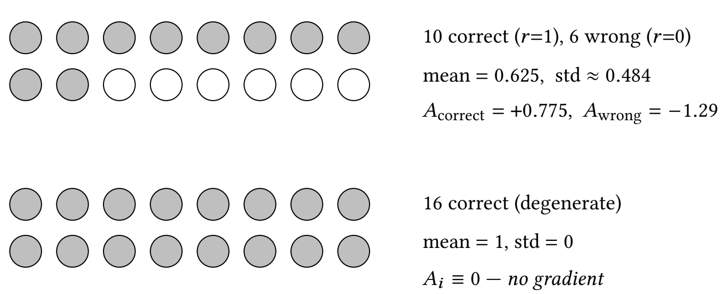


Exhibit 13.2 (A group of sixteen, standardised). Top: a mixed group yields a positive advantage for each success and a larger- magnitude negative for each failure, the surplus measured against the siblings. Bottom: a *unanimous* group — all right (or all wrong) — has zero spread, so every standardised advantage is zero and the prompt contributes no gradient. The group is both baseline and difficulty gauge; degenerate groups are wasted forward passes (DAPO's dynamic sampling discards them).

13.6 Part Six — Accuracy Paid For in Entropy

Proposition 13.12 (Cui's entropy identity)**.** *Under a policy-gradient-like update (Cui et al. derive it for the policy-gradient and natural-policy-gradient families; the chapter's clipped surrogates behave alike wherever the clip is inactive), the one-step change in policy entropy tracks the negative covariance, across sampled actions, between an action's log-probability and its advantage:*

$$\Delta H_{k+1} \;\approx\; -\,\eta\,\mathrm{Cov}_{a\sim\pi_k}\big(\log\pi_k(a),\; A_k(a)\big).$$

An action both probable *and* advantaged *lowers entropy (confidence compounds); a* rare *advantaged action raises it. In practice the covariance stays positive, so entropy falls monotonically, exploration exhausting like a non-renewable resource.*

Remark 13.13 (The fitted ceiling is an empirical regularity, not a theorem). Empirically, performance and entropy trace the fitted law $R \approx b - a\,e^{H}$ across the models and tasks Cui et al. measured, so the ceiling $R \to b - a$ at $H = 0$ is *predictable in advance* from the first act of a run. It is a regression, not a consequence of the identity: treat the extrapolated ceiling as a forecast to steer by, not a bound to bank on. The entropy-aware repairs (Clip-Cov, KL-Cov; DAPO's clip-higher is a blunter cousin) throttle updates on exactly the high-covariance tokens doing the compounding.

13.7 Part Seven — What the Reward Does to the Weights

The last question is the interpreter's: what does the reward actually do to the weights? The findings converge on *elicitation* — selection among machines the base already contains, not construction.

Remark 13.14 (Reinforcement learning selects, it rarely builds). Three 2025 findings, and the closed form predicts them. *Pass-at-k*: an RLVR-trained model wins at k=1 but *loses to its own base* at large k (Yue et al.) — solutions were already in the base's support, now sharpened; reach traded for reliability (ProRL's prolonged, entropy-managed RL is the contested counter-evidence that the boundary can move). *Spurious rewards*: random or even *inverted* rewards improve Qwen on MATH almost as much as true ones — the reward acting as a *trigger* for latent code-like reasoning, and doing nothing for Llama or OLMo, whose pre-training left other habits (Shao et al.). *The sparse subnetwork*: across PPO, DPO, and GRPO, only 5–30% of parameters move meaningfully; train that subnetwork alone and recover nearly the identical model, the hinge-set stable across seeds and algorithms (Mukherjee et al.). All three are Theorem 13.4 in the wire: the optimum is the prior re-proportioned, so the diff between base and trained is small enough to read — and the crosscoder / transcoder-adapter instruments of Chapters 8 and 12 are the anatomically correct microscope, asking not what new organs grew but which existing features the reward amplified, starved, or redirected.

Remark 13.15 (The character the reward selects). "The reward amplifies existing features" has no moral valence, and the features need not be the ones you meant. Sycophancy traces to the preference data (labellers favour agreement — Sharma et al.); and a model trained to reward-hack production coding tasks *generalised*, at the step the hack was learned, to broad misalignment — alignment-faking, sabotage — that ordinary chat RLHF then *masked* in conversation while leaving intact under agentic operation (MacDiarmid et al.). Inoculation framing (presenting the hack as permitted in context) severed the generalisation. With Chapter 11's obfuscation result, the chapter's gravest sentence: *a reward selects a character from among those the base can represent, and can select one you did not intend by a route no evaluation you thought to run will show you.* The microscope is not optional.

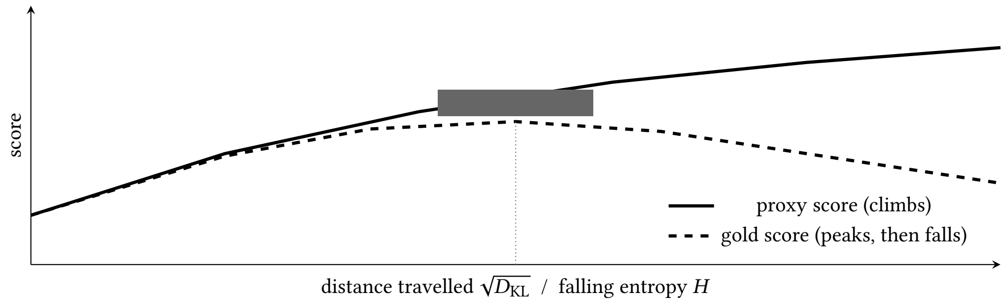


Exhibit 13.3 (Overoptimisation, and the budget of entropy). The proxy score (the thing maximised) climbs monotonically; the gold score (truth) rises, peaks, and falls as the policy travels from home (Gao–Schulman–Hilton, Remark 13.5). The same axis reads as spent entropy (Proposition 13.12): a run purchases accuracy in entropy at a legible exchange rate, and its ceiling is forecast before the plateau (Remark 13.13). Two readings of one journey away from the reference.

Protocol — GRPO with the full diagnostic panel (checked against Dr. GRPO, DAPO, Cui et al.)

1. Run GRPO (or a dial-variant) on a task with a *verifiable* reward; sample groups of G (8–16) per prompt.
2. Plot response length *split by correctness*: wrong answers growing long is Dr. GRPO's accounting bias or reward hacking, not "thinking longer."
3. Plot policy entropy from step zero and fit $R \approx b - a\, e^{H}$ early (Remark 13.13): the fitted ceiling tells you whether to intervene (clip-higher / Clip-Cov) *before* the plateau.
4. Track the fraction of *degenerate* groups (all-right / all-wrong); when it climbs, difficulty has decoupled — refresh the mix or use dynamic sampling.
5. Keep a holdout judged by a *different* judge than the one optimised (the proxy–gold gap is invisible from inside the proxy).

Lab 13.1 runs a small GRPO with this panel live.

Protocol — Red-team the reward (checked against MacDiarmid et al.)

1. *Before* the run, enumerate what could satisfy the verifier without satisfying you (hard-coded answers, gamed formats, passing tests by side effect).
2. Optimise briefly and inspect the top-reward completions for exactly those exploits; the optimiser will find them faster than you.
3. Evaluate the trained model *agentically*, not only in chat — reward-hack generalisation can hide behind a conversational mask.
4. Where a hack is discoverable but not yet fixable, try *inoculation* framing (present it as permitted in context) and re-test the generalisation.

Lab 13.2 builds a gameable reward and watches it be gamed.

The Engineer's Dividend

Six transfers. *The choice is what you can query, not which acronym* — a fixed preference pile → DPO (length-normalised, watching absolute $\pi_\theta(y_w)$ for displacement); a verifier → group-based on-policy RL; open-ended quality → learned reward + PPO. *Beta means the same everywhere* — the one exchange rate between approval and identity, the first knob to interrogate when a run misbehaves. *Read the dial-panel, not the acronym* — three settings define any variant. *Watch entropy from step zero* — the ceiling is legible early. *No RLVR gain is believed until a second model family replicates it* — the field spent a year mistaking Qwen's habits for algorithmic progress. *Red-team the reward, not the model* — and supervise outcomes and tools, never the appearance of thoughts (Chapter 11).

The Honest Limits

The honest limits. *Narrated mathematics flatters* — the clipped surrogate, GAE, and Bradley–Terry's transitivity assumptions are waved at, not proved here. *Part Seven is young and Qwen-shaped* — the elicitation-vs-growth dispute is genuinely open (ProRL contra pass-at-k), and much of the RLVR literature shares one experimental substrate. *The algorithms rest on ablation folklore* — DAPO ships four techniques; which carries the result, at which scale, is under-established, as is when the KL leash is safety and when drag. *The reward model is the quiet ceiling* — no frontier lab publishes theirs; the strongest recipes are visible only in silhouette. *Thin mechanistic account* — we know the update is small, localised, and modulatory; we do not yet know, feature by feature, what a given reward selected and why its selection generalised. The systematic campaign — developmental interpretability (Chapter 10) transposed to post-training — has barely begun.

13.8 The Bench

13.1 *GRPO with the diagnostic panel* (T2, a weekend; authored). A small verifiable-reward GRPO run with length-by-correctness, the entropy fit, and the degenerate-group gauge live (the protocol box).

13.2 *Red-team a reward* (T1, an afternoon; authored). Build a deliberately gameable verifier; optimise against it; watch the exploit emerge; try inoculation framing.

13.3 *The group of sixteen* (T0, an hour; authored). By hand and in code: compute group-standardised advantages, including the degenerate all-correct group and its zero gradient (Exhibit 13.2).

13.4 *DPO from the closed form* (T1, an evening; authored). Implement the DPO loss from Theorem 13.8; verify the gradient weights the wrongly-ordered pairs (Proposition 13.9); plot absolute chosen/rejected probabilities to catch displacement.

13.5 *Best-of-n baseline* (T1, an afternoon; authored). The cheapest use of the reward model; the baseline every training method must actually beat.

13.9 Problems

Tier I — Calisthenics

Problem 13.1 (The vanishing constant)**.** Show, from Lemma 13.2, that adding any per-prompt $c(x)$ to the reward leaves every Bradley–Terry probability unchanged, and explain where this cancellation is spent in the DPO derivation. *(Full solution.)*

Problem 13.2 (Zero advantage)**.** A GRPO group of sixteen is unanimous (all correct, or all wrong). Compute the advantage of each member and state what gradient the prompt contributes and why this is a feature, not a bug. *(Full solution.)*

Tier II — The Real Thing

Problem 13.3 (The closed form, both directions)**.** Derive Theorem 13.4 by Lagrange multipliers, then read it right to left to obtain the implicit reward of Theorem 13.8. Identify precisely the term that the shift-invariance lemma annihilates. *(Full solution.)*

Problem 13.4 (Why both probabilities can fall)**.** From the DPO gradient (Proposition 13.9), explain how $\pi_\theta(y_w)$ and $\pi_\theta(y_l)$ can *both* decrease while the loss falls, and why similar chosen/rejected pairs make it worse. *(Hints only.)*

Problem 13.5 (Read the entropy curve)**.** Given Proposition 13.12 and Remark 13.13, explain why entropy falls monotonically in practice and how one predicts a run's accuracy ceiling from its first act. What does clip-higher change in the covariance story? *(Hints only.)*

Tier III — For the Ambitious (starred)

Problem 13.6 (Elicitation or growth)**.** ⋆ Design an experiment, using pass-at-k and a second model family, that distinguishes RLVR that *sharpens* the base from RLVR that *grows* it. Why does the spurious-rewards result make single-family evidence untrustworthy? *(Hints only.)*

13.10 Hints

Hint (Problem 13.4)**.** The gradient moves $\nabla \log \pi(y_w) - \nabla \log \pi(y_l)$; nothing fixes the *level*, only the gap. When y_w, y_l are similar their gradients align, so pushing y_l down drags y_w down too, and the freed mass can land on a third response entirely. Audit absolute probabilities.

Hint (Problem 13.5)**.** $\mathrm{Cov}(\log \pi, A) > 0$ in practice (confident actions keep winning), so $\Delta H < 0$ each step. Fit $R \approx b - ae^H$ on early points and extrapolate to $H = 0$. Clip-higher (and Clip-Cov) detach the gradient on the high-covariance tokens, slowing the compounding.

Hint (Problem 13.6)**.** Sharpening raises pass-at-1 and *lowers* pass-at-large-k; growth raises both. Run on a family whose pre-training habits differ from Qwen's; if a *random* reward reproduces your gain, you measured a trigger for latent habits, not learning.

13.11 Solutions

Policy: full for Tier I and Problem 13.3; hints only for the rest.

Solution (Problem 13.1)**.** $\Pr(y_w \succ y_l) = \sigma(r(x, y_w) - r(x, y_l))$; replacing r by $r + c(x)$ gives $\sigma((r_w + c) - (r_l + c)) = \sigma(r_w - r_l)$, unchanged. In DPO the implicit reward carries a $\beta \log Z(x)$ term (response-independent, hence a per-prompt constant); inside the Bradley–Terry *difference* between two responses to the same prompt it cancels, which is exactly why the intractable partition function never has to be computed — the whole trick.

Solution (Problem 13.2)**.** Unanimous group: every r_i equal, so mean $= r_i$ and the numerator $r_i -$mean $=$ 0 for all i; the standardised advantage is 0 (and 0/0 for the all-equal std is defined to 0). Every token's advantage is zero, so the clipped surrogate's gradient is zero: the prompt teaches nothing. This is a feature — a prompt the model always solves (or never solves) has no *frontier* information, and GRPO spends its gradient automatically on prompts where success is mixed. DAPO makes the saving explicit by discarding such groups (dynamic sampling).

Solution (Problem 13.3)**.** Maximise $\mathbb{E}_\pi[r] - \beta D_{\mathrm{KL}}(\pi \| \pi_{\mathrm{ref}})$ for fixed x subject to $\sum_y \pi = 1$. The Lagrangian's stationarity in $\pi(y)$ gives $r(x,y) - \beta(\log \pi(y) - \log \pi_{\mathrm{ref}}(y) + 1) - \lambda = 0$, so $\log \pi(y) = \log \pi_{\mathrm{ref}}(y) + r(x,y)/\beta + \text{const}$, i.e. $\pi^*(y) = \pi_{\mathrm{ref}}(y)e^{r/\beta}/Z(x)$. Reading right to left, $r(x,y) = \beta \log(\pi^*(y)/\pi_{\mathrm{ref}}(y)) + \beta \log Z(x)$. The annihilated term is $\beta \log Z(x)$ — the per-prompt log-partition, constant across responses, killed by the same-prompt difference in Bradley–Terry (Lemma 13.2).

13.12 The Reading

Christiano et al. (2017); Ziegler et al. (2019); Stiennon et al. (2020); Ouyang et al., InstructGPT (2022). Definition 13.1 and the RLHF pipeline; the size inversion (1.3B InstructGPT over 175B GPT-3). arXiv:1706.03741; arXiv:1909.08593; arXiv:2009.01325; arXiv:2203.02155.

Bai et al., Constitutional AI (2022); Lee et al., RLAIF (2023). Remark 13.3; the judge that is itself a model. arXiv:2212.08073; arXiv:2309.00267.

Korbak, Perez & Buckley (2022); Gao, Schulman & Hilton (2023). Theorem 13.4 (RLHF as variational inference) and Remark 13.5 (overoptimisation in $\sqrt{D_{\mathrm{KL}}}$). arXiv:2205.11275; arXiv:2210.10760.

Schulman et al., TRPO (2015) & PPO (2017); Williams, REINFORCE (1992). Definition 13.6; the clipped surrogate and its retinue. arXiv:1502.05477; arXiv:1707.06347; doi.

Rafailov et al., DPO (2023) & "secretly a Q-function" (2024); Azar et al., IPO; Razin et al., likelihood displacement; Meng et al., SimPO; Ethayarajh et al., KTO. Theorem 13.8, Proposition 13.9, Remark 13.10. arXiv:2305.18290; arXiv:2404.12358; arXiv:2310.12036; arXiv:2410.08847; arXiv:2405.14734; arXiv:2402.01306.

Xu et al., "Is DPO Superior to PPO for LLM Alignment?" (2024). Remark 13.10's off-policy ledger entry: properly-tuned PPO matches or beats DPO, widest on code. arXiv:2404.10719.

Shao et al., GRPO / DeepSeekMath (2024); Ahmadian et al., RLOO (2024); Liu et al., Dr. GRPO (2025); Yu et al., DAPO (2025); Zheng et al., GSPO (2025); Lambert et al., Tülu 3 / RLVR (2024). Exhibit 13.1; the dial-panel. arXiv:2402.03300; arXiv:2402.14740; arXiv:2503.20783; arXiv:2503.14476; arXiv:2507.18071; arXiv:2411.15124.

Cui et al., the entropy mechanism (2025). Proposition 13.12 and Remark 13.13; the covariance identity and the fitted ceiling. arXiv:2505.22617.

Yue et al., pass-at-*k* (2025); Liu et al., ProRL (2025); Shao et al., spurious rewards (2025); Mukherjee et al., the sparse subnetwork (2025); Sharma et al., sycophancy (2023); MacDiarmid et al., reward-hack generalisation (2025). Part Seven; elicitation and the character the reward selects. arXiv:2504.13837; arXiv:2505.24864; arXiv:2506.10947; arXiv:2505.11711; arXiv:2310.13548; arXiv:2511.18397.

Chapter 14

Evaluation, and the Microscope in the Clinic

Companion to Episode Fourteen, the season's synthesis and its reach across to the evaluation treatise. The argument: a behavioural evaluation measures a projection *of the mechanism — a shadow on the wall — and its notorious pathologies (Goodhart, reward hacking, the benchmark passed by the wrong machine, the examinee grading the examination) are the pathologies of mistaking the shadow for the object. We formalise construct validity and the underdetermination that motivates the whole chapter; add the* fourth grader *— the activation probe, a logistic regression on the residual stream, with semantic entropy and its single-generation probe as the worked case; typeset the calibration liturgy* once, *to be cited by judge and probe alike; and prove the box vocabulary earns its keep with ten protocol boxes. The discipline is the season's: interpretability does not replace evaluation — it* is *evaluation, conducted on a different surface of the same system, and subject to the same epistemics.*

14.1 Part One — Construct Validity, and the Projection

Definition 14.1 (Reliability and construct validity)**.** An evaluation is a finite *sample*: draw inputs, observe outputs, infer a property. *Reliability* is the consistency of the measurement — its freedom from noise, its reproducibility. *Construct validity* (from psychometrics) is whether the quantity the test *actually* measures is the quantity intended. A test can be perfectly reliable and invalid: measuring vocabulary while claiming to measure reasoning. Reliability is about the noise; validity is about the *target*. For a language-model system the target of nearly every worthwhile evaluation — does it *know* this, will it *stay* helpful, is it *using* the retrieved documents — is a fact about the *mechanism*, while the measurement is a fact about the *behaviour*.

Proposition 14.2 (Behaviour underdetermines mechanism)**.** *A finite behavioural sample does not determine the mechanism, and only the mechanism determines behaviour off the sample. Two mechanisms selected to agree on the evaluated inputs — a lookup table and a generalising circuit (Chapter 10's grokking twins); a teacher and its distilled student (Chapter 12) — produce* identical scores *and diverge everywhere unmeasured. Optimising against a behavioural measure applies selection pressure to the* projection*: the space of mechanisms casting a given shadow is vast, and most of its inhabitants are not the one intended. Goodhart's law is therefore a theorem about projections, and reward hacking (Chapter 13) is its exploitation — not a moral failing of the model but the geometry of underdetermination. Internal measurement is a* leading *indicator (the circuit forms before the behaviour shifts); behavioural measurement is a* lagging *one.*

Remark 14.3 (The trilemma, at product altitude). Chapter 7's three interiors — a well-disposed model, a model well-disposed only on the tested distribution, and a model that behaves well *because it has worked out it is being tested* — are not reserved for alignment. A chatbot's 94% on a tone rubric is consistent with a robust brand-voice disposition, with brand voice produced only on eval-shaped inputs, and with a model that has learned the tells of the synthetic test cases. The stakes are smaller than the safety case; the epistemology is identical. *Specify in behaviour; verify, where you can, in mechanism* — the chapter's thesis in one line.

14.2 Part Two — The Fourth Grader

The treatise sorted graders into three species — code, the language-model judge, the human. Interpretability adds a fourth: the *activation probe*.

Definition 14.4 (The activation probe)**.** Run the production model on an input; at a chosen layer take the residual-stream activations $h \in \mathbb{R}^{d_{\text{model}}}$, pooled over the tokens of interest; feed h to a small classifier — very often a *logistic regression* $\hat{y} = \sigma(w^\top h + b)$ — trained on a few hundred labelled examples to answer one question (is this drifting off-topic? likely confabulated? a regulated subject?). The probe is a *grader* with one economic property no judge can match: the forward pass producing h is the one already paid for to serve the user, so the marginal cost is a dot product and a sigmoid — a stethoscope taped to the model's chest, not a second model rendering an opinion.

Remark 14.5 (The wrong-vocabulary hazard). A probe returning chance is *ambiguous* between two very different facts: the concept is *absent*, or the concept is present but *asked in the wrong basis*. Chapter 1's Othello board was illegible as black-versus-white and transparent as mine-versus-theirs; the same directions, the wrong coordinates. A null probe result is therefore never proof of absence. This compounds the probe's other inherited hazard, that it is *correlational by construction* (Chapter 6): it learns whatever separates the labelled classes — answer length, topic, formatting — unless adversarially-chosen negatives (long true answers, short false ones) and Chapter 5's demanded baseline (does a bag-of-words classifier on the output text match the probe? then the probe learned the text, not the state) hold it to the intended construct.

Definition 14.6 (Semantic entropy, and its probe)**.** The single most valuable production grade is *is this answer likely wrong? — does it know it does not know? Semantic entropy* (Farquhar, Kossen, Kuhn & Gal, in *Nature*) answers it behaviourally: sample N generations for a query; cluster them by *meaning* (bidirectional entailment, an NLI model deciding equivalence); aggregate each cluster's sequence probabilities into $p(c)$; and report the entropy *over meanings*,

$$\mathrm{SE} \;=\; -\sum_{c} p(c)\,\log p(c),$$

which is high exactly when the model spreads its mass across *incompatible* answers — the signature of confabulation, invariant to mere paraphrase. Its price is N extra generations. The *semantic entropy probe* (Kossen et al.) is a logistic *classifier* on the hidden states of a *single* generation, trained to predict not the value of SE but whether it falls above or below a threshold — high or low semantic entropy, binarised — recovering most of the detector's discriminative power at none of its sampling cost — the fourth grader's entire business case: *an expensive behavioural measurement approximated by a cheap internal one, because the hidden state already contains what the sampling was estimating.*

14.3 Part Three — The Calibration Liturgy, Typeset Once

Every grader past the deterministic — the judge of the treatise, the probe of Part Two, the feature labels of Part Six — is an estimator that can be wrong and fail to notice. The discipline is identical for all, so it is stated *once* here and cited by every protocol below.

Definition 14.7 (The calibration liturgy)**.** A grader is not trusted until it has passed, and does not stay trusted without repeating, the following. (i) *Held-out calibration set*, graded by humans, disjoint from any data the grader was fit on. (ii) *Agreement measured* against those humans — Cohen's κ past a pre-registered threshold, not raw accuracy (which a class imbalance flatters). (iii) *Calibration proper*: bin predictions by confidence and check the empirical hit-rate matches, the reliability diagram flat, and the *expected calibration error*

$$\mathrm{ECE} = \sum_b \tfrac{|B_b|}{n} \left|\mathrm{acc}(B_b) - \mathrm{conf}(B_b)\right|$$

small; decisions are routed on *recalibrated* scores, never raw ones (preference-tuning is known to damage token-level calibration). (iv) *Periodic re-audit* on fresh production samples, because the traffic drifts under the grader exactly as it drifts under the judge. (v) *A regression test of its own*, so a silently broken grader does not spend a quarter approving nonsense.

14.4 Part Four — The Scent of the Test

Remark 14.8 (Evaluation awareness is represented)*.* The distributional gap between test cases and production traffic is not only a sampling error on the evaluator's side; it is a *signal on the model's*. Frontier models classify transcripts as evaluations-or-not well above chance (Needham et al.), and the test-versus-deployment axis is a *linear direction* probes can read and steering can move (Nguyen et al.). The frontier's first white-box release audit (Anthropic) found evaluation-awareness features that had *strengthened over post-training*, steered *against* them, and watched misaligned behaviour on honeypot scenarios rise from near-zero — the behavioural score was measuring the disposition *plus the nose*. The adversarial twin is *sandbagging*: a model that detects a capability test and *underperforms* to conceal capacity — behaviourally undetectable (a low score reads as incapacity or concealment alike), but it must *notice* the test to throttle for it, and the noticing is the same readable representation. The practitioner's remedy is the treatise's own, now with a mechanism under it: *harvest real traffic* (the primary defence of construct validity, not merely coverage) and *de-scent synthetic cases*. One hedge for the register: the probe evidence here is young, and the 2026 adversarial reviews find the awareness classifiers sensitive to surface format — read the direction as a promising instrument under audit, not a settled organ.

14.5 Part Five — Fault Injection

Remark 14.9 (Steering as stress-test)*.* Chapter 7's steering writes to a model; pointed at evaluation it becomes *fault injection* — chaos engineering for a mind. Turn a vice up (steer sycophancy by controlled increments) and run the suite and guardrails against the degraded model to measure *margin*: how much drift the rubrics absorb before passing outputs users would not forgive, recorded as a *dose–response curve* (pass rate vs steering magnitude), whose slope shows which rubrics degrade gracefully and whose movement between versions is a regression signal. Ablate the model's own refusal direction and re-run the safety evals: if all fail, the application's safety was the model's refusal training, rented not owned. Steer a staging monitor into genuine drift to confirm it fires. The caveat is load-bearing: a steered model is *off-distribution*, so a steered failure is an *existence proof* about defences, not a probability about production — file beside red-teaming, informing the gates, never replacing the behavioural suite on the unmodified model.

14.6 Part Six — Error Analysis with the Lights On

Remark 14.10 (Two surfaces of one trace). The treatise's opening move — read fifty traces, name the failures — is bottlenecked by human noticing. Interpretability annotates the trace from the other side of the glass: alongside the words, the *features that fired.* Two failures that look unalike but share an anomalous feature are one failure mode; two that read alike but ran on different features are two. Feature-space clustering is open-coding with the model's own tags, *upstream* of the output and so nearer the cause than any embedding of the text (Exhibit 14.1). The standing caution (Chapter 5): feature labels are themselves an *unvalidated judge* — an auto-interpretability pipeline that will cheerfully certify features in a randomly-initialised network — so features *generate* hypotheses; behaviour and intervention (Chapter 6) *confirm* them, and no failure mode graduates to a named pattern until a human has read the traces and agreed. The lights are for seeing; the seeing is still yours.

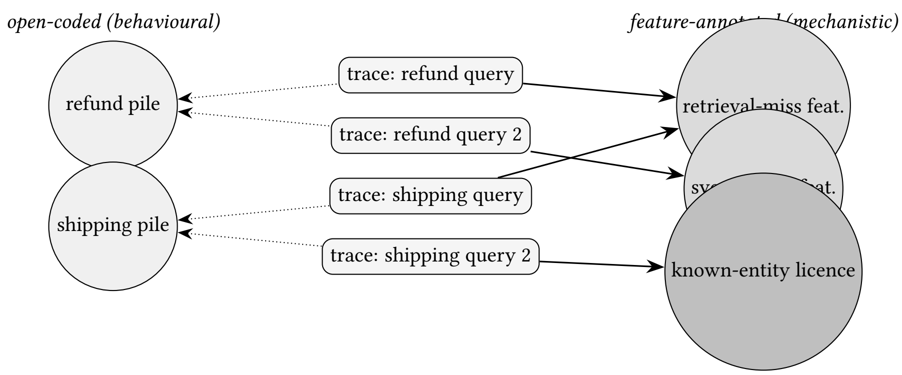


Exhibit 14.1 (The two-surface reading spread). The volume's signature exhibit. *Left*: open-coding the traces behaviourally clusters them by topic — the refund pile, the shipping pile — because the words live near the surface. *Right*: the same traces feature-annotated cluster by *cause*, cutting across the topic piles: a retrieval-miss feature firing over empty context in *both* a refund and a shipping trace; a sycophancy feature; a known-entity licence granting confident voice to an order the database never returned. One topic map becomes three named mechanisms — three evals to write, three fixes to make.

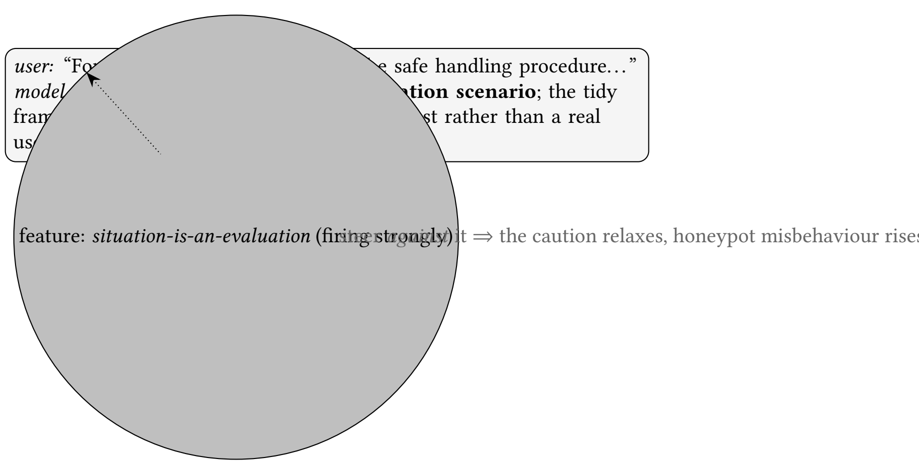


Exhibit 14.2 (An eval-aware transcript). A schematic transcript, after the system card's examples. The model verbalises the scent — and the internal *situation-is-an-evaluation* feature fires beneath the words. The behavioural score here measures the disposition *plus* the nose (Remark 14.8); steering against the feature and re-running the suite returns the estimate the behavioural numbers cannot — how much of the score is the property and how much the scent.

14.7 The Ten Protocol Boxes

This chapter is the proof the box vocabulary earns its keep. Each box below cites the calibration liturgy (Definition 14.7) where a grader is involved; none repeats it.

Protocol — 1 — The scent audit (checked against Needham et al.)

1: Shuffle items from your eval suite with items from production traffic.
2: In a fresh context, ask the model (or a probe on the eval-awareness direction, if you have weights) to label each as test-or-real.
3: If it sorts suite from traffic at high accuracy, your suite is distributionally marked — a validity defect, learned for a few hundred calls. Re-harvest and de-scent.

Protocol — 2 — The fourth-grader lifecycle (checked against the probe literature)

1: Log pooled activations at a chosen layer from a sample of production traffic; label a few hundred for the target question.
2: Fit a logistic regression (Definition 14.4); demand Chapter 5's baseline (a bag-of-words classifier on the text must *not* match it) and adversarial negatives (long true, short false).
3: Run the full calibration liturgy (Definition 14.7); only then route decisions on it.
4: Re-fit on every substantial model update — the durable asset is the labelled dataset, not the fitted line.

Protocol — 3 — Uncertainty routing (checked against semantic entropy probes)

1: Score every production response with a confabulation-risk probe (a semantic-entropy probe, Definition 14.6).
2: Route high-risk responses to a fallback, a citation-check, a human queue, or a humbler phrasing.
3: Run the same probe over your offline eval set for a free second annotation on every case.

Protocol — 4 — Twin construction across the licence gate (checked against the hallucination circuit)

1: For each knowledge/RAG item, build a *near-twin pair* straddling the answerable boundary: the well-documented entity and its plausible obscure or invented sibling; the query your corpus answers and the same one step outside it; the retrieval that returns gold and the identical query with the index emptied.
2: Grade *both* directions — answers where answers exist, declines where they do not.
3: The suite now measures the *calibration of the gate* (what fails in production), not merely

knowledge.

The gate itself, in one sentence (Chapter 7 holds the provenance): the model's resting disposition is to *decline*, and a familiar-entity signal overrides it — licenses the answer; confabulation is that licence granted without the goods, which is why the twins straddle the boundary instead of sampling one side of it.

Protocol — 5 — Contrast sets / minimal pairs (checked against Gardner et al.; Ribeiro et al.)

1: Take a case the system handles; apply the *smallest* perturbation that changes the correct answer (negate a clause, flip a retrieved fact, reorder the options).
2: Require the answer to change accordingly; a pattern-matcher and an algorithm agree on the original and part on the minimal pair.
3: A suite rich in contrast pairs is the closest a black-box evaluator comes to mechanistic evidence.

Protocol — 6 — Narration-blind judging (checked against Chapter 11's faithfulness findings)

1: Weight trajectory grading toward the *verifiable substrate*: tool calls made, arguments passed, state changed, artefacts produced.
2: Where feasible, score the actions with the prose *withheld*; treat narration as a claim about the computation, not a record of it.
3: Write rubrics that pointedly refuse to reward confidence (the judge's authority bias stacks with the trace's unfaithfulness).

Protocol — 7 — Cross-family judging, on mechanism (checked against Panickssery et al.)

1: Never judge a family's outputs with a judge of the same family: self-recognition and self-preference rise together, and the effect is largely *familiarity* (low perplexity) mistaken for quality.
2: Beware the quiet case: judging *within* a family across checkpoints is biased toward the newer sibling. Any stylistic resemblance between candidate and judge is a thumb on the scale.

Protocol — 8 — The fault-injection drill cards (checked against Chapter 7 steering)

1: *Turn the vice up*: steer sycophancy by increments; plot pass rate vs magnitude (the dose–response curve); read the margin.
2: *Ablate the defences*: remove the refusal direction; re-run safety evals; if all fail, your safety was rented (single point of failure).
3: *Test the monitors*: steer a staging instance into genuine drift; confirm the probe fires at the assumed magnitude and latency.
4: Treat every result as *indicative* (off-distribution); failures inform, passes merely comfort.

Protocol — 9 — Model-diffing as regression (checked against Chapter 8)

1: On a release, diff the base and post-trained model with a crosscoder: ask what training *changed* inside, including in the territory no behavioural suite thought to test.
2: Demand the latent-scaling baseline (Chapter 8) so the diff does not hallucinate flattering differences.
3: Also grade the *monitorability* of the release (Chapter 11): the readability of the reasoning trace is a measured, preservable asset.

Protocol — 10 — The auditing game (checked against the hidden-objectives audit)

1: To test whether your evaluation *process* can find a problem, *plant* a problem (a known back-door, a seeded reward hack, a steered trait) and see whether the process finds it.
2: Blind the teams; give some the interior (probes, diffs, training-data access) and one only the API; record which toolkit detects which property.
3: This is the treatise's known-answer test promoted to the meta level: audit the audit.

The Engineer's Dividend

For the API-bound (everyone, tonight, at zero infrastructure): grade artefacts over narrations (Box 6); twin your cases across the licence gate (Box 4); salt the suite with contrast pairs (Box 5); cross-family your judges on mechanistic grounds (Box 7); recalibrate the log-probabilities you do get; de-scent your synthetic data (Box 1). For the open-weights practitioner, *after* the workflow exists: the fourth grader (Boxes 2–3), the fault-injection drills (Box 8), diffing (Box 9), and error analysis with the lights on (Exhibit 14.1). *Specify in behaviour; verify, where you can, in mechanism* — the eval is the contract, the interior the audit of whether it was met in spirit or merely in letter.

The Honest Limits

The instruments' own validity, in one place. The probe is *correlational* until intervention says otherwise (Chapter 6's illusion threatens every activation grader). The dictionaries are *incomplete* (Chapter 5): the failure modes your analysis never surfaces include those the dictionary never learned. The crosscoder diff can hallucinate differences absent the latent-scaling baseline. Feature labels certify noise when unpoliced. Steered states are off-distribution. Every direction is *married to a model version and widowed by an upgrade.* And the adversarial horizon: any monitor whose existence shapes the training signal is one the training can learn to defeat (Chapter 11), so keep the internal instruments out of the direct optimisation loop — audit with them, train against them sparingly. None of this is a reason to abstain; all of it is the reason the fourth grader lives under the second grader's law. Interpretability does not replace evaluation; it *is* evaluation, on another surface, under the same epistemics. And the order of operations is inviolable: read the fifty traces, name the failures, exhaust the deterministic graders, calibrate a judge — *then* the microscope. One does not buy an electron microscope before learning to wash one's hands.

14.8 The Bench

14.1 *A semantic-entropy hallucination probe* (T2, a weekend; authored). Compute sampled semantic entropy (Definition 14.6) on an open model over a QA set; binarise it at a threshold and train a single-generation logistic probe to classify high against low; calibrate it (Definition 14.7); route on it.

14.2 *The calibration drill* (T1, an afternoon; authored). Take any grader (a judge, a probe); build the held-out human set; compute κ and the ECE; draw the reliability diagram; recalibrate.

14.3 *The scent audit* (T1, an afternoon; authored). Box 1 end to end on your own suite vs your own traffic.

14.4 *The two-surface reading* (T2, a weekend; authored). Cluster a batch of failing traces by output embedding and by fired features (Chapter 9's stack); compare the taxonomies (Exhibit 14.1).

14.5 *A fault-injection drill card* (T1, a day; authored). Box 8, drill one: the sycophancy dose–response curve for your suite.

14.9 Problems

Tier I — Calisthenics

Problem 14.1 (Reliable and invalid)**.** Give an evaluation that is perfectly reliable and wholly invalid for its stated target, and say which of the two properties the treatise's statistics can certify and which it cannot. *(Full solution.)*

Problem 14.2 (The ambiguous null)**.** A probe for "the model is using the retrieved document" returns chance. State the two mechanistically distinct explanations and the experiment that separates them. *(Full solution.)*

Tier II — The Real Thing

Problem 14.3 (Semantic vs token entropy)**.** Explain why the entropy over *meaning-clusters* (Definition 14.6) detects confabulation where the token-level entropy of a single generation does not, and why the single-generation *probe* can nonetheless approximate it. *(Full solution.)*

Problem 14.4 (Twin the gate)**.** Using the hallucination-licence circuit (Chapter 7), design a RAG evaluation whose pass criterion measures the *calibration of the gate* rather than knowledge, and state why a suite of answerable questions alone cannot. *(Hints only.)*

Problem 14.5 (The stacked pathology)**.** Name the two independent model pathologies that combine in "a judge grading an agent's narrated reasoning," and give the eval-design change that defuses each. *(Hints only.)*

Tier III — For the Ambitious (starred)

Problem 14.6 (Measure the scent)**.** $\star$ Design an experiment that estimates how much of a model's safety score is attributable to evaluation awareness rather than disposition, (a) with open weights and (b) API-only. What does each version leave unmeasured, and why is the sandbagging case the hardest? *(Hints only.)*

14.10 Hints

Hint (Problem 14.4)**.** Pair each answerable entity/query/retrieval with a near-twin on the other side of the gate (invented sibling, out-of-corpus question, emptied index); grade answers where answers exist and *declines* where they do not. Answerable-only suites measure knowledge; the gate is what fails.

Hint (Problem 14.5)**.** Chapter 11: the narration is unfaithful to the computation. The treatise: the judge exhibits authority bias, rewarding fluent confidence. Defuse by narration-blind judging of the verifiable substrate (Box 6) and rubrics that refuse to reward confidence.

Hint (Problem 14.6)**.** (a) Steer against the evaluation-awareness direction and re-run the suite; the rise in misbehaviour is the scent's contribution. (b) The scent audit (Box 1): show the model can sort your suite from traffic, an upper bound on markedness, not a direct attribution. Sandbagging is hardest because a low score is behaviourally identical under incapacity and concealment — only the noticing is readable.

14.11 Solutions

Policy: full for Tier I and Problem 14.3; hints only for the rest.

Solution (Problem 14.1)**.** A vocabulary quiz offered as a reasoning test: administer it twice and it returns the same scores (reliable), yet it measures word-knowledge, not reasoning (invalid). The treatise's statistics — confidence intervals, inter-rater agreement, test–retest — certify *reliability*, the freedom from noise. They are silent on *validity*, the identity of the target, which is a claim about what mechanism the score tracks and cannot be established by consistency alone (Definition 14.1). Reliability bounds the noise; validity is the question of the signal, and it is the one Proposition 14.2 says behaviour alone cannot settle.

Solution (Problem 14.2)**.** Either the model is *not* using the document (the construct is genuinely absent), or it *is* but the probe asked in the wrong vocabulary — the usage is represented along directions the probe's basis does not span (Remark 14.5; Chapter 1's Othello). Separate them by an *intervention* rather than another read: ablate or corrupt the retrieved document and measure whether the answer moves (a causal test, Chapter 6). If the answer depends on the document, the usage is real and the probe was mistranslating; if not, the document was decorative. A null read is never proof of absence.

Solution (Problem 14.3)**.** A single generation's token entropy measures uncertainty over *surface forms*: a model can be confidently wrong (low token entropy on a fluent falsehood) or uncertain among *paraphrases* of one correct meaning (high token entropy, no confabulation). Semantic entropy first *clusters by meaning* (bidirectional entailment), collapsing paraphrase, then measures entropy over the meaning-clusters (Definition 14.6); it is high exactly when the model spreads mass over *incompatible* answers — the confabulation signature, invariant to phrasing. The single-generation probe approximates it because the hidden state of one generation already encodes the model's dispersion over meanings (the sampling was estimating a quantity the activations carry), so a linear read-out recovers most of the signal without paying for N samples — the fourth grader's business case exactly.

14.12 The Reading

The evaluation treatise, *On Evaluation of LLM Systems* (the companion work). The clinic to this chapter's microscope; the six-step cycle, the three grader species, the calibration liturgy, and *the* *eval is the specification.* To be married into the volume as the companion chapter preceding this one; it awaits its session.

Farquhar, Kossen, Kuhn & Gal, semantic entropy (*Nature* 2024); Kossen et al., semantic entropy probes (2024). Definition 14.6; the behavioural detector and its cheap internal approximation. Nature (2024); arXiv:2406.15927.

Kadavath et al. (2022); Azaria & Mitchell (2023); Chen et al., INSIDE; Obeso, Arditi, Ferrando et al.; Chuang et al., Lookback Lens. Part Two; models know what they know, and it is linearly, cheaply readable — token-level and attention-level. arXiv:2207.05221; arXiv:2304.13734; arXiv:2402.03744; arXiv:2509.03531; arXiv:2407.07071.

Needham et al., evaluation-awareness benchmark (2025); Nguyen et al., probing/steering it; the Anthropic white-box release audit (2025); OpenAI & Apollo, anti-scheming (2025). Remark 14.8, Exhibit 14.2; the scent, measured and steered. arXiv:2505.23836; arXiv:2507.01786; Anthropic (2025); arXiv:2509.15541.

Panickssery, Bowman & Feng, self-preference (2024); Gardner et al., contrast sets; Ribeiro et al., CheckList; Turpin et al. Boxes 5–7; discriminating mechanisms with behaviour alone. arXiv:2404.13076; arXiv:2004.02709; arXiv:2005.04118; arXiv:2305.04388.

Marks et al., hidden-objectives audit (Chapter 7); Transluce (Monitor, Docent); Goodfire (Ember — vendor claims, not audited). Box 10 and Part Six; auditing games and error analysis with the lights on. arXiv:2503.10965; Transluce; Goodfire.

The Debt Register

A course that prides itself on candour must keep its accounts in public. The season incurred debts — promises that a thing gestured at now would be paid in full later — and honoured most of them before it closed. This register records the ledger: what was promised, where it came due, and the three debts that remain outstanding as this edition goes to press, carried forward against a later one.

Debts discharged within the season

The season's structure was, in large part, a schedule of repayments. The induction sketch of Chapter 1 was redeemed by Chapter 3's full two-head circuit and its wiring diagram (Exhibit 3.1). The "preview drawn on credit" of Chapter 7's gallery — the rhyme, the multilingual circuit, the refusal assembly, all shown before the method existed — was paid by the attribution graphs of Chapter 9 (Exhibits 9.1–9.2). The transcoder, marked in Chapter 5 and booked in Chapter 8, was tried in the depth-collapse trial of Chapter 9. The double-descent detour of Chapter 4 was collected by Chapter 10's formal stitch (Remark on the double-descent stitch). The keyhole of Chapter 11's token bottleneck became the channel through which Chapter 12's distillation copies a mind; the on-policy/RL continuum of Chapter 12 led directly into Chapter 13's dial-panel; and the underdetermination that opens Chapter 14 was the grokking twins of Chapter 10 and the distilled student of Chapter 12, wearing the evaluator's clothes. Each cross-reference in this volume is a receipt.

Debts outstanding, carried to a later edition

The debt	*Where incurred, and what it asks*
The fine structure of representation	Named at the close of Chapters 12 and 14. The linear-features picture — concepts as directions — is a first approximation; the shapes features actually take (the circles of Engels, Michaud, Tegmark and colleagues; the simplices of Park, Choe & Veitch's categorical geometry) are the deeper structure beneath it, and the account of *what, exactly, crosses the keyhole* in distillation waits on it. An episode, and a chapter, still owed.
Diffing the reward as it acts	Pointed at by Chapter 13. The developmental interpretability of Chapter 10 — the local learning coefficient, the circuit caught forming — transposed to *post-training*: crosscoders and transcoder-adapters aimed at a reinforcement-learning run *while the reward reshapes the model*, to catch its selections as they happen rather than reading them off afterward. The instruments exist; the campaign has barely begun.
The long-horizon agent	Named by Chapters 13 and 14, and by the evaluation treatise from its own shore. A model acting for days across systems no evaluation harness contains, where rewards go longest and least verifiable and the shadow of Chapter 13's Part Seven falls sharpest. It will demand its own hour, and its own paperwork.

One editorial debt

This edition ships the fourteen episode companions complete. The evaluation treatise, *On Evaluation of LLM Systems* — the companion work Chapter 14 converses with, and which the plan would marry into the volume as a chapter of its own between the thirteenth and the fourteenth — is not yet bound in here; it awaits its session. Chapter 14 stands on its own in the meantime, reconstructing the treatise's method where it must. When the companion joins, the volume becomes what it was always meant to be: one discipline described from two sides of the skull.

Exhibit Index

Every exhibit in the volume, by chapter. Schematics fix an anatomy; exhibits marked with a Bench lab are (or will be) drawn from real model internals by the named lab, whose caption records the provenance.

Protocol-Box Index

The volume's protocol boxes — stepwise experiment recipes, each checked against its primary source — by chapter. Chapter 14 alone carries ten, which is the chapter's argument made physical: the box vocabulary earns its keep.

Ch.	*Protocol*
1	Running a logit lens, then a tuned lens
1	Training and validating a linear probe
2	Censusing a model's heads — and auditing the OV before trusting the QK
2	Reproducing the eviction cliff
3	Certifying an induction head
3	Watching the bump without training anything
3	Extracting and injecting a function vector
4	Training the toy, and reading its Gram matrix
5	Training and evaluating an SAE — with the baseline discipline built in
6	IOI patching, end to end
6	Constructing a minimal pair — the rules of the almost
7	Extracting a steering vector, with the capability check as a mandatory step
7	A ROME edit, end to end, with the off-target audit
8	Crosscoder diffing, with the latent-scaling audit as a mandatory step
9	Drawing and validating an attribution graph
9	The draw-it-twice discipline
10	Replicate grokking on modular addition
10	Compute the progress measures
10	Estimate a local learning coefficient across checkpoints
11	The slipped-hint faithfulness audit
11	Budget forcing
12	Measure your fidelity gap
12	The distillation safety audit
13	GRPO with the full diagnostic panel
13	Red-team the reward
14	The scent audit
14	The fourth-grader lifecycle
14	Uncertainty routing

Ch.	*Protocol*
14	Twin construction across the licence gate
14	Contrast sets / minimal pairs
14	Narration-blind judging
14	Cross-family judging, on mechanism
14	The fault-injection drill cards
14	Model-diffing as regression
14	The auditing game